

**BIOCHEMICAL BASIS OF RESISTANCE IN
CHICKPEA (*Cicer arietinum* L.) AGAINST DRY
ROOT ROT CAUSED BY *Macrophomina phaseolina*
(Tassi) Goid. AND ITS MANAGEMENT**

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PALB 3073

**DEPARTMENT OF PLANT PATHOLOGY
UNIVERSITY OF AGRICULTURAL SCIENCES
BENGALURU – 560 065**

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Thesis submitted to the

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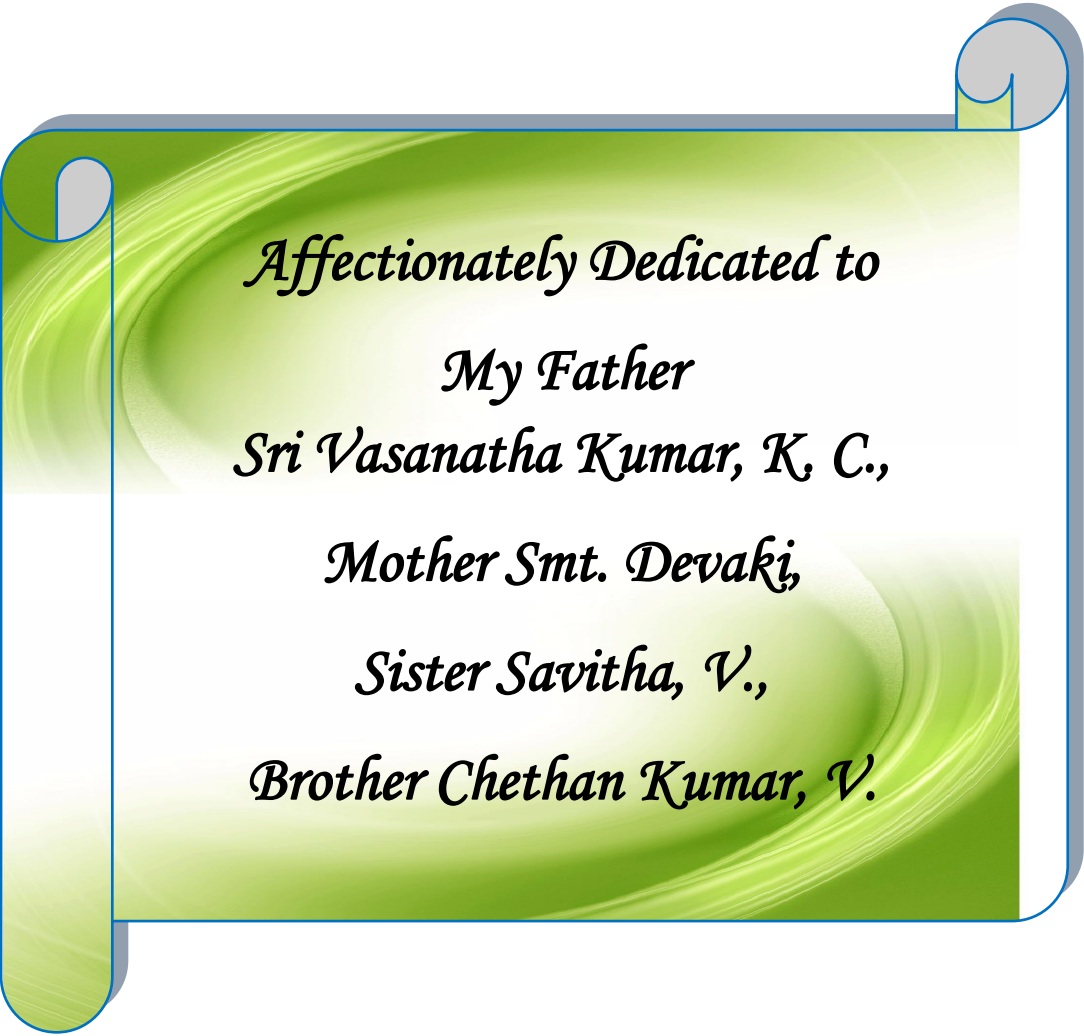
DOCTOR OF PHILOSOPHY

IN

PLANT PATHOLOGY

BENGALURU

DECEMBER, 2016



Affectionately Dedicated to
My Father
Sri Vasanatha Kumar, K. C.,
Mother Smt. Devaki,
Sister Savitha, V.,
Brother Chethan Kumar, V.

**DEPARTMENT OF PLANT PATHOLOGY
UNIVERSITY OF AGRICULTURAL SCIENCES
GKVK, BENGALURU-560065**

CERTIFICATE

This is to certify that the thesis entitled **“BIOCHEMICAL BASIS OF RESISTANCE IN CHICKPEA (*Cicer arietinum* L.) AGAINST DRY ROOT ROT CAUSED BY *Macrophomina phaseolina* (Tassi) Goid. AND ITS MANAGEMENT”** submitted by **Ms. JYOTHI, V., ID. No. PALB 3073** in partial fulfillment of the requirements for the award of the degree of **DOCTOR OF PHILOSOPHY** in **PLANT PATHOLOGY** to the University of Agricultural Sciences, Bengaluru, is a record of research work done by her during the period of her study in this University under my guidance and supervision and that no part of the thesis has been submitted for the award of any other degree, diploma, associateship, fellowship or any other similar titles.

BENGALURU
DECEMBER, 2016


M. SAIFULLA

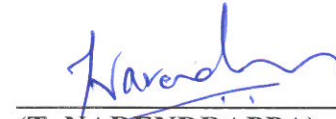
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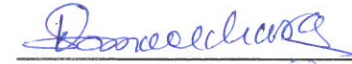
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“With regardful memories”

Destiny is not a matter of chance; it is a matter of choice. It is not a thing to be waited for; it is a thing to be achieved. Any specific endeavour is not a result of individual's effort. It would certainly be wrong on my part if I claim I have carried entire research on my own without anybody's help. The gratitude has been a courtesy of rich culture towards all who shares burden of my work and help to succeed in my endeavour. Hence, it is a matter of pleasure and a sincere duty to acknowledge all the persons and spirits in the form of teachers, friends, near or dear ones who directly or indirectly gave their supporting hands while doing the research or preparation of this present manuscript. It is the most difficult task to accomplish and to express all the feelings and sense of gratitude in words as “Gratitude being the memory of mind and feelings of heart”.

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*Bengaluru
December, 2016*

(Jyothi, V.)

BIOCHEMICAL BASIS OF RESISTANCE IN CHICKPEA (*Cicer arietinum* L.) AGAINST DRY ROOT ROT CAUSED BY *Macrophomina phaseolina* (Tassi) Goid. AND ITS MANAGEMENT

JYOTHI, V.

THESIS ABSTRACT

Dry root rot (DRR) caused by *Macrophomina phaseolina* is an important and destructive disease of pulses. Survey for chickpea DRR in southern parts of Karnataka showed the disease incidence ranged from 8.65-69.30 per cent with highest mean disease recorded in Chitradurga (56.10 %). Five isolates viz., KAMP-2, KAMP-3, KAMP-4, TNMP-10 and PUMP-14 caused cent per cent disease and were highly virulent. Based on host differentials reaction against *M. phaseolina* isolates screened at five locations, there were prevalence of three strains viz., I-(Bengaluru, Coimbatore and Raipur), II-(Durgapur), III-(Sabour). The phylogenetic tree obtained by comparing the nucleotide sequences of ITS-region in twenty *M. phaseolina* isolates resulted clusters A and B indicating variability and existence of two pathotypes. Among 202 chickpea germplasm screened for DRR resistance in field condition, 25 were moderately-resistant. Increase in activity of peroxidase, catalase, total-phenols, Phenylalanine-ammonia-lyase, total free amino-acids and decrease in total-sugar, reducing-sugar and non-reducing sugar was recorded in genotypes when inoculated with *M. phaseolina*. Among microbial antagonists, maximum inhibition (81.44 %) was observed in *Trichoderma harzianum* (Th-55) followed by *Bacillus subtilis* (75.56 %). The systemic fungicides viz., tebuconazole, carbendazim, difenoconazole and propiconazole inhibited cent per cent at 50 ppm concentration. Whereas, thiophanate methyl inhibited cent per cent at 500 ppm. The contact fungicide mancozeb, inhibited cent per cent at 250 ppm. Combi-products viz., carboxin 37.5 % + thiram 37.5 % WP and carbendazim 12 % + mancozeb 63 % WP showed cent per cent inhibition at 250 ppm. The DRR was effectively managed during *rabi* and *kharif* season under field condition by seed treatment with tebuconazole (2 g/kg of seed) which recorded lowest DRR incidence (8.70 %) with highest yield (726.05 kg/ha) and cost benefit ratio (1:2.37) compared to control which recorded 43.91 per cent disease incidence with yield of 284.70 kg/ha.

December, 2016
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ಜೀವರಾಸಾಯನಿಕಗಳ ಆಧಾರದ ಮೇಲೆ ಕಡಲೆ (ಸೈಜರ್ ಅರಿಯಾಟಿನಮ್ ಎಲ್.) ಒಣ ಬೇರು ಕೊಳೆರೋಗ (ಮ್ಯಾಕ್ರೋಫೋಮಿನ ಪ್ಯಾಸಿಯೋಲಿನ (ಟಪ್ಸಿ) ಗೋಯ್ಡ್. ನಿರೋಧಕತೆ ಮತ್ತು ಅದರ ನಿರ್ವಹಣೆ

ಜ್ಯೋತಿ, ವಿ.

ಸಂಶೋಧನ ಸಾರಾಂಶ

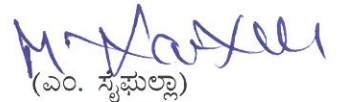
ಮ್ಯಾಕ್ರೋಫೋಮಿನ ಪ್ಯಾಸಿಯೋಲಿನ (ಎಂ.ಪಿ.) ಉಂಟುಮಾಡುವ ಕಡಲೆಯ ಒಣ ಬೇರು ಕೊಳೆರೋಗವು ಒಂದು ಪ್ರಮುಖ ರೋಗವಾಗಿದ್ದು ದ್ವಿಧಳ ಧಾನ್ಯಗಳಲ್ಲಿ ಹೆಚ್ಚಾಗಿ ಕಂಡುಬರುತ್ತದೆ. ಸಮೀಕ್ಷೆಯ ಪ್ರಕಾರ ಈ ರೋಗವು ದಕ್ಷಿಣ ಕರ್ನಾಟಕ ಜಿಲ್ಲೆಗಳಲ್ಲಿ ಶೇ. ೮.೬೫ ರಿಂದ ೬೯.೩೦ ರಷ್ಟು ಕಂಡುಬಂದಿದ್ದು, ಅತಿ ಹೆಚ್ಚು ರೋಗದ ಪ್ರಮಾಣವು ಚಿತ್ರದುರ್ಗ ಜಿಲ್ಲೆಯಲ್ಲಿ (ಶೇ. ೫೬.೧೦) ಕಂಡುಬಂದಿದೆ. ಐದು ಎಂ.ಪಿ. ಮಾದರಿಗಳಾದ, ಕೆ.ಎ.ಎಮ್.ಪಿ-೨, ಕೆ.ಎ.ಎಮ್.ಪಿ-೩, ಕೆ.ಎ.ಎಮ್.ಪಿ-೪, ಟಿ.ಎನ್.ಎಮ್.ಪಿ-೧೦ ಹಾಗೂ ಪಿ.ಯು.ಎಮ್.ಪಿ-೧೪ ಶೇ. ೧೦೦ ರಷ್ಟು ಸೋಂಕನ್ನು ಉಂಟುಮಾಡಿದೆ. ಮ್ಯಾಕ್ರೋಫೋಮಿನ ಪ್ಯಾಸಿಯೋಲಿನ ಆಶ್ರಿತ ಗಿಡಗಳ ಪ್ರತಿಕ್ರಿಯೆ ಆಧಾರದ ಮೇಲೆ ಇವುಗಳನ್ನು ಮೂರು ಗುಂಪುಗಳನ್ನಾಗಿ ವಿಂಗಡಿಸಲಾಗಿದೆ, ಸ್ಟ್ರೈನ್-೧ (ಬೆಂಗಳೂರು, ಕೊಯಂಬತ್ತೂರು, ರಾಯ್‌ಪುರ), ಸ್ಟ್ರೈನ್-೨ (ದುರ್ಗಾಪುರ), ಹಾಗೂ ಸ್ಟ್ರೈನ್-೩ (ಸಾಬೋರ್). ಐಟಿಸ್ ಸರಣಿಯ ಅಣುಶಾಸ್ತ್ರೀಯ ವೈವಿಧ್ಯತೆಯ ಜೀವವಿಕಾಸ ಚರಿತ್ರೆಯ ಅಧ್ಯಯನದ ಮೇಲೆ ಇಪ್ಪತ್ತು ಎಂ.ಪಿ. ಪ್ರತ್ಯೇಕಕರಣಗಳು “ಎ” ಮತ್ತು “ಬಿ” ಸಮೂಹಗಳಾಗಿ ವಿಂಗಡಣೆಯಾಗಿದ್ದು ಎರಡು ಉಪಜಾತಿಗೆ ಸೇರಲ್ಪಟ್ಟಿದೆ. ೨೦೨೨ ಕಡಲೆಯ ನಮೂದುಗಳನ್ನು ಕ್ಷೇತ್ರ ಪರತ್ತು ಅಡಿಯಲ್ಲಿ ಪರೀಕ್ಷಿಸಿದಾಗ ೨೫ ನಮೂನೆಗಳು ಒಣ ಬೇರು ಕೊಳೆರೋಗಕ್ಕೆ ಸಾಧಾರಣ ನಿರೋಧಕತೆಯನ್ನು ತೋರಿಸಿದೆ. ಜೀವರಾಸಾಯನಿಕಗಳಾದ ಪೆರಾಕ್ಸೈಡ್ಸ್, ಕೆಟಲೀಸ್, ಫೀನ್ಯಲ್‌ಅಲೆನಿನ್ ಅಮೋನಿಯಾಲ್ಯೆಯಸ್, ಅಮೈನೋಆಸಿಡ್‌ಗಳು ಹೆಚ್ಚಿನ ಪ್ರಮಾಣದಲ್ಲಿ ಹಾಗೂ ಶುಗರ್, ರೆಡೋಸಿನ್-ಶುಗರ್, ನಾನ್-ರೆಡೋಸಿನ್ ಶುಗರ್‌ಗಳು ಕಡಿಮೆ ಪ್ರಮಾಣದಲ್ಲಿ ಎಲ್ಲಾ ಎಂ.ಪಿ. ಉಪಚರಣೆಗೆ ಒಳಪಡಿಸಿದ ತಳಿಗಳಲ್ಲಿ ಕಂಡುಬಂದಿದೆ. ಪ್ರಯೋಗಾಲಯದಲ್ಲಿ ಟ್ರೈಕೋಡರ್ಮ ಹಾಜೀಯಾನಮ್ (ಟಿ.ಹೆಚ್-೫೫) ಜೈವಿಕ ಶಿಲೀಂಧ್ರವು ಎಂ.ಪಿ. ಸೂಕ್ಷ್ಮಾಣು ಜೀವಿಯ ಶೇ. ೮೧.೪೪ ರಷ್ಟು ಹಾಗೂ ಬ್ಯಾಸಿಲಸ್ ಸಬ್ಟಿಲಿಸ್ ಶೇ. ೭೫.೫೬ ರಷ್ಟು ತಂತ್ರುಜಾಲದ ಬೆಳವಣಿಗೆಯನ್ನು ತಡಿಯುವಲ್ಲಿ ಪರಿಣಾಮಕಾರಿಯಾಗಿದೆ. ವಿವಿಧ ಶರೀರ ಶಿಲೀಂಧ್ರನಾಶಕಗಳಾದ ಟೆಬುಕೊನೊಜೋಲ್, ಕಾರ್ಬೆಡೈಜಿಮ್, ಡೈಫೆನೊನೊಜೋಲ್, ಪ್ರೊಪಿಕ್ಲೊನೊಜೋಲ್ ೫೦ ಪಿ.ಪಿ.ಎಮ್ ನಲ್ಲಿ ಹಾಗೂ ಥಯೋಫಿನೈಟ್ ಮೀಥೈಲ್ ೫೦೦ ಪಿ.ಪಿ.ಎಮ್ ನಲ್ಲಿ ಶೇ. ೧೦೦ ರಷ್ಟು ತಂತ್ರುಜಾಲದ ಬೆಳವಣಿಗೆಯನ್ನು ನಿಯಂತ್ರಿಸಿದೆ. ಸಂಪರ್ಕ ಶಿಲೀಂಧ್ರನಾಶಕವಾದ ಮ್ಯಾಕೋಜೆಬ್ ೨೫೦ ಪಿ.ಪಿ.ಎಮ್ ನಲ್ಲಿ ಶೇ. ೧೦೦ ರಷ್ಟು ಹಾಗೂ ಸಂಯೋಜಕ ಶಿಲೀಂಧ್ರನಾಶಕಗಳಾದ ಕಾರ್ಬಾಕ್ಸಿನ್ ೩೭.೫ % + ಥೈರಾಮ್ ೩೭.೫ % ಡಬ್ಲ್ಯೂ.ಪಿ, ಕಾರ್ಬೆಡೈಜಿಮ್ ೧೨ % + ಮ್ಯಾಕೋಜೆಬ್ ೬೩ % ಡಬ್ಲ್ಯೂ.ಪಿ ಶೇ. ೧೦೦ ರಷ್ಟು ಬೆಳವಣಿಗೆಯನ್ನು ೨೫೦ ಪಿ.ಪಿ.ಎಮ್ ನಲ್ಲಿ ಕುಂಠಿತಗೊಳಿಸಿದೆ. ಮುಂಗಾರು ಹಾಗೂ ಹಿಂಗಾರಿನ ಕಡಲೆಯ ಒಣ ಬೇರು ಕೊಳೆರೋಗಕ್ಕೆ ಅಳವಡಿಸಿದ ನಿರ್ವಹಣಾ ಕ್ರಮಗಳಲ್ಲಿ ಟೆಬುಕೊನೊಜೋಲ್ ೨ ಗ್ರಾಂ./ಕೆ.ಜಿ. ಬೀಜೋಪಚಾರಣೆಯು ಅತಿ ಕಡಿಮೆ ಪ್ರಮಾಣದ ರೋಗ (ಶೇ. ೮.೭೦), ಅಧಿಕ ಇಳುವರಿ (ಶೇ. ೭೨೬.೦೫ ಕೆ.ಜಿ./ಹೆ.) ಹಾಗೂ ಖರ್ಚು ಲಾಭದ ಅನುಪಾತವು (೧:೨.೩೭) ಹೆಚ್ಚಾಗಿದ್ದು ನಿರ್ವಹಣಾ ಕ್ರಮ ಕೈಗೊಳ್ಳದ ತಾಕುಗಳಲ್ಲಿ ಅತಿ ಹೆಚ್ಚು ರೋಗ (ಶೇ. ೪೩.೯೧) ಹಾಗೂ ಕಡಿಮೆ ಇಳುವರಿಯು (ಶೇ. ೨೮೪.೭೦ ಕೆ.ಜಿ./ಹೆ.) ಕಂಡುಬಂದಿದೆ.

ಡಿಸೆಂಬರ್, ೨೦೧೬

ಸಸ್ಯ ರೋಗಶಾಸ್ತ್ರ ವಿಭಾಗ

ಕೃಷಿ ವಿಶ್ವವಿದ್ಯಾಲಯ

ಜಿ.ಕೆ.ವಿ.ಕೆ., ಬೆಂಗಳೂರು-೫೬೦೦೬೫



(ಎಂ. ಸೈಫುಲ್ಲಾ)

ಮುಖ್ಯ ಸಲಹೆಗಾರರು

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I INTRODUCTION

Chickpea (*Cicer arietinum* L.), also known as Gram or Bengal gram, is the second most important pulse crop in the world, India accounting for 60 to 75 per cent of the world's chickpea production. It is known by a variety of names that includes, poischiche (French), homos (Arabic), chickpea (English), grao-de-bico (Portuguese), garbanzo (Spanish), chana (Hindi, Punjabi, and Urdu) and kondaikadalai (Tamil). Chickpea seeds contain high quality easily digestible protein (25 %) and carbohydrates (20 %) making it an important source of protein for the vegetarians of the country and thus it is also called "Poor man's meat." Chickpea is widely used either as grain or vegetable in Indian cuisine and an important source of vegetable protein. The origin of chickpea is thought to have been in South Eastern Turkey and neighbouring Northern Syria (Van der Maesen, 1972). It has since spread to many other geographical regions of the world because of its ability to grow in diverse environmental conditions. There are two main commercial types of chickpea, the "Desi" type with smaller and darker coloured seeds which may vary from yellow to black and the "Kabuli" type with large, smooth and light coloured seeds.

The global area under chickpea is 14.80 million ha, with production of 14.23 million tonnes and productivity of 962 kg/ha. In India, it is grown in an area of about 10.74 million ha with production of 9.88 million tonnes and productivity of 920 kg/ha. In Karnataka, it is grown in 0.95 m ha with production of 0.72 million tonnes and productivity of 757 kg/ha (Anon., 2015).

The per hectare production is low in spite of high yielding varieties and new agronomic practices. The reasons for low yield is due to incidence of diseases. The crop is known to be affected by number of soil-borne pathogens, some of which may be devastating. Chickpea suffers from about 172 pathogens consisting of fungi, bacteria, viruses and nematodes. Soil borne diseases such as wilt (*Fusarium oxysporum* f. sp. *ciceri*), dry root rot (*Macrophomina phaseolina*), black root rot (*Fusarium solani*), collar rot (*Sclerotium rolfsii*), and stem rot (*Sclerotinia sclerotiorum*) are important in reducing the yield of the crop. The foliar diseases viz., Ascochyta blight (*Ascochyta rabiei*) and grey mould (*Botrytis cinerea*) are more severe in chickpea crop.

Diseases with limited distribution are economically important locally because of continuous changes in cultural practices, human interventions and climate change. The dry root rot (*M. phaseolina*) is a major constraint in the chickpea production as it is emerging as a potential threat to chickpea cultivation in semi-arid regions due to moisture stress and high temperatures during the flowering to pod filling stage (Sharma *et al.*, 2010). The annual yield loss due to this disease alone is 10-20 per cent (Vishwadhar and Chaudhary, 2001).

The dry root rot disease generally appears around flowering and podding time. The disease may also appear at seedling stage, however, the susceptibility of the plant increases with age. The disease generally appears when day temperature is more than 30 °C and soil moisture content of 60 per cent. Drooping of petioles and leaflets is confined

to those at the very top of the plant. Sometimes when rest of the plant is dry, the top most leaves are chlorotic. The leaves and stems of affected plants are usually straw colored. The lower portion of the tap root usually remains in the soil when plants are uprooted. The tap root is dark and is devoid of most of its lateral and finer roots. Dark, minute sclerotial bodies can be seen on the roots or inside the wood (Nene *et al.*, 2012). *M. phaseolina* is primarily seed and soil-borne fungal pathogen. In chickpea, infected seeds and microsclerotia surviving in the soil are the major source of primary inoculum. The pathogen also has wide host range. Since 75 per cent cultivation of chickpea in India is under rainfed, the crop faces severe moisture stress at flowering to podding stage which predisposes the crop to dry root rot development.

It is necessary to conduct survey for dry root rot disease to get comprehensive information on disease distribution, level of severity, extent of spread and to locate hot spots of disease. The identification of isolates of *M. phaseolina* is usually based on morphological criteria, but due to wide variations in the phenotype of the isolates these criteria are often not reliable. Therefore study of molecular pathogenic variability is essential for breeding disease resistance in crop improvement programme. Prolonged saprophytic survival ability of the pathogen in soil makes chemical control and crop rotation ineffective. Cultivation of resistant varieties is the most practicable, feasible, and economical approach for the management of dry root rot of chickpea, but only a few sources with low level of genetic resistance are available, so there is a need to identify the resistant sources in chickpea through screening of germplasm against dry root rot pathogen. Biochemical analysis is important to understand the resistant mechanism before and after infection of plant and to isolate the resistance sources. The information on the management of disease is negligible. Hence detailed investigation was undertaken to evaluate the bio-agents and fungicides. The improved practices for disease management will pave a way to mitigate losses caused by dry root rot and improves livelihoods of the poor farmers. Hence, the present study was undertaken with the following objectives:

1. Survey for dry root rot of chickpea in southern parts of Karnataka.
2. Assessment of molecular variability in *Macrophomina phaseolina* isolates of chickpea from different locations of India.
3. Screening of chickpea germplasm against dry root rot and biochemical analysis of cultivars.
4. Management of *Macrophomina phaseolina* causing dry root rot in chickpea

II REVIEW OF LITERATURE

The fungi, *Macrophomina phaseolina* (Tassi) Goid., causing dry root rot disease of chickpea is a soil borne pathogen. The severity of the disease affect the pulse production as it influences major source of dietary proteins. The literature pertaining to chickpea dry root rot disease spread, resistance source, variability among pathogen population, biochemical analysis, *in-vitro* evaluation of bioagents, chemicals and its field management through an integrated approach was reviewed and presented in this chapter to support the present investigation.

2.1 Economic importance of dry root rot

Chickpea is susceptible to many diseases (Nene *et al.*, 1984) and among them dry root rot (DRR) is one of the major limitations in chickpea production causing 10 to 20 per cent annual loss (Dhar and Chaudhary, 2001). With the drastic reduction of area in northern India and its quantum increase in hot, drier rainfed central and southern India, the dry root rot is emerging as the most destructive constraint to chickpea cultivation. The disease is more prevalent between flowering and podding stages especially during hot (temperatures of 30 to 35 °C) and dry (deficit soil moisture) environmental conditions (Sharma *et al.*, 2012).

2.2 The causal organism: *Macrophomina phaseolina*

Haware and Nene (1978) first reported *Rhizoctonia bataticola* as the causal organism of dry root rot of chickpea. Later, appearance of *Rhizoctonia* root rot in other pulses was reported by Tripathi and Sharma (1983), Pandey and Singh (1990).

Dry root rot is caused by a necrotic fungus *Macrophomia phaseolina* (Tassi) Goid which is a Pycnidial stage. It belongs to Sub kingdom : Mycota, Division : Eumycota, Sub-division : Deuteromycotina, Class : Coelomycetes, Order : Sphaeropsidales, Family : Sphaeropsidaceae, Genus : *Macrophomina*, Species : *M. phaseolina*. It is highly variable, with isolates differing in microsclerotial size and presence or absence of pycnidia. The pycnidial stage is not common on soybean, but is on peanut. Pycnida are initially immersed in host tissue, then erumpent at maturity. They are 100 - 200 µm in diameter, dark to grayish, becomes black with age, globose or flattened globose, membranous to sub carbonaceous with an inconspicuous or definite truncate ostiole. The pycnida bear simple, rod-shaped conidiophores, 10-15 µm long. Conidia (14-33 × 6-12 µm) are single celled, hyaline, and elliptic or oval. Secondary spread is through wind-borne pycnidiospores (Mehrotra *et al.*, 1990)

Microsclerotia are jet black in colour and appear smooth, round to oblong or irregular, across isolates and on different substrates they vary on size and shape. The microsclerotia formed from aggregates of hyphal cells joined by a melanin material with 50 to 200 individual cells. Primary spread is through seed-borne and soil-borne sclerotia. Sclerotia survives in the soil or in the seed initiate the infection. They germinate and penetrate the host directly. Fungus produces cellulolytic, pectinolytic and other enzymes, which kill and disintegrates the tissues in advance of penetration, resulting in rotting of

tissues. It is a facultative saprophyte and it lives saprophytically in the dead tissue (Alexopoulos *et al.*, 1996)

Colonies in culture varies in colour from white to brown or gray and darken with age. The Mycelium is septate, stout and hyphae branches generally at right angles to parent hyphae, but branching is also common at acute angles. Aerial mycelium with completely or partially appressed growth may or may not be produced in culture. Some isolates may form concentric growth rings (Manjunatha, 2014).

2.3 Symptomatology of dry root rot

Infected plants, at first do not showed above ground symptoms. However, in advanced stages, leaves turned yellow and wilt, but remained attached to the plant. When the epidermis was removed sclerotia appeared as grayish black on the tissue. In the absence of the plants, it survived over many seasons as small black sclerotia in diseased root and stem debris or in soil after decay of the plant material. Thus, the primary source of inoculum was sclerotia in the soil (Smith, 1969).

Ilyas and Sinclair (1974) observed intra-xylem sclerotia of *R. bataticola* that totally plugged the xylem vessels of soybean infected plants. Later, Singh and Mehrotra (1982) in their histopathological studies reported that, roots inoculated by the pathogen showed only discoloration but no necrosis.

The disease generally appeared around flowering and podding time as scattered dried plants in field, where the leaves and stems of affected plants remained straw-coloured. The roots dry and the tap root became brittle, showed shredding of bark and was broken easily. The disease appeared suddenly when the ambient temperatures was above 30 °C. The fungus lacks fruiting bodies and spores. The mycelium was light-brown and thick in which black sclerotia were formed (Haware, 1990).

Singh *et al.* (1990b) observed that roots inoculated with *R. bataticola* (*M. phaseolina*) showed disintegration of cortical tissues while mycelium and sclerotial bodies plugging the xylem were seen in longitudinal sections. The sudden collapse of infected plants at the pod-filling stage was attributed due to the plugging of the xylem vessels.

Prashanthi (1994) reported that, safflower plants infected by *R. bataticola* were characterized by gradual yellowing and drying of leaves followed by loss of vigour and premature death. The bark of such plants was easily peeled off along with extensive shredding of affected bark.

2.4 Host range of dry root rot pathogen and distribution of disease

The pathogen infected over 500 plant species and had wide geographic distribution. Major cultivated hosts included: *Arachis hypogaea* (peanut), *Beta vulgaris* (beet), *Brassica oleracea* (cabbage), *Capsicum annuum* (pepper), *Cicer arietinum* (chickpea), *Citrus* spp., *Corchorus* sp., *Cucumis* sp., *Fargaria* sp., *Glycine max*

(soybean), *Gossypium* sp. (cotton), *Helianthus annuus* (sunflower), *Ipomoea batatas* (sweet potato), *Medicago sativa* (alfalfa), *Phaseolus* sp., *Pinus* sp., *Prunus* sp., *Sesamum indicum* (sesame), *Solanum tuberosum* (potato), *Sorghum bicolor* (sorghum), *Vigna unguiculata* (bean) and *Zea mays* (corn) (Purkayastha *et al.*, 2006).

The dry root rot disease caused by *M. phaseolina* was one of the most destructive pathogens in arid regions and had wide host range (Waller, 1976). In India, it was reported from Andhra Pradesh, Bihar, Gujarat, Karnataka, Maharashtra, Madhya Pradesh, Rajasthan, Tamil Nadu, Uttar Pradesh and West Bengal (Uppal, 1931; Likhite and Kulkarni, 1934; Patel *et al.*, 1949; Venkataram, 1950; Thirumalachar, 1955; Chattopadhyay and Bhattacharya, 1968; Deshpande *et al.*, 1969 and Phillip *et al.*, 1969). Dry root rot due to *R. bataticola* was reported in Austria, Ethiopia, India, and Turkey on chickpea (Nene, 1978), on chilli (Mathur and Gurjar, 1999; Bunker and Mathur, 2001) and on urdbean (John *et al.*, 2008).

Raut and Ingle (1989) reported that the plant species of Leguminaceae, Pedaliaceae and Compositae encouraged the growth and survival of conidia, sclerotia and pycnidiospores of *R. bataticola*. Singh *et al.* (1990a) found that all the commonly cultivated crops in the semi-arid tropics were found to be infected. Su *et al.* (2001) noted that *M. phaseolina* readily infected soybean, corn, sorghum and cotton.

2.5 Survey for dry root rot of chickpea in southern parts of Karnataka

Sharma *et al.* (1983) recorded occurrence of *Rhizoctonia* root rot and periodic surveys on wilt and root rot incidence of chickpea in northern region of Madhya Pradesh for two consecutive years which showed that incidence of wilt (15.90 - 37.99 %) and root rot (3.58 - 20.63 %) respectively in the districts surveyed. The highest wilt and root rot incidence was 27.79 and 14.17 per cent respectively and the lowest was 20.81 and 8.76 per cent.

Pandey and Singh (1990) conducted a survey of chickpea diseases in Allahabad region and reported less infection by *Rhizoctonia* and *Fusarium oxysporum* f. sp. *ciceri*. Nearly 83 per cent of villages showed less than 25 per cent disease intensity and remaining ones recorded 26 - 50 per cent disease intensity. Dolar (1996) conducted surveys in main chickpea growing areas of Turkey and noted the most common root rot complex pathogens viz., *Rhizoctonia*, *Fusarium* and *Sclerotium* which included 49 per cent disease incidence.

Singh and Agarwal (2002) conducted a survey in the Vindhya plateau zone of Madhya Pradesh (Bhopal, Raisen, Sagar, Seshore and Vidish) to evaluate the prevalence of dry root rot in chickpea and reported that the incidence of *R. bataticola* ranged from 8 to 20 per cent.

Prajapati *et al.* (2003) reported that the incidence of dry root rot of chickpea caused by *R. bataticola* was highest in bold-seeded cultivars. Singh and Sirohi (2003) evaluated the incidence of chickpea diseases in the Himachal Pradesh and reported that the incidence of dry root rot was highest in Una (4.86 %) and Sirmour (3.04 %) districts.

Veerendra Kumar (2004) noticed an incidence of 9.97 per cent dry root rot in chickpea from Gulbarga followed by Raichur (5.16 %) and Bidar (4.28 %). The highest incidence was 12.24 per cent in Chittapur of Gulbarga and least was in one of the locations of Raichur (3.21 %). Gurha and Trivedi (2008) conducted a survey in all the chickpea growing districts of Karnataka and reported that *R. bataticola* was found to be the predominant pathogen which infected 60 per cent plants in the fields of Gulbarga.

The mean maximum dry root rot incidence was noticed in Gulbarga district (9.8 %) followed by Raichur (7.60 %) and the least (6.18 %) in Bidar (Manjunatha *et al.*, 2011). Dry root rot disease was found at all sites and incidence ranged from 8.90 per cent to 10.30 per cent irrespective of cultivar type and locations (Raju Ghosh *et al.*, 2013). The result indicated that dry root rot was highly distributed in all surveyed chickpea growing areas of central and southern parts of India

To understand the spatial distribution of soybean charcoal rot incidence and *M. phaseolina* populations in Golestan Province, 172 soybean fields were surveyed for population density, in two successive years integrated with Geographic Information System (GIS). In the seasons of 2009-2010 and 2010-2011, disease incidence ranged from 0 to 97 per cent and 3 to 91 per cent with the highest in Gorgan and Aliabad, respectively. Total mean of disease incidence was 21.01 and 35.84 percent in the province. In the two sampling years, Sclerotia were recovered from 73.33 and 93.57 per cent of the total fields. The average population density per gram of soil ranged from 0.65 to 14.31 and 4.70 to 16.90, respectively, with the highest levels in Aliabad in both the years. Geostatistical analyses of the survey data showed that the influence range of propagule density and disease incidence was between 8,000 to 14,000 m (Taliei, 2013).

A survey was conducted in January, 2014 *rabi* cropping season in different chickpea growing locations of central (Madhya Pradesh and Maharashtra) and southern (Andhra Pradesh, Telangana and Karnataka) India. Due to diversified weather conditions, variation in sowing dates, different crop growth stages from seedling (20 days old) to podding stage, the disease was observed in all the stages. The maximum dry root rot incidence was observed in Telangana (18.28 %) and the least in Maharashtra (5.38 %). Disease occurrence was observed irrespective of cropping system, soil types and cultivars. The disease incidence was low in the irrigated fields compared to rainfed fields (Srinivas, 2016).

2.6 Assessment of molecular variability in *Macrophomina phaseolina* isolates of chickpea from different locations of India

The identification of isolates of *M. phaseolina* is usually based on morphological criteria, but due to wide variations in the phenotype of the isolates these criteria are often not reliable. Pathogenic variability among the isolates or sensitivity to chlorate are some other characters utilized to categorize the isolates (Cloud and Rupe 1991).

2.6.1 Collection, isolation and maintenance of *M. phaseolina* isolates

M. phaseolina was isolated from different plant parts viz., diseased seeds and seedlings (Rajeevpant and Mukhopadhyay, 2001; Bagri *et al.*, 2004; Vinod Kumar *et al.*,

2007), roots (Sajeena *et al.*, 2004; Nagamma and Saifulla, 2012), stem (Pan and Bhagat, 2008) and leaves (Sarkar and Pradhan, 1999).

Goidanich and Camici (1947) collected isolates of *M. phaseoli* from bean and Lavender (*Lavandula angustifolia*) in three widely separated parts of Italy. Similarly, Dhingra and Sinclair (1973b) collected nine isolates of *M. phaseoli* from various parts of United States. Shekhar *et al.* (2006a) obtained seven isolates of *M. phaseolina* incitant of charcoal rot of maize from different agro-ecological zones of India.

Aghakhani and Dubey (2009b) collected 23 isolates of *M. phaseolina* from chickpea plants showing characteristic dry root rot symptoms from different chickpea growing states of India and isolated the associated fungi from the roots of diseased plants on PDA medium. Manjunatha and Naik (2011) collected thirty isolates of *R. bataticola* causing dry root rot of chickpea collected from three major chickpea growing regions from Karnataka.

Sharma *et al.* (2012) collected fifty isolates of *R. bataticola* in different years (2004 to 2010) from different chickpea growing regions of India covering eight states including Uttar Pradesh, Rajasthan, Tamil Nadu, Andhra Pradesh, Madhya Pradesh, Delhi, Himachal Pradesh and Uttarakhand and the pathogen was isolated by plating surface sterilized (0.8 % sodium hypochlorite for 2 min) pieces of dry root rot infected roots of chickpea plants on potato dextrose agar medium. Twenty different isolates were collected from major chickpea growing states of India and *M. phaseolina* were isolated from infected root samples (Manjunatha, 2014).

Potato dextrose agar (PDA) was found to be the best supporting medium for *R. bataticola* (Vinod Kumar *et al.*, 2007; Konde *et al.*, 2008). The pathogen *R. bataticola* was stored in an Agar Papa dextrose culture medium in a bioclimatic chamber at 4 °C (Valiente *et al.*, 2008).

2.6.2 Pathogenicity and virulence of *M. phaseolina*

Prameela Devi and Singh (1998) reported 10 per cent inoculum density causing 71 per cent seedling mortality in green gram and black gram. Kataria *et al.* (2007) assessed nine isolates of *R. bataticola*, the incitant of dry root rot in chickpea and revealed that all nine isolates were found pathogenic on C-235 chickpea cultivar in both sterilized and unsterilized soil. The root rot incidence varied from 26 to 100 per cent in unsterilized soil and 20 to 80 per cent in sterilized soil.

Jaiman and Jain (2008) conducted pathogenicity test for *M. phaseolina* in cluster bean seeds and observed that seed inoculation with *M. phaseolina* resulted in pre and post emergence mortality and loss of seed germination (59 %) as compared to control (93 %).

Aghakhani and Dubey (2009a) evaluated the pathogenicity and virulence of twenty-three isolates of *R. bataticola* against 10 chickpea cultivars viz., ICC 12441, ICC 11332, ICC 11224, ICC 12450, Pusa 362, BGD 112, Pusa 1103, Pusa 212, Pusa 1088 and

BCP 17 under sick pot. Based on the dry root rot incidence they revealed that the isolates were highly variable in their virulence. Cultivar Pusa 212 showed maximum dry root rot incidence (31.7 - 100 %) against the isolates tested followed by ICC 12441 and ICC 11332. whereas, root rot incidence was minimum on cultivar Pusa 1103. Among the isolates, Rb1 from Bengaluru proved to be highly virulent and caused 16.6 - 100 percent (76.4 % mean) root rot in different cultivars. Faridkot (Rb5) and Badanapur (Rb11) isolates were comparatively less virulent.

2.6.3 Mass multiplication of *M. phaseolina*

The pathogen *R. bataticola* was mass multiplied on sand maize medium (Sajeena *et al.*, 2004; Kataria *et al.*, 2007). Sterilized sorghum grains were used for mass culturing of *R. bataticola* (Durai, 2004).

2.6.4 Molecular identification of the fungus

Babu *et al.* (2007) designed two species-specific primers MpKFI and MpKRI from the specific nucleotide region and one oligonucleotide probe MpKH1 (19 mer) from the conserved region, adjacent to ribosomal 5.8 S gene that yielded single amplified product of 350 bp with all the *M. phaseolina* isolates. The specificity of the primers was tested on representative species of common soil borne microbes and found to be specific for *M. phaseolina* as none of the other soil microbes tested could yield any amplification product under identical conditions of amplification.

Mahdizadeh *et al.* (2011) confirmed that the fifty two *M. phaseolina* isolates recovered from 24 host plant species through the 14 Iranian provinces using species-specific primers MpKFI and MpKRI.

Asran-Amal (2012) employed the microsatellite primers (ATG)₅, (TAGG)₄ and a primer derived from the intergenic spacer regions (T3B) for identification of *M. phaseolina*. The three primers tested amplified species-specific microsatellites that unambiguously distinguished *M. phaseolina* from other fungal species tested.

2.6.5 Internal Transcribed Spacer (ITS)

Ribosomal DNA (rDNA) in eukaryotes typically present in several hundred tandemly repeated multiple copies. They were one of the most extensively used molecular markers. The region was a part of the rDNA cistron which consists of 18S, ITS1, 5.8S, ITS2, and 28S sequences. Both the ITS1 and ITS2 non coding regions that lie between the 18S and 5.8S (ITS1) and between the 5.8S and 28S (ITS2) coding regions. The entire rDNA unit was transcribed by RNA polymerase-I as a single 45S precursor molecule (Atkins and Clark, 2004; Brunner *et al.*, 2007). Multiple copies made it easy to amplify rDNA even from small amounts of samples. Because of each of the two internal transcribed spacer (ITS1 and ITS2) were flanked by rDNA highly conserved coding regions, primers can be designed for the amplification of the first internal transcribed spacer (ITS1) and the second transcribed spacer (ITS2).

Molecular methods involving the use of the PCR was recently described to resolve genetic variation among the *M. phaseolina* isolates (Jana *et al.*, 2005; Das *et al.*, 2006). Moreover, these techniques were laborious, time consuming, not reliable and useful for grouping rather than identification. Therefore, a rapid diagnostic test that can exactly identify and detect *M. phaseolina* isolates, both in *in-vitro* and *in-vivo* conditions are required. Nuclear rDNAs, particularly the internal transcribed spacer (ITS) region, are good targets for phylogenetic analysis in fungi (Bruns *et al.*, 1991).

Oligonucleotide specific primers or probes targeted the ITS region demonstrated selective detection of several agriculturally important fungi like *Trichoderma*, *Verticillium* (Nazar *et al.*, 1991), *Phytophthora* (Lee *et al.*, 1993), *Fusarium* (Edel *et al.*, 2000) and *Hypocrea* (*Irina et al.*, 2005) etc.

Naseema *et al.* (2005) used six RAPD primers for molecular characterization of seven isolates of *Fusarium* spp. The results of UPGMA cluster analysis revealed the existence of two main groups among these isolates.

Liu Jingmei *et al.* (2006) analysed 18 isolates of *F. oxysporum* f. sp. *cubense* using 200 RAPD oligonucleotide primers. Based on the cloning and sequencing of the RAPD marker fragments, the Sequence Characterized Amplified Region (SCAR) primers were designed. The PCR amplification of the 18 *F. oxysporum* f. sp. *cubense* isolates and 9 new field isolates with SCAR primers showed that 4 SCAR markers could be specifically used to separate race 1 and race 4.

Honnareddy and Dubey (2006) studied genetic variability of 24 *F. o. f. sp. ciceri* isolates collected from different parts of India. Seven races of *F. o. f. sp. ciceri* was assessed by RAPD, with a set of 40 primers. UPGMA cluster analysis divided the isolates into seven distinct clusters at 0.55 genetic similarities.

The genomic DNA amplification of ITS region of different isolates yielded a fragment of approximately 650 bp. The restriction enzymes EcoR I, Sau3A I, Alu I and Cla I gave identical pattern for all isolates and produced two fragments of different sizes at single cut. The enzymes Hap II, Msp I and Taq I had multiple cleavage sites in ITS amplified product and produced fragments less than 100 bp in size. Sequence comparison of ITS region encompassing ITS-1, 5.8S rRNA gene and ITS-2 of *M. phaseolina* and other related fungal species revealed three regions that were conserved among the *M. phaseolina* isolates. The complete sequence was divided into five regions from ITS-1 to ITS-2 (Babu, 2007).

Aghakhani and Dubey (2009b) determined genetic diversity among six isolates of *R. bataticola* representing five different ITS-RFLP groups with respect of their nucleotide sequences of ITS I, 5.8S and ITS II regions. The ITS of Karnataka isolate (Rb1) was largest (670 bp), whereas it was smallest (499 bp) in Haryana (Rb21) isolate. After sequencing the ITS sequences of the isolates were deposited to Gen-Bank nucleotide database. The phylogeny tree constructed from the nucleotide sequence, similarity of these six isolates along with 27 other ITS sequences of fungi which showed 96 - 100 per

cent similarity during blast analysis clustered them into three major groups. The isolates Rb16 (Jharkhand), Rb21 (Haryana) and Rb26 (Madhya Pradesh) clustered in one group and showed 96 - 100 per cent ITS nucleotide sequences similarity with the nucleotide sequences of *M. phaseolina* (*R. bataticola*) available in the GenBank.

El-Fadly *et al.* (2008) used RAPD-PCR technique as a useful tool for differentiating between species and formae speciales of the genus *Fusarium* either alternatively or complementary to methods based upon morphological and pathological characteristics.

Genetic diversity among 27 isolates (23 from chickpea and 4 from other host crops) of *Rhizoctonia bataticola* representing 11 different states of India was determined by random amplified polymorphic DNA (RAPD), internal transcribed spacer restriction fragment length polymorphism (ITS-RFLP) and ITS sequencing. The restriction enzymes DraI, MboI, RsaI, and AluI were found to be suitable for differentiating the isolates into five categories by showing isolate specific ITS-RFLP patterns. The isolates were variable in their nucleotide sequences of the ITS regions (Aghakhani and Dubey, 2009b).

High levels of pathogenic variability and genetic diversity were observed between *R. bataticola* isolates collected from different geographical origins after characterization with different markers [Random Amplified Polymorphic DNA (RAPD), Internal Transcribed Spacer Restriction Fragment Length Polymorphism (ITS-RFLP), ITS sequencing and Amplified Fragment Length Polymorphism (AFLP)]. The phylogenetic tree based on rDNA-ITS analysis showed that diversity in *R. bataticola* isolates was independent on their geographic origin (Sharma *et al.*, 2012).

Molecular characterization of twenty isolates included identification of *M. phaseolina* isolates with species specific primers MpKFI and MpKRI and molecular variability was studied by employing PCR-RAPD and ITS sequence. Among both the markers used for the study RAPD helps to know only variability but ITS region sequence was found best to identify the pathogen and also to study the molecular variability among the twenty isolates (Manjunatha, 2014).

Among 10 RAPD primers used to study genetic variation against *Fusarium oxysporum* f. sp. *ciceri* causing wilt of chickpea, nine primers showed 100 per cent, whereas OPA-12 showed 87.5 per cent polymorphism among 12 isolates and similarity ranged from 43 to 76 per cent. Among 12 isolates, MPFOC-9 caused cent per cent wilt incidence on chickpea host differentials (Chaithra, 2014).

2.7 Screening of chickpea germplasm against dry root rot and biochemical analysis of cultivars

2.7.1 Screening

Nene *et al.* (1981) developed pot culture techniques for greenhouse screening and a paper towel technique for laboratory screening against *M. Phaseolina*.

Gangwar *et al.* (2002) screened thirty-five chickpea cultivars for resistance to dry root rot caused by *R. bataticola* in a field experiment conducted in Uttar Pradesh, India. Disease intensity was recorded 30 days after sowing up to maturity at 15 day intervals. All cultivars exhibited disease symptoms. Ten cultivars viz., ICC 2644, 10384, 10630, 112244, 11332, ICCL 81002, 810810, ICC 12263, 12441 and ICCV 90254 were found resistant. Whereas, five cultivars viz., Phule G 9504, 96020, 96105, 96313 and GL 91059 were moderately resistant to dry root rot.

Out of 47 lines screened against dry root rot of chickpea under blotter paper method, three genotypes showed resistant, 22 moderately resistant, 19 susceptible and three highly susceptible reaction (Pande *et al.*, 2004).

Om Gupta and Babbar (2006) screened 233 desi and Kabuli chickpea genotypes along with 38 promising lines with susceptible check JG 62 against collar rot, dry root rot and vascular wilt in multiple disease sick plot at Jabalpur, Madhya Pradesh, India. Genotypes H 99-264, PG 9425-5, PG 9425-9 (desi) and HK 00297, PG 97-313 (Kabuli) exhibited resistance against all the 3 pathogens. WCG 97-29, WCG 97-37, IPC 97-72 and H 00-256 (desi) showed less than 10 per cent mortality for dry root rot. GF 88240, MPJG 98-9032 and MPJG 98-11551 (desi) and HK 94-134 (Kabuli) showed resistance to wilt and dry root rot diseases.

Jayalakshmi *et al.* (2008) conducted pot culture experiment for screening of chickpea promising entries against dry root rot. Out of 12 genotypes tested against dry root rot disease, four entries were found resistant, two entries were moderately susceptible, three entries were susceptible and three were highly susceptible.

Al Shareef *et al.* (2009) observed that out of the forty chickpea genotypes screened for resistance to wilt and root rot diseases in the sick-plot. Thirty five genotypes were found resistant while, the other five genotypes were found to be moderately susceptible. Chaturvedi and Dua (2009) reported 25 resistant cultivars including KPG-59, Radhey and K-50 against dry root rot of chickpea.

Saifulla *et al.* (2011a) screened 196 entries against dry root rot and collar rot diseases in sick plot of which 21 entries viz., GNG 1861, Phule G07112, BGD 1056(R), RSG 931, Phule G 07101, Vijay, RSG 888, GNG 0825, PG 064, GNG 1947, JG 1307, GBC6 (AVT-1), GNG 2002, GNG 1936, CSJ 313, JG 18, HIR 60 and JG 11, ICCV 07107 and ICCV 08323 were resistant and 36 entries were moderately resistant to dry root rot disease.

Saifulla *et al.* (2011b) screened 16 chickpea genotypes for collar rot, dry root rot, wilt and rust diseases at GKVK, Bengaluru during the year 2010-11. For dry root rot, HIR 60, BGD 256, A-1, JG 11, BBG 1, BBG 2, JG 62, and L 550 genotypes were found resistant to moderately resistant, whereas HIR 60, BGD 256, A1, BBG 1 and BBG 2 showed multiple disease resistance to collar rot, dry root rot, wilt and rust diseases.

Om Gupta *et al.* (2012) screened 170 chickpea genotypes for dry root rot at JNKVV, Jabalpur. Sixty eight genotypes exhibited resistant reaction (< 10 % mortality), out of which 26 were the promising lines viz., JG 1-14, 2-125, 2-4-110, 14-11, 14-10, 2001-13, 2001-13, 2001-18, 2001-80, 2001-115, 2002-20, 2003-95, 2003-14-16, 2004-110, 210, 9605, 1-9, 99-115, 2001-04, 2003-14-2, JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas. These lines were further evaluated for their performance in sick field for three consecutive years and six lines viz., JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas were found to be resistant exhibiting < 10 per cent mortality, however 14 lines showed moderately resistance reaction.

Nagamma and Saifulla (2012) screened sixty four Kabuli genotypes against dry root rot, out of which six genotypes viz., Phule G 04305, IPCK 07-62, RVSSG 12, HK 08-212, Phule G 09305 and AKG 2002-1K showed resistant reaction, nine genotypes viz., HK 94-134, GNG 1888, HK 06-152, GNG 2112, HK 06-171, Phule G 09316, CSJK 74, JGK 13 and CSJK 70 showed moderately resistant reaction.

2.7.2 Biochemical analysis

A number of phenolic acids were synthesized or significant increase in the concentration of phenolic acids was observed after inoculation with *M. phaseolina* (Bray, 1954). It was supposed that these phenolic compounds might already be present at low concentration in plant, but their rapidly accumulation upon attack as induced compounds Kuc (1982).

Retig (1974) stated that peroxidase (PO) activities were tested in resistant (Cv. Hosen Eilon) and susceptible (Cv. Supermarmande VR) tomato plants inoculated with *F. oxysporum* f. sp. *lycopersici*. PO increased in the roots of resistant type shortly after inoculation, whereas in the susceptible type, a similar increase in PO activity occurred 24 hr later.

Kaur and Dhillon (1989) they also noticed susceptible cultivars of groundnut showed a rapid decrease of sugars but the decrease was relatively slow in resistant cultivars.

The increase in protein content may be because of increase in PR protein after infection with pathogen (Bailey and Mansfield, 1992). Induction of PR proteins following elicitor treatment, wounding or tissue infection was accompanied by a massive alteration in the pattern of gene expression and activation of other defense responses such as phytoalexin accumulation, lignin deposition and synthesis of cell wall hydroxyproline rich glycoproteins (Lawton, 1996).

Gupta and Khare (1992) carried out biochemical analysis of cultivars of chickpea resistant and susceptible to vascular wilt, caused by *F. oxysporum* f.sp. *ciceris*. They established that the quantities of total amino acids, total phenols were greater in the roots of resistant cultivars.

Chandniwala (1996) reported that the activity of peroxidase was generally higher in the infected tissues of the resistant varieties than susceptible ones. The activities of soluble proteins increased in both resistant and susceptible cultivars. The activities were higher in resistant cultivars than the susceptible cultivars. The results obtained in this investigation indicated differences in both enzymes as well as soluble proteins in wilt resistant and susceptible cultivars and lines.

Sindhan *et al.* (1999) screened two hundred and sixty genotypes of mungbean (*Vigna radiata*) against *Cercospora* leaf spot, the total sugar, reducing sugar and non reducing sugar content was decreased in diseased leaves of susceptible varieties than resistant varieties.

Fungal elicitors 34 was known to induce the production of phenyl ammonia lyase (PAL) and peroxidase which were involved in the synthesis and depolymerization of lignin precursors. The rapid increase and higher levels of PAL and peroxidases activity was found in resistant cultivars as compared to the susceptible cultivars of chickpea against *Fusarium* wilt (Aguilar *et al.*, 2000).

A number of phenolic compounds were involved in resistance reaction, though they were also present in normal plants but synthesis and accumulation occurs at a faster rate after infection in the resistance varieties than in susceptible ones (Duthie and crozier, 2000).

Shreenivas (2000) reported that resistant varieties (ICPL 8863 and ICPL 87119) showed higher levels of phenolics contents in both the control and inoculated seedlings over the susceptible (TTB 7 and AKT 9221) at all the growth stage.

Mandavia *et al.* (2002) collected root exudates from ten day old seedlings and analyzed for total phenol content. A significant inverse relationship was found between wilt susceptibility and total phenol content in root exudates of chickpea seedlings. Resistant genotypes (0-30 % wilt) had the highest amount of phenol, followed by moderately resistant (31-70 % wilt) and susceptible (71-100 % wilt) genotypes.

Biochemical changes in banded leaf and sheath blight affected maize plants caused by *Rhizoctonia solani* f. sp. *sasakii* were studied. There was an increase in phenolic content in maize leaf sheaths inoculated with *Rhizoctonia solani* f. sp. *sasakii*. Peroxidase (PO), and phenylalanine ammonia lyase (PAL) activities increased when leaf sheaths were inoculated with the pathogen which may play a role in defense mechanism in plants against pathogen (Sivakumar and Sharma, 2003).

Prasad *et al.* (2003) estimated the peroxidase activity (PEO) in both leaf and root tissues of four pigeonpea cultivars (ICP 8863, Selection-I, ICP 2376 and LRG-30) at 1st, 4th and 7th day after inoculation (DAI) in healthy and *F. udum* infected tissues. Higher PEO activity in both leaf and root was observed and at 4th DAI in susceptible cultivars.

Khan *et al.* (2005) reported increased total phenolic contents in roots of susceptible and resistant cultivars of chickpea after inoculation with the virulent and hypovirulent isolates of *F. oxysporum* f.sp. *ciceris*. The highest increase was observed against the highly virulent isolate 9718 in the roots of Aug - 424 and CM 98, while least increase was observed in less virulent isolates, namely 2002 and 2014. The increases in total phenols was more in CM 98 than in Aug- 424.

Madhukeshwara *et al.* (2004) observed marked difference in the quantity of total phenols and sugars in root exudates of resistant and susceptible genotypes. Resistant genotypes had higher phenol and sugar content than susceptible genotypes. The resistant genotypes had higher total phenols in the root and leaf samples whereas the phenol content of the seed did not show any trend between susceptible and resistant types.

Phenolics played an important role in reducing the susceptibility of a plant to pathogens. A distinct correlation between the degree of a plant resistance and phenolics present in plant tissues was demonstrated (Onyilagha and Grotewold, 2004). Among the diverse roles of polyphenols, they protect the cell constituents against destructive oxidative damage thus limiting the risk of various degenerative diseases associated with oxidative stress (Villano *et al.*, 2005). Increase in phenolics was higher in resistant cell line than in susceptible cultivars after treatment with the cultural filtrate, indicating that phenols reach an inhibitory level to the fungus in the resistant cell line. Peroxidase followed a similar pattern. (Singh *et al.*, 2005).

In order to elucidate the elicitor properties of cell-wall protein (CWP) of *Fusarium oxysporum* f. sp. *ciceri* and *Macrophomina phaseolina*, CWPs were extracted from fungus and tested against chickpea. The chickpea seedlings exposed to CWPs showed enhanced synthesis of phenol, pathogenesis-related protein and activities of phenylalanine ammonia lyase and peroxidase relative to water-treated controls. Both *Fusarium* wilt and charcoal rot diseases of chickpea were significantly reduced in seedlings following CWP treatment. The results suggest that CWPs of *Fusarium* and *Macrophomina* had elicitor properties and effectively induced resistance in chickpea (Ratul Saikia *et al.*, 2006).

Changes in phenolic acid content in tomato fruits in response to pathogen attack was determined by Ruelas *et al.* (2006). Plants phenolics are the most important group of secondary plant products that had implicated in both constitutive and induced resistance (Cleberon *et al.*, 2006). A number of phenolic and salicylic acids were synthesized or significant increase in the concentration of phenolic acids was observed after inoculation with inoculums of *M. phaseolina*. The induction of phenolic compounds as part of the defense system against pathogen using elicitors had also been demonstrated by Pearce *et al.* (1998). Phenolic compounds may accumulate in plants as low molecular weight compounds called phytoalexins as a result of microbial attack (Mattila and Hellstrom, 2007).

Generally total phenols ortho-dihydroxy phenols and the enzyme activity were invariably high in resistant parents and hybrids irrespective of growth stages, while, in

case of susceptible parents of chilli the phenols content and enzyme activities were comparatively less. There existed a positive correlation between the host resistance and the amount of phenols and increased enzyme activities while it was almost the opposite in susceptible lines (Jabeen *et al.*, 2009).

The peroxidase increased in resistant cultivars and RIL lines than in susceptible ones in inoculated seedlings over control. In vijay, peroxidase activity was increased from 2.06 units to 8.54 units (un-inoculated over inoculated) and in JG-62 activity was increased from 1.68 units to 1.96 units (un-inoculated over inoculated) seedlings. Soluble proteins ($\mu\text{g/gm}$) were increased in inoculated seedlings as compared to control. In JG-62 (wilt susceptible) the soluble proteins increased from 45.96 $\mu\text{g/gm}$ to 62.25 $\mu\text{g/gm}$ from un-inoculated to inoculated seedlings. Similarly in vijay (wilt resistant) the soluble proteins increased from 48.28 to 64.23 $\mu\text{g/gm}$ from un-inoculated over inoculated seedlings. This clearly indicated that the soluble proteins increased in seedlings infected with *Fusarium oxysporum* f. sp. *ciceri* (Dalvi *et al.*, 2011).

The biochemical response of guar against *M. phaseolina* infection revealed the maximum accumulation of phenolic acids in infected plants 23 per cent higher than in control after 120 hours of infection in all the four cultivars. Similarly, PAL activity was also significantly increased by 37 per cent after 96 or 120 hours of infection depending upon the cultivar in comparison to control. Phenolic acid accumulation and enhanced PAL activity in the compatible host-pathogen combination presumes that both participated actively in the guar resistance to root rot (Sharma *et al.*, 2011).

Amino acids content increased in 70 per cent of genotypes after injury or infection by *Phytophthora megakarya* in cocoa. A significant positive relationship was observed between amino acid contents and the severity of necrosis (Pierr *et al.*, 2011). The resistant varieties ICP 8863 had highest total sugar content compared to BDN 2010 (moderately resistant), whereas, TTB 7 (susceptible) has lowest total sugar content (Prasad, 2003).

Wilt resistant Pigeonpea variety ICP 8863 had highest total phenol, Peroxidase, polyphenol oxidase, sugar content and protein content activity at different days after sowing compared to moderately resistant (WRP 1) and susceptible variety (TTB 7) when inoculated with pigeon pea sterility mosaic virus (Asha, 2012).

Cassava leaf extract infected with the African cassava mosaic virus and uninfected were prepared and used as crude catalase extract. The total protein was determined by Biuret method and activity of the crude catalase was compared using hydrogen peroxide as substrate. Infected leaf extract had 141.02 ± 3.536 mg/mL protein, apparent Michealis constant (K_{mapp}) of 26.7 μM and maximum rate of reaction (V_{max}) of 54.50 $\mu\text{M/min}$ compared to the uninfected leaf extract with 75.04 ± 0.560 mg/mL protein, K_{mapp} of 39.61 μM and V_{max} of 143.06 $\mu\text{M/min}$. The activation energy of the infected extract was 0.1578 J/mol compared to 0.2181 J/mol obtained for the uninfected extract. Activity of the crude catalase in the viral infected leaf extract from the study was

higher than that in the uninfected one and confirms the response to the stressful condition imposed by the viral infection (Amoako *et al.*, 2015).

2.8 Management of *Macrophomina phaseolina* causing dry root rot in chickpea

2.8.1 In-vitro evaluation of Bioagents

Sanford (1926) was the first to realize that soil microorganisms themselves exert a natural biological control on root diseases.

The potential use of *Trichoderma* spp. as a bio control agent was suggested more than 70 years ago by Weindling (1932) who was first to demonstrate the parasitic activity of members of this genus against soil borne fungal pathogens.

Biological control of plant diseases was defined as any condition under which practice, where survival and activity of pathogen was reduced through agency of other living organism as a result reduction in the disease caused by pathogen (Garret, 1956)

Cook and Baker (1983) defined biological control as reduction of the amount of inoculum or disease producing activity of a pathogen accomplished by one or more organisms other than man.

The mechanism proposed to explain the biocontrol of plant pathogens by *Trichoderma* or *Gliocladium* are presumptive. The suggested mechanisms for biocontrol was antibiosis, lysis, competition and mycoparasitism (Cook and Baker, 1983; Hardar *et al.*, 1984).

Mechanism of mycoparasitism includes interaction like coiling of hyphae around the pathogen, penetration by haustoria and lysis. *Trichoderma* spp. recognizes and attaches to the pathogenic fungus and begins to excrete extra cellular lytic enzymes like β -1,3-glucanase, chitinases, protease and lipase. The recognition mechanism was the basis for the specificity of the antagonist and lectins (glycoproteins) produced by soil borne pathogenic fungi (Elad and Chet, 1983; Barak *et al.*, 1985 and Barak and Chet, 1985).

There was a dramatic increase in research efforts on biological control of soil borne diseases (Cook, 1984; Naik and Sen 1995, Bineeta Sen, 2000; Jangid *et al.*, 2004; Roy *et al.*, 2005; Naik and Raju, 2006; Rini and Sulochana, 2007) had come up, covering the use of specific microbes, their mechanisms and commercial evaluation. *Trichoderma* spp. represents the most thoroughly studied fungi that showed antagonistic activities towards soil borne plant pathogens.

Jana *et al.* (2000) studied the agglutination potential of *P. fluorescens* in relation to energy stress and colonization of *M. phaseolina*. Among 172 isolates of fluorescent *Pseudomonas* tested, 12 isolates showed maximum (73 %) agglutination and 30 isolates showed minimum (10 %) and the remaining were in the range of 10 - 73 per cent agglutination.

Karunanithi *et al.* (2000) observed the production of antibiotic components *i.e.*, pyrrolnitrin by *P. fluorescens* which inhibited the mycelial growth of *M. phaseolina* in dual culture by producing an inhibition zone of 12 mm. *P. fluorescens* isolate 4 induced the systemic resistance against charcoal rot pathogen in chickpea by accumulation of pathogenesis related proteins such as chitinases and glucanases (Srivastava *et al.*, 2010).

Kumar *et al.* (2002) observed that *P. fluorescens* produced siderophores and antifungal metabolites which are involved in the control of phytopathogenic fungi.

Sindhan *et al.* (2002) evaluated the efficacy of *T. viride*, *T. harzianum*, *A. flavus*, *A. niger*, *A. ochraceus*, *P. citrinum*, *B. subtilis*, and *P. fluorescens* against *R. bataticola* in *in-vitro* conditions. All antagonists inhibited mycelial growth and sclerotia production, with *P. fluorescens* being the most effective at 65.0 and 90.5 per cent respectively.

Indra *et al.* (2003) evaluated fungal antagonists like *T. viride*, *T. harzianum*, *T. hamatum*, *T. koningii*, *T. pseudokoningii*, *T. longibrachiarum* and *Gliocladium virens* against black gram root rot fungus *M. phaseolina*. All the antagonists were on par in controlling *M. phaseolina*.

Kaswate *et al.* (2003) evaluated the antagonism of *T. viride*, *P. fluorescens* and *B. subtilis* to isolates of *R. bataticola* (*M. phaseolina*) collected from pigeonpea, chickpea, green gram, cluster bean, field pea, cotton, okra and safflower in *in-vitro*. *T. viride* was the most effective in the inhibition of the various isolates of *R. bataticola* (100 % inhibition), followed by *B. subtilis* (87.41 - 92.89 %) and *P. fluorescens* (73.98 - 78.94 %).

The effectiveness of 4 isolates of *B. subtilis* (B1, B2, B3 and B4) in controlling *R. bataticola* (*M. phaseolina*) infecting chickpea was investigated in the laboratory. Results showed that among these isolates, B4 was highly antagonistic to the fungi (Om Gupta and Sharma, 2003).

Suriachandraselvan *et al.* (2004) evaluated known fungal antagonists against *M. phaseolina*. Among them, *T. harzianum* recorded the maximum growth inhibition (74.0 %) followed by *T. virens* (73.0 %). They also reported that both bioagents produced non-volatile antibiotics in culture filtrate which showed the maximum fungi toxic effect against the pathogen.

Patil and Raut (2004) reported the efficacy of the biological control agents *T. harzianum* and *T. viride* against *R. bataticola* (*M. phaseolina*) in *in-vitro* using the dual culture technique. *T. viride* + *R. bataticola* recorded the highest growth inhibition of *R. bataticola* (45.4 %), followed by *T. harzianum* + *R. bataticola* (32.2 %).

Kumar *et al.* (2007) reported the inhibition of the mycelial growth of *M. phaseolina* by *P. fluorescens* under *in-vitro* conditions, reduction in disease severity, significant increase in the biomass, and increased production of IAA in culture medium by chickpea plants. *P. fluorescens* was reported to produce antifungal metabolites against

fungus pathogen of rice and completely inhibit the mycelial growth of *Pythium* spp., *Fusarium* spp., and *Rhizoctonia* spp (Reddy *et al.*, 2004).

Paramasivan *et al.* (2007) tested the fungal and bacterial antagonists against dry root rot pathogen in which *T. virens* (Isolate 4) resulted in maximum growth inhibition followed by *T. harzianum* and least growth inhibition by *P. fluorescens*. The per cent inhibition by all other fungal and bacterial antagonists ranged between 41 and 56 per cent respectively.

Salunke *et al.* (2008) tested the antagonistic activity of *T. viride*, *T. harzianum* and *P. fluorescens* against *R. bataticola* by dual culture method. *T. viride* was found most efficient bioagent which suppressed mycelia growth ranging from 71.11 to 75.92 per cent followed by *T. harzianum* (67.29 to 74.93 %) and *P. fluorescens* (60.37 to 70.00 %).

Surendra *et al.* (2008) reported that the antagonistic activity of *T. harzianum* against *M. phaseolina* showed maximum growth inhibition (75.5 %) followed by *T. virens* (65.93 %) and *G. virens* (57.77 %) and the least inhibition (45.74 %) by *Aspergillus* sp.

Konde *et al.* (2008) observed that *T. viride* inhibited the growth of *R. bataticola* to the extent of 96.39 per cent in pot culture studies. Maximum seed germination (84.45 %) and per cent disease reduction (52.39 %) was recorded in the seed treatment with thiram + carbendazim (2 + 1 g/kg).

Ammajamma *et al.* (2009) evaluated the biocontrol agents and indicated that maximum inhibition of mycelial growth of *R. bataticola* (76.63 %) was noticed in *T. harzianum* (Darwad isolate).

Anand *et al.* (2009) tested eight native isolates of *P. fluorescens* against *R. solani*, cotton root rot pathogen *in-vitro* and observed the maximum growth inhibition was by Pf 4 isolate followed by Pf 6 and minimum inhibition was by Pf 2.

Manjunatha (2009) reported that *in-vitro* studies of bioagents against *R. bataticola*, *T. viride* (Tv-R) showed maximum inhibition of mycelial growth (78.50 %) followed by *T. viride* (Tv-32) with 74.92 per cent and least inhibition (73.83 %) was recorded in *T. viride* (Tv-16) isolate. However, among two isolates of *T. harzianum*, Th-10 showed maximum inhibition (75.56 %) of mycelial growth than *T. harzianum* -R isolate.

Rajput *et al.* (2010) reported that *T. harzianum*, *T. viride*, *P. fluorescens* and *B. subtilis* exhibited antagonistic effect against *R. bataticola*. *T. harzianum* inhibited the maximum radial growth of *R. bataticola* (89.67 %).

Among fungal and bacterial antagonists, the maximum inhibition of mycelial growth was observed in *T. harzianum* (Th-55) isolate with 81.48 per cent followed by

Bacillus subtilis (75.85 %) and minimum inhibition was observed in *T. harzianum* isolate with 62.96 per cent (Nagamma, 2012).

2.8.2 *In-vitro* evaluation of Fungicides

Seed treatment and soil drenching of fungicides has been recommended for the control of dry root rot disease.

Peshney *et al.* (1992) reported that growth and sclerotial germination of *R. bataticola* were effectively controlled by thiram (0.2 %), captan (0.2 %), mancozeb (0.2 %), Iprodione (0.2 %), carbendazim (0.2 %) and tridemorph (0.2 %).

Chattannavar *et al.* (1988) isolated *R. bataticola* (*M. phaseolina*) from roots of *Casurina equisetifolia* in pure culture on potato dextrose agar and evaluated 11 fungicides against four concentrations (100, 250, 500 and 1000 ppm). Rovrol inhibited growth by 80 - 90 percent at all concentrations tested. Similarly, bayleton, penoxtine and penotil at 500 - 1000 ppm and vitavax at 1000 ppm were also effective. Whereas, alliette was least effective.

Prashanthi *et al.* (2000) reported efficacy of systemic fungicides (carbendazim, metalaxyl MZ and propiconazole) at 250, 500 and 1000 ppm and non-systemic fungicides (chlorothalonil SC, chlorothalonil WP, copper oxychloride, captafol and iprodione) at 1000, 2000 and 3000 ppm against *R. bataticola in-vitro*. Carbendazim and propiconazole were found highly effective with the mean mycelial growth inhibition of 100 and 93.70 per cent, respectively. Carbendazim showed 100 per cent inhibition at all concentrations tested. Among the non-systemic fungicides, chlorothalonil SC at 3000 ppm was found most effective (82.45 % inhibition of mycelial growth).

Prajapati *et al.* (2002) conducted laboratory studies to determine the efficacy of 11 fungicides, 8 non-target pesticides, and 2 biological control agents against *R. bataticola* (*M. phaseolina*) of dry root rot in chickpea and reported carbendazim, carbendazim + thiram, carboxin, and topsin-M (thiophanate-methyl) completely inhibited the growth of *R. bataticola*.

Four systemic and three non-systemic fungicides were evaluated against the *M. phaseolina in-vitro*. Among systemic fungicides benomyl significantly inhibited the growth of the fungus at 250 ppm followed by carbendazim. Among non-systemic fungicides, mancozeb and thiram at 500 ppm significantly inhibited fungal growth followed by captan (Lokesha, 2003).

Amaraya *et al.* (2004) conducted laboratory experiments to study the effect of 7 seed dressing fungicides against 2 isolates *sclerotinia sclerotiorum* and one isolate of *R. bataticola* (*M. phaseolina*) affecting chickpea and lentil by following poisoned food techniques at three concentrations (normal, double, and three-fold dose). Thiabendazole and carboxin-thiram were found best in inhibiting the radial growth of the different isolates.

Konde *et al.* (2008) studied the efficacy of fungicides against *R. bataticola*, *in-vitro* which revealed that carbendazim + thiram (0.1 + 0.2 %), penconazole (0.1 %) and thiophanate-M (0.1 %) completely inhibited (100 %) the growth of pathogen.

Salunke *et al.* (2008) carried out an experiment to test the efficacy of seven fungicides viz., carbendazim 12 % + mancozeb 63 % WP (0.2 %), zineb (0.2 %), tricyclazole (0.2 %), curzate M8TM (0.2 %), copper oxychloride (0.3 %), carbendazim (0.1 %) and thiram (0.3 %) against *R. bataticola* by poisoned food technique. Out of these, fungicides companion and zineb each at 0.2 per cent concentration completely inhibited growth of all isolates and the isolates showed tolerance to copper oxychloride.

Among five systemic fungicides evaluated, hexaconazole, carbendazim, difenoconazole and propiconazole recorded maximum mycelial inhibition of cent percent at all concentrations (50, 100, 250 and 500 ppm). Among the contact fungicides mancozeb recorded cent per cent inhibition of mycelial growth at all the concentrations viz., 250, 500, 750 and 1000 ppm. The combi products carboxin 37.5 % + thiram 37.5 % WP and carbendazim 12 % + mancozeb 63 % WP showed cent per cent inhibition of mycelia at all concentrations viz., 250, 500, 750 and 1000 ppm (Nagamma, 2012).

2.8.3 Management of dry root rot under greenhouse condition

The germination of chickpea in pot was found to be increased by treating seeds with secure 600 WG (51.03 %) and folicure 200 W (51.03 %) followed by BAU-biofungicide (48.98 %) over soil inoculated with pathogens. Disease incidence decreased by treating seeds with secure 600 WG (74.04 %) which was statistically similar to the treating seeds with heprodin 50 WP (73.94 %), BAU-biofungicide (73.64 %), provex 200 WP (73.32 %), gaivet (72.17 %) and du-du 70 SL (71.75 %). Plant stand were increased by seeds treating with secure 600 WG (98.67 %) which was statistically similar to seed treated with heprodin 50WP (98.53 %), BAU-biofungicide (98.13 %) and provex 200 WP (97.71 %). Shoot length (38.41 %), root length (66.83 %), vigour index (122.44 %), shoot dry weight (117.78 %) and root dry weight (158.33 %) were increased when seeds were treated with BAU-biofungicide over soil inoculated with pathogens (Gulam Morshed, 2013).

Studies on chickpea revealed that benomyl was found effective followed by *Pseudomonas fluorescens* in the inhibition of mycelial growth of *Fusarium solani*. Maximum germination percentage was recorded with benomyl (Benomyl 500 WP) @ 2 g/kg seed (treated check) and tebuconazole @ 1 ml/kg seed (95 per cent) followed by treatment with *P. fluorescens* @ 10 g/kg seed along with soil application of *P. fluorescens* @ 3 kg/acre (94 per cent). Maximization of growth parameters like root length, shoot length and total seedling length were observed highest with benomyl @ 2 g/kg seed as 17.0 cm, 10.3 cm and 27.3 cm, respectively (Bharathi *et al.*, 2014)

2.8.4 Management of dry root rot in field condition.

Jayasree *et al.* (2000) investigated that the *Pseudomonas fluorescens* strain pf1 effectively inhibited the mycelia growth of *M. phaseolina*, the pathogen causing dry root rot in blackgram and sesame. Among various methods of pf1 application, seed treatment + soil application recorded maximum yield in blackgram (1238 kg/ha) and sesame (1200 kg/ha) followed by seed treatment + soil drenching of carbendazim, which recorded grain yield to the extent of 1205 kg/ha in blackgram and 820 kg/ha in sesame.

Patel and Anahosur (2001) enumerated rhizosphere microflora of chickpea under irrigated conditions and found that carbendazim as seed treatment and soil drench coupled with soil application of *T. harzianum* reduced maximum fungal population of *Fusarium* sp. and *M. phaseolina*. FYM also reduced pathogen population and stimulated all other microflora more than other treatment there by showing antagonism against *Fusarium* sp. and *M. phaseolina*.

Gaikwad *et al.* (2002) studied the management of charcoal rot of sorghum by using three biocontrol agents viz., *T. viride*, *T. harzianum* and *P. fluorescens* and the fungicides thiram and carbendazim. They found that seed treatment with biocontrol agents and fungicides were highly effective against *M. phaseolina*. Similarly Sindhan *et al.* (2002) reported satisfactory disease control when *P. fluorescens* was used as seed treatment along with carbendazim against *M. phaseolina* in chickpea.

Many of the workers reported that soil application of *Trichoderma* sp. and *P. fluorescens* effectively controlled the diseases caused by soil borne plant pathogens and also synergistic effect in the growth of plants was observed (Suriachandraselvan, 1997; Manoranjitham *et al.*, 2000; Bharati *et al.*, 2004 and Poddar *et al.*, 2004) in several crops.

Khan and Gangopadhyay (2008) evaluated the combined seed treatment of *P. fluorescens* and fungicides against the dry root rot of chickpea and revealed that *P. fluorescens* isolate PFBC-26 was effective against the pathogen when used in combination with carbendazim.

Konde *et al.* (2008) studied the efficacy of fungicides against *R. bataticola*, *in-vitro* study revealed that carbendazim + thiram (0.1 + 0.2 %), penconazole (0.1 %) and thiophanate- M (0.1 %) completely inhibited (100 %) the growth of pathogen.

Nagamani *et al.* (2011) reported that seed treatment with carbendazim @ 2 g/kg of seed + *T. viride* @ 4 g/kg of seed + soil application of FYM fortified with *T. viride* recorded least dry root rot disease incidence (4.33 %) compare to the control (35.03 %). It also recorded highest growth parameters like shoot length (50.20 cm), root length (7.20 cm) and also highest yield (893 kg/ha). Whereas, in control least shoot length (26.50 cm), root length (3.50 cm) and yield (660 kg/ha) was recorded.

Manjunatha (2014) reported that seed treatment with tebuconazole 2 g/kg recorded lowest per cent disease incidence (26.61 %), with a highest yield of 722.81

kg/ha, that was on par with seed treatment with tebuconazole 2 g + carbendazim 1 g/kg which recorded 38.59 per cent disease incidence with a yield of 596.74 kg /ha followed by seed treatment with carbendazim 2 g + captan 1g/kg of seed recorded 46.27 per cent disease incidence with a mean yield of 519.92 kg/ha. They were significantly superior to control which recorded 95.77 per cent disease incidence with a lowest mean yield of 362.02 kg/ha.

III MATERIALS AND METHODS

In the present investigation, the following experiment were carried out in laboratory and field condition during 2014 - 2016. The survey for dry root rot incidence in chickpea, assessment of molecular variability in *M. Phaseolina*, screening of germplasms against disease, biochemical analysis of chickpea cultivars and management of disease in *in-vitro* as and *in-vivo* condition were carried out. The experiment was conducted at All India Co-ordinated Research Project on Chickpea, University of Agricultural Sciences, Gandhi Krishi Vignan Kendra (GKVK), Bengaluru located in Eastern dry zone (Zone-5), geographically at 12° 58' latitude north and 77° 35' longitude east. The details regarding the materials used and methodologies followed during the study are presented in this chapter:

3.1 Glassware cleaning

For all the laboratory experimental studies, corning and Borosil glasswares were used. The glasswares were kept in cleaning solution for 24 hrs, containing 60 g of potassium dichromate ($K_2Cr_2O_7$), 60 ml of concentrated sulphuric acid (H_2SO_4) in 1000 ml of water. They were washed with vim powder followed by washing in running tap water and then rinsed in distilled water before use.

3.1.1 Sterilization

All glassware, solid and liquid media were subjected to sterilization by autoclaving at 1.1 kg/cm² (121.6 °C) for 20 min. For isolation of the fungus, plant tissues were surface sterilized in 1: 1000 sodium hypochlorite solution followed by washing in sterile water for three times. All cultural studies were conducted in aseptic conditions under laminar flow. The tip of inoculation needle and forceps were sterilized over flame.

3.2 Survey for dry root rot of chickpea in southern parts of Karnataka

An intensive roving survey on the incidence of dry root rot caused by *M. phaseolina* was conducted in chickpea growing areas of Karnataka viz., Tumakuru, Chitradurga, Bengaluru rural, Bengaluru urban, Chikkaballapur, Chikkamagaluru, Hassan and Mysuru districts. In each district, minimum two taluks, from each taluk 2-9 villages and from each village minimum two chickpea fields were selected. During survey per cent disease incidence was recorded and infected plant samples were collected for further isolation of pathogen. The following formula was used to calculate disease incidence

$$\text{Per cent Disease Incidence} = \frac{\text{Number of disease plants}}{\text{Total number of plants}} \times 100$$

3.3 Assessment of molecular variability in *Macrophomina phaseolina* isolates of chickpea from different locations of India

3.3.1 Collection of *Macrophomina phaseolina* isolates from different locations of India

The chickpea plants showing typical symptom of dry root rot were collected from different states viz., Karnataka, Andhra Pradesh, Telangana, Tamil Nadu, Maharashtra, Madhya Pradesh, Punjab, Uttar Pradesh, Bihar, Uttarakhand, West Bengal and Himachal Pradesh. Totally twenty infected chickpea samples were brought to the laboratory and washed thoroughly in running water. The plants were dried in shade for 3-4 days and preserved for further studies.

The isolation of the fungus was done by following standard tissue isolation method under aseptic condition. The infected tissues of the root were cut into small bits of 1-2 mm size and surface sterilized in 1:1000 sodium hypochlorite solution for a minute and then washed thrice in sterile distilled water to remove the traces of sodium hypochlorite before transferring them to sterile potato dextrose agar [PDA] slants under aseptic conditions and incubated at 27 ± 1 °C.

3.3.1.1 Hyphal tip culture

Hyphal tip isolation was done on two per cent water agar plates. Dilute mycelial suspensions (10^{-6} ml⁻¹) of the fungus were prepared in sterile distilled water. One ml of such (10^{-6} ml⁻¹) suspension was spread uniformly on water agar plates and observed for mycelial growth initiation under microscope such part was marked with marker and that mycelial bit was transferred on PDA plates under aseptic conditions. The PDA plates were incubated at a temperature of 27 ± 1 °C and observed for fungal growth. This method was followed for maintaining pure culture of the fungus.

3.3.1.2 Maintenance of culture

The fungus was sub cultured on PDA containing petriplates and allowed to grow at 27 ± 1 °C. Such plates were stored in refrigerator at 4 °C and maintained by sub culturing at 20 to 25 days interval. This culture was used throughout the study.

3.3.1.3 Identification of the fungus

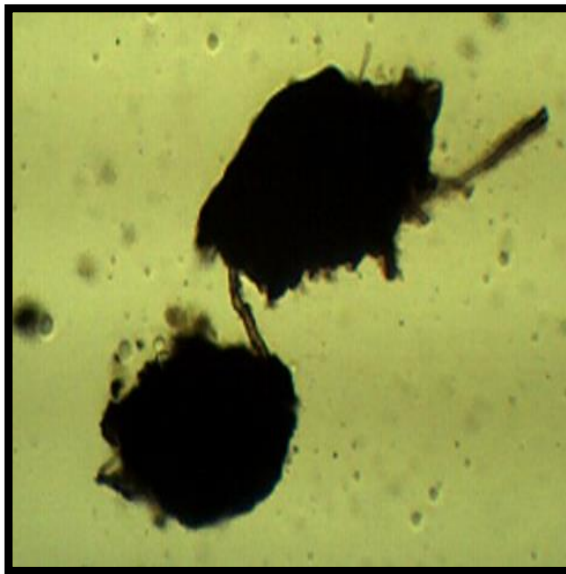
The morphological characters (sclerotial bodies, right angle branching of mycelia on PDA) and cultural characters (growth pattern, mycelial colour) were the principal characters considered for identification of pure culture isolates of *Macrophomina phaseolina*. The pure culture of the pathogen was maintained on PDA slants (Plate 1).

3.3.1.4 Mass multiplication of *M. phaseolina* isolates

Giant culture of *M. phaseolina* isolates was prepared by soaking chickpea seeds in tap water for 12 h. About 500 g of seeds were taken in 1000 ml conical flasks (500 g/flask) and were sterilized at 15 lb pressure, at temperature of 121.6 °C for 20 min. These flasks were inoculated with 7 day old culture of *M. phaseolina* isolates under laboratory condition and incubated at a temperature of 28 ± 1 °C for 30 days. The flasks



Colony characters of *M. phaseolina*



Sclerotial bodies on PDA



Right angle branching of mycelium

Plate 1. Morphological characters of *M. phaseolina*

were shaken at alternate days to get uniform growth of the culture. The giant cultures obtained were used for proving pathogenicity and virulence of these isolates.

3.3.2 Pathogenicity and virulence test

The pathogenicity and virulence of 20 *M. phaseolina* isolates was determined by modified sick pot method on susceptible chickpea cultivars GNG 2237, BG 1053, DBGV 151, L-550 and BG-212 (Nene *et al.*, 1981). Surface sterilized alluminium trays (45 cm × 30 cm) were filled with sterilized soil which was treated with 0.1 per cent formalin for a week under air tight polythene cover and inoculated 5 days prior to sowing with 8 day old inoculum of *M. phaseolina* multiplied on chickpea seed. Ten seeds of chickpea cultivars were sown in each tray in three replications and observed for symptoms development. Dry root rot incidence was recorded after 60 days of sowing at the time of maturity of the crop on the basis of the number of diseased plants out of the total number of plants. Re-isolation of the isolates were made from the roots of diseased plants on PDA medium, and isolated fungus was confirmed to its identity.

3.3.3 Identification of *M. phaseolin* strains

Ten host differentials viz., HK 11-104, DBGV 151, BGD 1094, AKG 1108, GJG 1205, H 11-22, DBGV 101, GNG 2226, DKG 964 and L550 were sown during the month of November 2014-15 at ZARS, GKVK, Bengaluru. At the time of sowing giant culture of *M. phaseolina* Bengaluru isolate was incorporated to the soil to improve sickness of the soil. The same host differentials were sown at Coimbatore, Raipur, Durgapura and Sabour (Anon., 2015). Observations on per cent dry root rot incidence was recorded from flowering to maturity stage.

The following formula was used to calculate dry root rot disease incidence:

$$\text{Disease Incidence} = \frac{\text{Number of plants infected}}{\text{Total number of plants}} \times 100$$

Reaction of the host differentials against dry root rot were categorized based on following scale given by Aghakhani and Dubey (2009b).

Sl. No.	Disease reaction	Dry root rot incidence
1.	Resistant	1-20 % disease incidence
2.	Susceptible	> 20 % disease incidence

3.3.4 Isolation of DNA from *M. phaseolina* isolates

The genomic DNA was extracted from *M. phaseolina* isolates by following CTAB extraction method (Doyle and Doyle, 1987).

1. Each fungal culture was inoculated in potato dextrose broth (PDB) and grown at a temperature of 30 °C for 7 days with sufficient shaking and aeration.

2. Approximately two grams of mycelium was harvested, frozen in liquid nitrogen and crushed in liquid nitrogen in a pre-cooled pestle and mortar to make fine powder.
3. The grinded powder was transferred to a 2 ml sterile polypropylene tube containing 1 ml of extraction buffer (2 % w/v CTAB, 1.4 M NaCl, 0.1 M Tris HCl, pH 8.0, 0.03 % β -Mercapto ethanol, 0.1 % PVP) pre-warmed to 65 °C.
4. The samples were incubated at 65 °C for 60 minutes with intermittent mixing of tubes at every 15 minutes.
5. After bringing back tubes to room temperature, equal volume (1 ml) of chloroform: Isoamyl alcohol (CIA) (24:1 v/v) was added to each tube containing sample and extraction buffer then mixed by inverting (no vortex).
6. Tubes were then centrifuged at 8500 rpm for 30 minutes.
7. Upper aqueous phase was carefully transferred to new sterile tube (CIA step was repeated once again if the upper layer did not look clear). Then equal volume of pre-chilled isopropanol was added and mixed gently.
8. Tubes were kept for 1-2 h at -20 °C / overnight and centrifuged at 8500 rpm for 15 minutes to pellet the DNA.
9. The supernatant was decanted and the pellet was washed with 1 ml of 70 per cent ethanol and then air dried completely for 1-2 hrs (there should not be any traces of alcohol).
10. The DNA pellet was suspended in 200-400 μ l of T₁₀E₁ (10 mM Tris-HCl; 1 mM EDTA maintained at pH 8) in sterile centrifuge tube. When DNA was fully dissolved, 1 μ l of RNase (stock 10 mg/ml) per 100 μ l volume of sample was added (Final RNase concentration 100 μ g/ml) and incubated at 37 °C for one hour or at room temperature for overnight.
11. Finally the samples were stored at -20 °C.

3.3.5 DNA quality and quantity estimation

To assess the quality of DNA, samples were run on 1 per cent agarose gel in 1x TAE buffer (For 250 ml 50x TAE – 60.5 gm of Tris base, 14.25 ml glacial acetic acid and 25 ml of 0.5M EDTA) and stained with 5 μ l ethidium bromide (10 mg/ml stock) per 100 ml of gel and checked for shearing of DNA and contamination of RNA.

3.3.6 PCR (Polymerase Chain Reaction)

Requirements:

1. Template: Purified genomic DNA extracts (20-30 ng/ μ l) of isolates were used as template DNA.
2. Buffers: 10x assay buffer from M/s Bengaluru Black biotech Pvt. Ltd. Bengaluru, India was used.

3. dNTP: Individual dNTPs, viz., dATP, dGTP, dCTP and dTTP were obtained from M/s Bengaluru Black biotech Pvt. Ltd., Bengaluru.
4. Thermal cycler: BIO-RAD thermal cycler was used for cyclic amplification of DNA.
5. Taq: Taq DNA polymerase (3 U/μl) was obtained from M/s Bengaluru Genei Pvt. Ltd., Bengaluru
6. Primers: Primers were synthesized from Sigma Pvt. Ltd., Bengaluru, India.

3.3.7 Molecular identification of *M. phaseolina* isolates

Molecular identification of isolates was done by using species specific primers MpKFI (5'-CCGCCAGAGGACTATCAAAC-3') and MpKRI (5'-CGTCCGAAGCGAGGTGTATT-3') in a PCR (Babu *et al.*, 2007). Reactions were carried out in a final volume of 15 μl containing 1.2 μl of template DNA (30 ng/μl), 0.3 μl of Taq polymerase (3 U/μl), 1.5 μl Taq buffer containing MgCl₂ (10x), 0.5 μl dNTP's (2.5 mM) each, 0.6 μl (10 pmol) each of forward and reverse primers and 10.3 μl of nuclease free water.

Amplification conditions (PCR protocol):

1. Initial denaturation : 95 °C for 5 min followed by 40 cycles
2. Denaturation : 94 °C for 45 sec
3. Annealing : 58 °C for 1 min
4. Extension : 72 °C for 2 min
5. Final extension : 72 °C for 8 min
6. Final hold : 4 °C

3.3.8 The ITS region amplification

The ITS regions and the 5.8S rDNA of 20 *M. phaseolina* isolates were amplified with a set of primers ITS1 (5'-TCCGTAGGTGAACCTGCGG-3') and ITS4 (5'-TCCTCCGCTTATTGATATGC-3') as described by White *et al.* (1990). The PCR-amplification reaction was optimized with various concentrations of MgCl₂, dNTPs, primer and template DNA. The amplification was carried out in a BIO-RAD thermal cycler in a 30 μl PCR reaction consisting of 3μl PCR buffer with MgCl₂ (10X), 0.6 μl of (3 U/μl) Taq polymerase (Bengaluru Genei, India), 1 μl (2.5 mM) dNTPs each, 1 μl (10 pmol) of each primer and 2.4 μl (30-50 ng) of genomic DNA. Initial denaturation of 95 °C for 5 min was followed by 40 cycles of 94 °C for 45 sec, 58 °C for 1 min (primer annealing), 72 °C for 2 min (primer extension). A final extension of 72 °C for 8 min was incorporated into the program, followed by cooling to 4 °C until recovery of the samples. PCR product (5 μl) was mixed with loading buffer (2 μl) containing 0.25 per cent bromophenol blue, 40 % w/v sucrose in water and then loaded in 1.5 per cent Agarose gel with 0.1 per cent ethidium bromide for examination with horizontal electrophoresis. Nucleotide base pairs of the amplicons were determined on the basis of its migration and confirmation relative to the molecular size marker of 1000 kb ladder.

3.3.9 The ITS region sequencing

The ITS region amplified PCR product produced bands (~550-600 bp) of 20 isolates on agarose gel were sent to sequencing to Genotypic Technology Pvt. Ltd., Bengaluru.

The sequenced PCR product was aligned with ex-type strain sequences from NCBI Gene Bank and established fungal taxonomy for identification and variability studies. Sequences were aligned following the Clustal W algorithm included in the Megalign module (DNASTAR Inc.) Multiple alignment parameters used were gap penalty = 10 and gap length penalty = 10. Both values are aimed to prevent lengthy or excessive numbers of gaps. The default parameters were used for the pair wise alignment. The use of Clustal W determines that, once a gap is inserted, it can only be removed by editing. Therefore, final alignment adjustments were made manually in order to remove artificial gaps. Phylogenetic analyses were completed using the MEGA package version 7 (Kumar *et al.*, 2015). Neither gaps (due to insertion-deletion events) nor equivocal sites were considered phylogenetically informative. Hence, complete deletion prevented the use of any of these sites in further analyses. Phylogenetic inference was performed by the neighbour joining method. Bootstrap tests with 1,000 replications were conducted to examine the reliability of the interior branches and the validity of the trees obtained (Felsenstein, 1985). An additional standard error test was performed with the data set using the same characters in order to evaluate the statistical confidence of the inferred phylogeny.

3.4 Screening of chickpea germplasm against dry root rot and biochemical analysis of cultivars

The chickpea germplasm were evaluated by two methods viz., field and blotter paper method for the identification of resistant sources against dry root rot disease.

3.4.1 Screening under field condition

A total of two hundred and two chickpea germplasm were screened in *M. phaseolina* sick plot field at G.K.V.K., Bengaluru, during *rabi* season of 2014-2015. Each germplasm was sown in five meter row length. After every five test germplasm one line of susceptible check L550 was sown. At the time of sowing giant culture of *M. phaseolina* Bengaluru isolate was incorporated to the soil to improve sickness of the soil. Observations on per cent dry root rot incidence were recorded from pod formation to maturity stage.

The following formula was used to calculate dry root rot disease incidence:

$$\text{Per cent Disease Incidence} = \frac{\text{Number of plants infected}}{\text{Total number of plants}} \times 100$$

After estimating disease incidence, the germplasm were categorized into different disease reactions as per standard AICRP scale mentioned below.

Categorization of chickpea germplasm into different disease reactions:

Sl. No.	Reaction	Dry root rot incidence (%)
1.	Resistant	0 -10
2.	Moderately resistant	11-20
3.	Moderately susceptible	21-30
4.	Susceptible	31-50
5.	Highly susceptible	51-100

3.4.1.1 Blotter paper technique

Ninety three chickpea germplasm were screened under blotter paper method for dry root rot (Nene *et al.*, 1981).

The 250 ml of PDB was prepared in 500 ml conical flasks and sterilized. The flasks were then inoculated with *M. phaseolina* and incubated for seven days. The mycelial mat from the flask was removed and macerated in a warring blender along with 100 ml of sterilized distilled water for a minute. The inoculum was later collected in a beaker. In the mean time, the chickpea seedlings were raised in plastic pots containing sterilized sand mixed soil. One week old seedlings were uprooted and the roots were immersed in sterile water in order to remove the adhered soil particles. The seedlings were then immersed completely in the inoculum in a beaker for a minute. The seedlings particularly the root portion were then placed side by side on a blotter paper (45 cm × 25 cm with one fold); in such a way that only the cotyledons and roots were covered and the green tops of seedlings remained outside and then blotter paper was folded. The folded blotter papers were then placed in trays, and kept in an incubator at 35 °C for eight days under 12 h artificial light, 80 per cent RH and the blotters were moistened with sterile water every day. At the end of the incubation period, the seedlings were examined for the extent of root damage. One folded blotter paper had seedlings of one test line and along with susceptible check (L550). Folded blotters were kept one above the others, in heaps of ten in a tray. One of these ten blotters had the seedlings of susceptible check. At the end of the incubation period (8 days), the seedlings were examined for the extent of root damage, and scored for the disease. Dry root rot severity was scored by using 1-9 rating scale on the 8th day after inoculation. Ten seedlings of each entry were considered as one replication, based on the disease score the germplasm were grouped into different reactions as mentioned below (Plate 2).

Categorization of chickpea germplasm into different disease ratings by using 1-9 scale (Nene *et al.*, 1981) and categorized as below:

Rating	Disease reaction	Symptoms
1	Resistant	No infection on roots
3	Moderately resistant	Very few small lesions on roots
5	Moderately susceptible	Lesions on roots clear but small, new roots free from infection
7	Susceptible	Lesions on roots many, new roots generally free from lesions
9	Highly susceptible	Roots infected and completely discolored

3.4.2 Biochemical analysis of cultivars

3.4.2.1 Drying and grinding

Chickpea roots were dried to a constant weight in an oven maintained at 60 °C. The dry weights of roots were recorded and the samples were powdered using an electric grinder. The grinded samples were used for various chemical estimation.

3.4.2.2 Protein extraction

Root portion of chickpea seedlings were washed under running tap water for 5 min, dried gently, weighed and grinded with a mortar and pestle. The grinded matter was homogenized (2 min, 4 °C) in phosphate buffer (1:2 w/v, pH 6, 0.05 M) by use of Corex tubes. The homogenate was centrifuged twice at 10,000 g and 4 °C, and the supernatant was collected and kept at -20 °C and was used in the enzyme assay.

3.4.2.3 Assay of peroxidase (PO)

The peroxidase activity was assayed spectrophotometrically (Hartee, 1955).

Reagents:

Hydrogen peroxide solution: 3.3 ml of H₂O₂ was mixed with 97.70 ml of distilled water to get 100 ml of 1 % H₂O₂ solution. The solution was prepared every time freshly.

Pyrogallol, 0.05 M: 6.3005 g of pyrogallol was dissolved in 100 ml of distilled water. The solution was prepared every time freshly.

Phosphate buffer (0.1 M, 6.5 pH): Solution A: 27.6 g of sodium phosphate monobasic (NaH₂PO₄·2H₂O Mol. wt. -156.01) was dissolved in small quantity of water and made up the volume to 1000 ml with distilled water.

Solution B: 28.4 g of sodium phosphate dibasic (Na₂HPO₄ Mol. Wt. 142 g) was dissolved in small quantity of water and made up the volume to 1000 ml with distilled water



Broth culture of *M. phaseolina*



Culture of *M. phaseolina*



Seedling inoculation



Placing on blotter



Incubation chamber



Healthy chickpea seedlings



Plate 2. Screening techniques for evaluation of chickpea germplasm under blotter paper

265 ml of solution A was mixed with 735 ml of solution B, finally pH was adjusted with NaOH.

Preparation of enzyme extract: One gram of plant root sample was homogenized in 3 ml of 0.1 M phosphate buffer, pH 6.5 at 4 °C. This mixture was filtered through 4 layer of muslin cloth. The filtrate was centrifuged at 12000 rpm at 4 °C for 20 minutes. The supernatant was collected and used for estimation of peroxidase activity.

Assay: The reaction mixture consisted of 1.5 ml of 0.05 M pyrogallol, 0.5 ml of the enzyme extract and 0.5 ml of one per cent H₂O₂. The reaction mixture was incubated at room temperature (28±1 °C). The change in absorbance was recorded at 470 nm at a time interval of 30 sec upto 3 min in Hitachi U-2900 spectrophotometer. The boiled enzyme preparation served as blank. The enzyme activity was expressed as change in the absorbance at 420 nm/min/g on fresh weight basis.

3.4.2.4 Assay of catalase

The activity of catalase was determined by the method of Aebi (1984) with slight modification by monitoring the disappearance of H₂O₂ at 240 nm. 3 ml H₂O₂ phosphate buffer was taken in cuvette and 100 µl enzyme extract was added. The absorbance was taken in the spectrophotometer at 240 nm. The absorbance was allowed to decrease from 0.45 to 0.40 from the time required in seconds (Δt) against diluted phosphate buffer as blank. Δt should be less than 60 seconds. The activity was calculated by using extinction coefficient 0.036 m/M/cm. One unit of the enzyme is the amount necessary to decompose 1 mmol of H₂O₂ per min at 25 °C.

3.4.2.5 Assay of phenylalanine Ammonia Lyase (PAL)

PAL activity was determined as the rate of conversion of L-phenyl alanine to trans-cinnamic acid at 290 nm as per the method described by Ross and Sederoff (1992).

Reagents:

Borate buffer (0.1M, 8.8 pH): 6.183 g of boric acid and 1 g of NaOH was dissolved in 800 ml of water and volume was made to 900 ml. To this solution 0.1 g of polyvinyl pyrrolidone (PVP) was added.

Substrate solution: (L-Phenyl alanine, 12 mM): 1.98 g of L-phenylalanine was dissolved in 1000 ml of distilled water. The solution was prepared freshly.

Trans-cinnamic acid: 29.64 mg of trans-cinnamic acid was dissolved in 10 ml of acetone. 100 µl of this solution was diluted to 10 ml with borate buffer to obtain 2 µl trans-cinnamic acid/ml working standard solution. The buffer was stored under refrigerated condition.

Trichloro acetic acid: (TCA, Mol. Wt. 163.39 g), 1 M: 16.339 g of TCA was dissolved in 100 ml of water.

Preparation of enzyme extract: one gram of root sample was homogenised with 5 ml of 0.1 M ice cold sodium borate buffer (pH 8.8). The homogenate was filtered through 4 layer muslin cloth. The filtrate was centrifuged at 15000 rpm at 4 °C for 20 min. The supernatant was collected and used for estimation of PAL activity.

Assay: Samples containing 0.4 ml of enzyme extract were incubated with 0.5 ml of 0.1 M borate buffer, pH 8.8 and 0.5 ml of 12 mM L-phenylalanine in the same buffer for 30 min at 30 °C. The reaction was arrested by adding 0.5 ml of 1M TCA and incubated at 37 °C for 5 min. The blank contains 0.4 ml of crude enzyme extract and 2.7 ml of 0.1 M borate buffer (pH 8.8) and absorbance was measured at 290 nm in Hitachi U-2900 spectrophotometer. Standard curve was drawn with graded amounts of cinnamic acid dissolved in acetone. The enzyme activity was expressed as μM of trans-cinnamic acid/min/g fresh weight of tissue.

3.4.2.6 Assay of total phenols

The total phenol content was estimated as per the procedure given by Sadasivam and Manickam (1996).

Reagents:

80 per cent ethanol: 80 ml of pure ethanol was diluted with 20 ml of distilled water.

FCR (Folin Cio-calteau reagent): Dilution was made in 1:1 volume with distilled water before use.

Saturated sodium carbonate solution: 100 g of solid sodium carbonate was dissolve in 400 ml of distilled water in a 1000 ml beaker. The solution was heated for few minutes to get completely dissolved solution. It was filtered and filtrate was stored.

Standard: 100 mg of catechol dissolved in 100 ml distilled water. The working standard was prepared by diluting 10 times with distilled water to get the concentration of 100 $\mu\text{g/ml}$.

Extraction

Two hundred milligram of powdered root sample was grinded with pestle and mortar 10 times the volume of 80 per cent ethanol. The homogenate was centrifuged at 10000 rpm for 20 minutes. The supernatant was collected and the residue was re-extracted with 5 times the volume of 80 per cent ethanol, centrifuged and supernatant was collected. The supernatant was kept in hot water bath until it gets dried. The residue was dissolved in 5 ml of distilled water.

Procedure of estimation:

1. The extract of 0.5 ml was pipetted out to a test tube
2. The volume was made to 3 ml in each tube with distilled water
3. To each tube 0.5 ml of FCR reagent was added and mixed thoroughly

4. Exactly three minutes later, 1 ml of saturated sodium carbonate solution was added
5. The tubes were placed in hot water bath exactly for 1 min.
6. After cooling of tubes the absorbance was measured at 625 nm against reagent blank.
7. The standard curve was prepared using different concentrations of catechol.
8. The phenol activity was calculated using a standard curve prepared with catechol and expressed as µg/g root.

3.4.2.7 Assay of total free amino acids

Total amino acids were determined by the ninhydrin method of Yemm and Cocking (1955) with some modifications. The incubation mixture (5±1 ml final volume) containing 100 µl of the ethanolic extract, 1 ml of 80 per cent ethanol, 1 ml of 0±2 ml citrate buffer pH 5, and 2 ml of an acetoninic ninhydrin solution (1 % ninhydrin and 0±06 % KCN in acetone), was incubated for 15 min at 100 °C. The mixture was cooled for 5 min in tap water before 8 ml of distilled water was added. The absorbance was recorded at 570 nm. Glycine equivalents were calculated from a standard curve obtained with pure analytical grade glycine.

3.4.2.8 Assay of total sugars

The total sugar content was estimated by Anthrone method and as per the procedure given by Hedge and Hofreiter (1962).

Reagents: (Anthrone, 0.2 %): 1 g of anthrone was dissolved in ice cold con. H₂SO₄. This reagent was prepared freshly before the analysis was carried out.

Glucose standard solution: 100 mg of D-glucose was dissolved in 100 ml of distilled water. 10 ml of this solution was diluted to 100 ml and used as a working standard. A few drops of toluene was added to this working standard. This solution was stored in refrigerator for further use.

Hydrochloric acid (HCl) 2.5 N: 201.8 ml of 38 % HCl (Sp. Gr. 1.19 g/ml) was dissolved in 798.8 ml of distilled water.

Extraction

1. One hundred mg of the powdered root sample was hydrolysed by keeping it in a boiling water bath for 3 hrs with 5 ml of 2.5 N HCl.
2. The tube was cooled to room temperature and the acid was neutralised with solid sodium carbonate until effervescence ceases.
3. The volume was made to 10 ml and this was centrifuged at 1000 rpm for 5 min.
4. The supernatant was collected and used for estimation of total sugars.

Estimation:

1. 0.4 ml of the extract was taken in a test tube and about 6 ml of anthrone reagent was added and the volume was made up to 6.5 ml.
2. Anthrone reagent was added gradually with constant stirring.
3. Tubes were immediately cooled.
4. They were then transferred to boiling water bath for 6-8 minutes.
5. After cooling to room temperature OD reading was taken at 620 nm with the HitachiU-2000 spectrophotometer.
6. The sugar concentration was calculated with the help of standard curve prepared by D-glucose and expressed as µg/g of root.

3.4.2.9 Assay of reducing sugars

The reducing sugars of powdered plant root samples was estimated by Nelson-Somogyi method (Nelson, 1944)

Reagents: Alkaline copper tartrate (Reagent A): 2.5 g of anhydrous sodium carbonate, 2 g of sodium bicarbonate, 2.5 g of potassium sodium tartrate, and 20 g of anhydrous sodium sulphate was dissolved in 80 ml of water and volume was made to 100 ml.

Reagent B: 15 g of copper sulphate was dissolved in small quantity of water and few drops of con. sulphuric acid was added and volume was made to 100 ml. The reagent A and B was mixed in 96:4 proportions before use.

Arsenomolybdate reagent: 2.5 g of ammonium molybdate was dissolved in 45 ml of water to this 2.5 ml of sulphuric acid was added and mixed well. 0.3 g of disodium hydrogen arsenate was dissolved in 25 ml of distilled water. Both the solutions were mixed and incubated in 37 °C for 24 to 48 hours. This was stored in a brown coloured bottle.

Standard glucose solution; stock: 100 mg of D-glucose was dissolved in 100 ml of water.

Working standard: 10 ml of stock was diluted to 100 ml with distilled water (100 µg/ml).

Extraction of plant root material (Loomis and Shull, 1957):

1. One hundred mg of powdered root sample was grounded with 5 ml of 80 per cent boiling ethanol.
2. The supernatant was collected and kept on hot water bath at 80 °C for complete evaporation of ethanol.
3. The sugars in test tube was dissolved by adding 5 ml of distilled water. This extract was used for further estimation.

Estimation:

1. 0.4 ml of the extract was taken in a test tube and the volume was made up to two ml with distilled water.
2. Two ml of alkaline copper tartrate reagent was added and boiled in boiling water bath for 20 minutes.
3. It was then cooled and 1 ml of Arsenomolybdate solution was added and mixed thoroughly.
4. The colour was measured in the Hitachi U-2000 spectrophotometer at 520 nm.
5. Standard curve was prepared by using known quantity of D-glucose and the sample was estimated for reducing sugars by comparing with standard curve and expressed as mg/g root.

3.5 Management of *Macrophomina phaseolina* causing dry root rot in chickpea**3.5.1 In-vitro evaluation of Bioagents**

The antagonistic microorganisms viz., *Trichoderma harzianum*, *T. viride*, *Pseudomonas fluorescens* and *Bacillus subtilis* were evaluated for their antagonistic effect in *in-vitro* conditions against *M. phaseolina* by dual culture technique (Table 1).

In dual culture technique, twenty ml of sterilized and cooled potato dextrose agar was poured into sterile petriplates. Fungal antagonists was evaluated by inoculating the pathogen at one side of petriplate and the antagonist inoculated at exactly opposite side of the same plate by leaving 3-4 cm gap. In case of bacterial antagonist evaluation, two mycelial discs of pathogen were inoculated and bacterial antagonist was streaked in the center of the plate. Each treatment was replicated four times. After required period of incubation i.e., after control plate reached 90 mm diameter, the radial growth of pathogen was measured. Per cent inhibition over control was worked out according to equation given by Vincent (1927).

Table 1: List of bioagents evaluated against *M. phaseolina* by dual culture technique

Sl. No.	Name of the antagonists	Source
1.	<i>Trichoderma viride</i>	Biocontrol lab, UAS, G.K.V.K, Bengaluru.
2.	<i>Trichoderma viride</i> (Tv-27)	IIHR*
3.	<i>Trichoderma harzianum</i>	IIHR
4.	<i>Trichoderma harzianum</i> (Th-55)	IIHR
5.	<i>Trichoderma harzianum</i>	IIHR
6.	<i>Pseudomonas fluorescens</i>	Department of Microbiology, UAS, G.K.V.K, Bengaluru.
7.	<i>Bacillus subtilis</i>	Department of Microbiology, UAS, G.K.V.K, Bengaluru.

*IIHR- Indian Institute of Horticulture Research, Bengaluru.

3.5.2 *In-vitro* evaluation of Fungicides

Different systemic fungicides viz. carbendazim, tebuconazole, thiophanate methyl, difenoconazole and propiconazole were evaluated at different concentrations of 50, 100, 250 and 500 ppm and contact fungicides viz., mancozeb, chlorothalonil, captan and thiram were evaluated at different concentrations of 250, 500, 750 and 1000 ppm and combi-products fungicides viz., tricyclozole 4 % + mancozeb 62 % WP, hexaconazole 4 % + zineb 68 % WP, carboxin 37.5 % + thiram 37.5 % WP and carbendazim 12 % + mancozeb 63 % WP were evaluated at different concentrations of 250, 500, 750 and 1000 ppm (Table 2).

The fungicides were tested against *M. phaseolina* by adopting 'Poisoned food technique'. The required concentrations of chemicals were prepared and incorporated into sterilized, cooled potato dextrose agar. Twenty ml of cooled medium was poured into petridishes and all plates were inoculated with actively growing five mm mycelia disc of pathogen.

Three replications were maintained for each treatment. These plates were incubated at 27±1 °C for seven days, and colony diameter was recorded. Per cent inhibition of mycelial growth over control was calculated by using the formula of Vincent (1927) as follows:

$$I = \frac{(C-T)}{C} \times 100$$

Where , I = Per cent inhibition of mycelium

C = Growth of mycelium in control

T = Growth of mycelium in treatment

3.5.3 Statistical analysis of experimental data

Analysis and interpretation of the experimental data was done by employing completely randomised design (CRD) method for laboratory studies suggested by Panse and Sukathme (1985).

3.5.4 Management of dry root rot in pot condition.

The experiment was conducted in the green house of ZARS, UAS, GKVK, Bengaluru, by following the CRD (Completely Randomized) design with three replications.

Mass culture of *M. phaseolina* was inoculated and mixed with the sterilized soil in pot except for control before 48 hrs of sowing the seeds. Data on different parameters viz., percent germination, shoot length, root length, percent disease incidence, shoot weight and root weight were recorded on 60 days after sowing. The recorded data on different parameters were subjected to statistical analysis using CRD to find out the significance of variation among different experimental treatments. The difference

Table 2: List of fungicides evaluated against *Macrophomina phaseolina* by poison food technique

Sl. No.	Common name	Chemical name	Trade name
Systemic fungicides			
1.	Carbendazim	Methyl -2-benzimidazole carbomate	Bavistin 50 % WP
2.	Tebuconazole	1-(4-Chlorophenyl)-4,4-dimethyl-3-(1,2,4-triazol-1-ylmethyl)pentan-3-ol	Raxil 2DS (2 % w/w)
3.	Thiophanate methyl	1,2-bis(3-methoxy carbonyl -2-thiouredo) benzene	Topsin M 70 WP
4.	Difenoconazole	1-2-(2- chloro-4-(4-chlorophenoxy) phenyl)-4-methyl-1, 3-dioxolan-2 ylmethyl)- 1H-1, 2, 4-triazole	Score 25 EC
5.	Propiconazole	1-2-(2- chloro-4-(4-dichlorophenoxy))-4-propyl -1, 3-dioxolan-2 yl methyl)- 1H-1, 2, 4-triazole	Tilt 25 EC
Contact fungicides			
1.	Mancozeb	Zinc-manganese ethylene bisdithiocarbamate	Indofil 75 % WP M-45
2.	Thiram	Tetramethyl thiuram disulphide (TMTD)	Thiram 75 %WP
3.	Chlorothalonil	Tetrachloride isophthalonitrile	Kavach 75 % WP
4.	Captan	N-trichloromethyl thio4-cyclohexene-1, 2-dicarboximide.	Captaf 50 % WP
Combi – products			
1.	Tricyclazole 4 % + mancozeb 62 % WP	5- methyl -1,2,4 triazole (3,4b)- benzothiazole + Zinc-manganese ethylene bis dithiocarbamate	Merger
2.	Hexaconazole 4 % + zineb 68 % WP	(RS)-2-(2,4-dichlorophenyl)-1-(1H-1, 2, 4-triazol-1yl) hexane -2-01 + Zinc ethylene bis dithiocarbamate	Avtar
3.	Carboxin 37.5 % + thiram 37.5 % WP	5,6 dihydro 2 methyl 1, 4 oxathin 3 carboxaniline + Tetramethyl thiuram disulphide	Vitavax power
4.	Carbendazim 12 % + mancozeb 63 % WP	Methyl benzimidazol 2-ylcarbamate + manganese ethylene bis dithiocarbamate	Companion

between the treatments means were evaluated for significance using Duncan's Multiple Range Test (DMRT) following the procedure as described by Gomez and Gomez (1984).

3.5.5 Management of dry root rot in field condition.

The field experiment was carried out during *kharif* 2014-15 and *rabi* 2014-15 seasons at fields of ZARS, UAS, GKVK, Bengaluru under sick plot condition. Three replications were maintained for each treatment with RCBD design. The per cent disease incidence and yield was recorded in individual treatments and the data was statistically analyzed. The treatment details were as follows.

3.5.5.1 Experimental details

Location	ZARS, GKVK
Season	<i>Kharif and Rabi</i>
Variety	JG-11
Design	RCBD
Plot size	3 × 2 sq.m
Spacing	30 × 10 cm
Treatments	15
Replications	3

3.5.5.2 Treatment details

T₁: Seed treatment with *T. viridae* @ 10 g/kg of seed

T₂: Application of *T. viridae* 5 kg in 50 kg of FYM/ha

T₃: Application of *T. viridae* 10 kg in 50 kg of FYM/ha

T₄: Seed treatment with carbendazim 2 g + thiram 1 g/kg of seed

T₅: Seed treatment with carbendazim 2 g + captan 1 g/kg of seed

T₆: Seed treatment with tebuconazole 2 g/kg of seed

T₇: Seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed

T₈: Seed treatment with vitavax power 2 g/kg of seed

T₉: Seed treatment with vitavax power 2 g + *T. viridae* 5 g/kg of seed

T₁₀: Neem cake 250 kg/ha

T₁₁: Pongamia cake 250 kg/ha

T₁₂: Neem cake 250 kg/ha+ *T. viridae* 5 g/kg of seed

T₁₃: Pongamia cake 250 kg /ha + *T. viridae* 5 g/kg of seed

T₁₄: Sorghum 1:3 ratio

T₁₅: Control

IV RESULTS AND DISCUSSION

The various laboratory and field experiments were carried out on dry root rot of chickpea caused by *Macrophomina phaseolina* (Tassi) Goid. with respect to survey for dry root rot of chickpea in southern parts of Karnataka, assessment of molecular variability in *M. phaseolina* isolates from chickpea, screening of chickpea germplasm against dry root rot and biochemical analysis of cultivars and management of dry root rot in chickpea in both *in-vitro* and *in-vivo* condition. The findings of various investigations with appropriate discussion are presented here under in details.

4.1 Survey for dry root rot of chickpea in southern parts of Karnataka

A survey was carried out in dry root rot growing areas of southern parts of Karnataka viz., Tumakuru, Chitradurga, Bengaluru rural, Bengaluru urban, Chikballapur, Chikmagalur, Hassan and Mysuru districts. The disease incidence was assessed based on the symptoms of dry root rot. The incidence of dry root rot disease varied with locations.

Dry root rot incidence varied from 8.65 per cent to 69.30 per cent in different villages (Table 3 and Plate 3). Among the districts the highest mean disease incidence was recorded in Chitradurga district (56.10 %) followed by Tumakuru (52.57 %), Bengaluru urban (51.13 %), Bengaluru rural (40.53 %), Chikballapur (36.41 %), Mysuru (32.09 %), Hassan (29.97 %) and Chikmagalur (24.73 %). The data is presented in Table 4 and Fig 1. This wide variation in disease incidence may be due to the change in the environment conditions, cultivars used, variation in date of sowing and cultural practices followed.

These results are in conformity with the findings of Khan *et al.*, (2012) who observed the highest DRR incidence (40.00 per cent) in village Shangus and lowest (4.11 per cent) in village Naina. The high incidence of the disease in such field might be due to the fact that the disease perpetuates through debris in field. The same type of observation was recorded by the survey conducted during *rabi* season of 2008 which revealed that dry root rot of chickpea varied from locality to locality due to different soil conditions (Black/Red soil conditions), cultivars used, cultivation practices and environmental conditions prevailing over these tracts. The higher incidence may be due to exposure of chickpea plants to moisture stress conditions evidenced during *rabi* season, which ultimately led to more production of sclerotia of *Rhizoctonia* sp. on chickpea plants roots. (Manjunatha *et al.*, 2011). This was supported by the findings of Tariq Hussain *et al.* (1997). The disease incidence of dry root rot of soybean was significantly high when inoculated seedlings were water stressed and grown at low soil moisture level (Wokocha, 2000).

Similarly, Srinivas (2016) conducted a survey in *rabi* cropping season in different chickpea growing locations of central (Madhya Pradesh and Maharashtra) and southern (Andhra Pradesh, Telangana and Karnataka) India. The maximum dry root rot incidence was observed in Telangana (18.28 %) and the least in Maharashtra (5.38 %). Disease occurrence was observed irrespective of cropping system, soil types and cultivars. The

Table 3. Survey for dry root rot disease incidence of chickpea in southern Karnataka

Sl. No.	District	Taluk	Village	Range	Disease incidence (%)	Mean disease incidence (Taluk)	Mean disease incidence (District)
1.	Tumakuru	Tumakuru	Begur	30-55	53.15	48.78	52.57
			Bisalahalli	30-45	43.52		
			Bommanahalli	20-30	26.76		
			Devlapura	50-62	60.00		
			G. Nagenalli	45-52	48.60		
			Hebbur	40-50	49.41		
			Hirehalli	40-62	60.00		
		Tiptur	Tiptur	20-50	49.54	55.72	
			Alur	35-55	50.00		
			Byrapura	50-70	65.50		
			Gopalpura	50-62	60.00		
			kallahalli	45-55	53.97		
			Alur	51-61	54.01		
			Yedehalli	40-60	57.05		
		Pavagoda	Venkatapura	35-43	41.22	53.21	
			Chikkahalli	38-62	60.00		
			Hosahalli	51-65	55.04		
			Kambalahalli	50-62	60.00		
			Yettinahalli	45-51	49.80		
2.	Chitradurga	Hiriyur	Khandikere	55-60	56.13	53.29	56.10
			Berenahalli	50-60	59.01		
			Kodihalli	45-55	52.31		
			Madenahalli	20-35	30.21		
			Nandihalli	60-70	69.30		
			Ambalagere	35-55	52.77		
		Hosadurga	Aralihalli	40-60	58.50	61.24	
			Honnenhalli	60-70	68.21		
			Hathihalli	51-70	69.26		
			Laxmidevarahalli	45-50	48.99		
		Challakere	Hulikunte	40-52	41.50	53.99	
			banjikere	50-65	50.30		
			Hiremadhure	60-70	61.45		

			Mahadevapura	60-70	68.70		
			Oblapura	35-50	48.00		
		Chitradurga	Bevinahalli	50-60	59.88	55.88	
			Haliyur	30-40	37.76		
			Lingadahalli	50-60	59.54		
			Surenahalli	60-70	66.32		
3.	Bengaluru Rural	Devanahalli	Bediganahalli	0-48	45.65	37.07	40.53
			Dasarahalli	50-60	54.22		
			Narayanapura	15-25	18.87		
			Hegganahalli	25-35	29.53		
		Dodballapura	Badrapura	50-61	56.77	45.71	
			Chikkamuddenahalli	23-32	30.75		
			Madhure	0-45	41.08		
			Melekote	50-55	54.23		
		Hoskote	Arehalli	45-50	41.88	38.82	
			Bagur	24-30	26.90		
			Chandrapura	30-35	33.24		
			Haralur	40-60	53.25		
4.	Bengaluru Urban	Anekal	Chikka hosahalli	55-70	65.31	42.97	51.13
			Gudnahalli	0-39	37.40		
			Shivanahalli	0-30	26.20		
		Bengaluru south	Anjanapura	50-60	59.29	59.29	
5.	Chikkaballapur	Gauribidanur	Basavanahalli	50-60	45.44	26.96	36.41
			Belthur	20-40	38.04		
			Gowdagere	0-20	16.85		
			Neralahalli	0-28	26.00		
		Chikkabalpura	Cheemanahalli	60-68	60.30	45.85	
			Dibbur	30-50	47.65		
			Gerehalli	50-60	53.11		
			Hiriyanaahalli	0-30	28.60		
			Thippenahalli	0-40	39.59		
6.	Chikkamagaluru	Chikkamagaluru	Nallur	0-20	18.42	18.61	24.73
			Ramanahalli	0-15	8.65		
			Kambihalli	20-39	32.96		
			Beeranahalli	0-15	14.07		
			Aldur	15-20	18.95		

		Kadur	Birur	30-40	36.11	28.79	
			Soorapura	25-36	24.21		
			Borapura	20-30	24.09		
			G. Thimmapura	20-30	28.07		
			Laxmipura	30-40	35.09		
			Kamasagara	35-40	38.99		
			Saraswathipura	20-30	27.93		
			Channapura	10-20	15.80		
			Bisalehalli	20-49	43.68		
		Tarikere	Channapura	20-30	29.96	26.79	
			Siddarahalli	0-15	13.31		
			Mallenahalli	30-40	35.09		
			Channapura	15-20	17.82		
			Baggavalli	20-40	35.50		
			Ajjampura	20-30	29.07		
7.	Hassan	Hassan	Arasihalli	20-30	26.66	27.90	
			Bylahalli	10-15	12.87		
			Hulihlli	40-60	55.42		
			Nayakarahalli	0-20	16.64		
		Arsikere	Bagivalli	40-50	44.80	40.21	
			Bandihalli	30-40	33.90		
			Chikkur	40-50	46.12		
			Pura	25-40	36.00		
		Chanarayapatana	Somlapura	0-15	13.04	21.81	
			Shettihalli	20-40	33.99		
			Hampanahalli	15-30	18.39		
		8.	Mysuru	K. R. Nagara	Sreerampura	0-20	18.52
Chikkahansogae	15-25				24.38		
Channamgere	40-50				45.89		
Saligrama	10-40				39.69		
Somanahalli	0-39				37.67		
Narachanahalli	40-50				43.39		
Hunsur	Athiguppe			17-32	19.53	29.26	
	Kempanahalli			0-20	14.10		
	Madahalli			20-40	36.43		
	Kangal			14-29	26.03		
	Hallikere			40-60	50.23		

Table 4. Mean dry root rot disease incidence of chickpea in different taluks and districts of southern Karnataka

Sl. No.	District	Taluk	Mean disease incidence (Taluk)	Mean disease incidence (District)
1.	Tumakuru	Tumakuru	48.78	52.57
		Tiptur	55.72	
		Pavagoda	53.21	
2.	Chitradurga	Hiriyur	53.29	56.10
		Hosadurga	61.24	
		Challakere	53.99	
		Chitradurga	55.88	
3.	Bengaluru Rural	Devanahalli	37.07	40.53
		Dodballapura	45.71	
		Hoskote	38.82	
4.	Bengaluru Urban	Anekal	42.97	51.13
		Bengaluru south	59.29	
5.	Chikkaballapur	Gauribidanur	26.96	36.41
		Chikkabalpura	45.85	
6.	Chikkamagaluru	Chikkamagaluru	18.61	24.73
		Kadur	28.79	
		Tarikere	26.79	
7.	Hassan	Hassan	27.90	29.97
		Arsikere	40.21	
		Chanarayapatana	21.81	
8.	Mysuru	K. R. Nagara	34.92	32.09
		Hunsur	29.26	

disease incidence was low in the irrigated fields compared to rainfed fields. The variation in disease incidence was due to diversified weather conditions, variation in sowing dates, different crop growth stages.

Similarly the mean maximum dry root rot incidence was noticed in Gulbarga district (9.8 %) followed by Raichur (7.6 %) and the least in Bidar (6.18 %) (Manjunatha *et al.*, 2011). Dry root rot disease was found at all locations and incidence ranged from 8.9 per cent to 10.3 per cent irrespective of cultivar type and locations. The result indicated that dry root rot was currently highly distributed in all surveyed chickpea growing areas of central and southern parts of India (Raju Ghosh *et al.*, 2013). The *R. bataticola* was found to be the predominant pathogen in all the chickpea growing districts of Karnataka which infected 60.0 per cent plants in the fields of Gulbarga (Gurha and Trivedi, 2008). The dry root rot incidence of 9.97 per cent in Gulbarga followed by Raichur (5.16 %) and Bidar (4.28 %) was recorded in chickpea. The highest incidence was 12.24 per cent in Chittapur of Gulbarga and least was in one of the locations of Raichur (3.21 %) as reported by Veerendra Kumar (2004).

4.2 Assessment of molecular variability in *Macrophomina phaseolina* isolates of chickpea from different locations of India

In nature, new strains may arise by mutation, hybridization, differential cytoplasmic inheritance (Hughes, 1956), heterokaryosis (Buxton, 1954) and by parasexual life cycle (Pontecarvo, 1949). Study of pathogenic variability is essential for breeding disease resistance in a crop improvement programme.

A potential pathogen is often blessed with biodiversity within its population. Basically, variation in pathogen is desirable trait for its existence in nature. This variability among the pathogens underlies their diverse nature and ability to withstand the host environment. Variability of pathogens can be studied by molecular variation in *M. phaseolina* isolates collected from different locations.

4.2.1 Collection of representative *M. phaseolina* isolates from different locations of India

Twenty dry root rot infected chickpea samples were collected from major chickpea growing states of India viz., Karnataka, Andhra Pradesh, Telangana, Tamil Nadu, Maharashtra, Madhya Pradesh, Punjab, Uttar Pradesh, Bihar, Uttarakhand, West Bengal and Himachal Pradesh. The *M. phaseolina* was isolated from infected samples by following standard tissue isolation procedure, the pure culture obtained was sub-cultured on petriplate and slants containing potato dextrose agar were stored in refrigerator for further studies. The isolates collected were designated as KAMP1, KAMP2, KAMP3, KAMP4, KAMP5, APMP6, APMP7, TEMP8, TEMP9, TNMP10, TNMP11, MHMP12, MPMP13, PUMP14, UPMP15, UPMP16, BHMP17, UKMP18, WBMP19, HPMP20 (Table 5 and Plate 4).

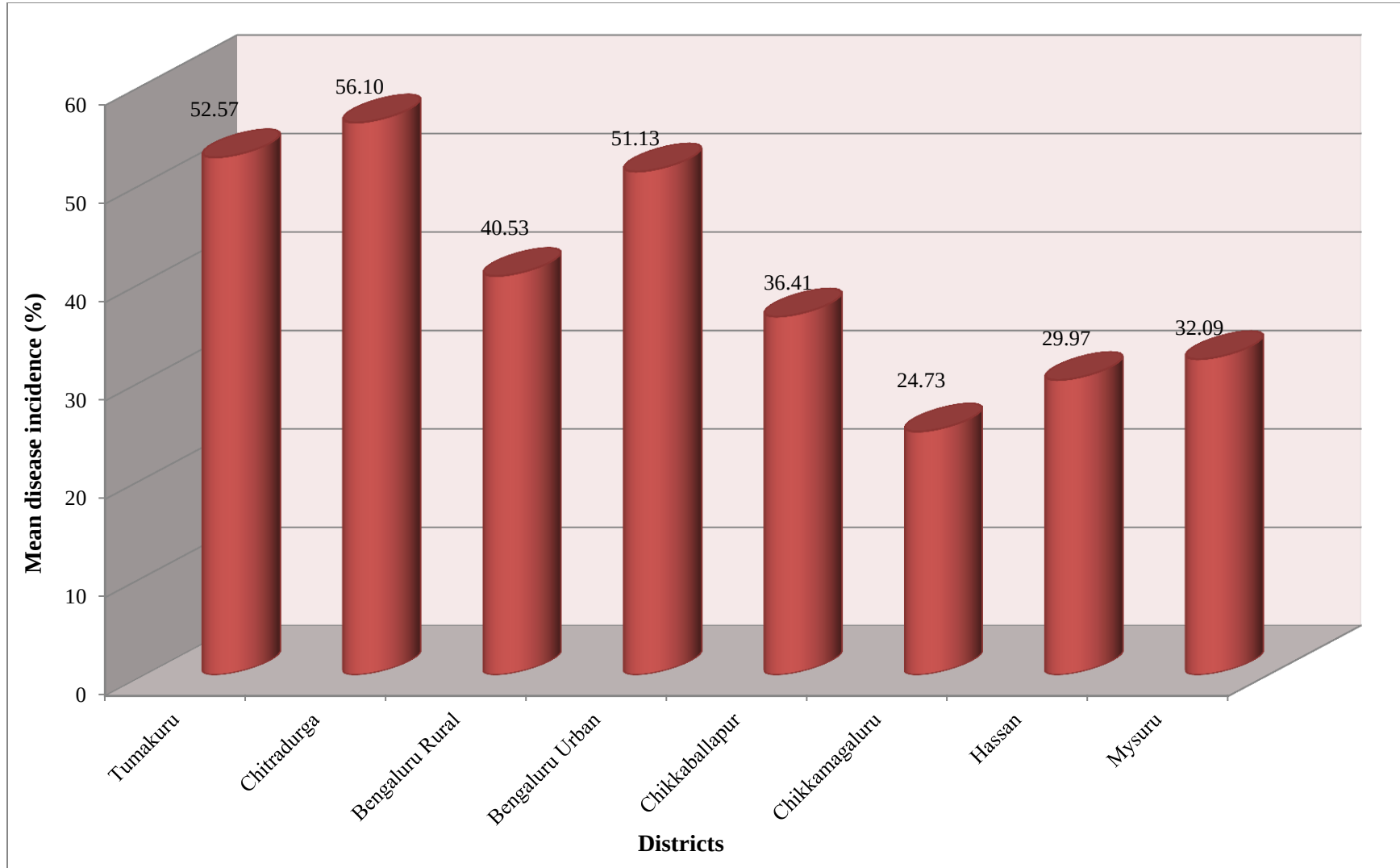


Fig. 1. Mean dry root rot disease incidence of chickpea in southern Karnataka

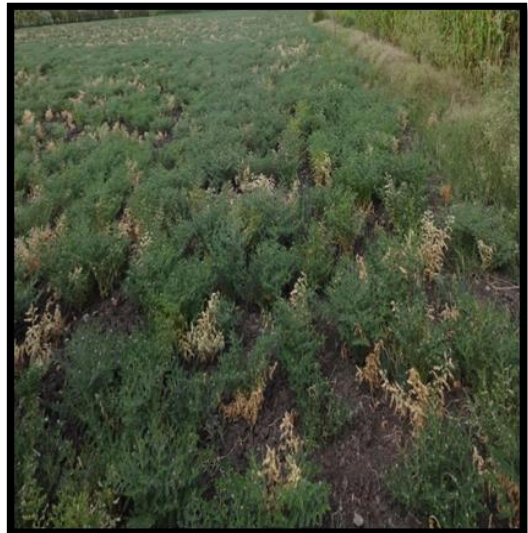


Plate 3. General view of farmer's fields infected with chickpea dry root rot disease

Table 5. List of *M. phaseolina* isolates collected from different geographical locations of India

Sl. No.	State	District	Place	Designation of isolates
1.	Karnataka	Bengaluru north	GKVK	KAMP-1
2.	Karnataka	Chitradurga	Hiriyur	KAMP-2
3.	Karnataka	Hubli-Dharwad	Dharwad	KAMP-3
4.	Karnataka	Gulbarga	Jevargi	KAMP-4
5.	Karnataka	Raichur	Sarjapur	KAMP-5
6.	Andhra Pradesh	Karnool	Karnool	APMP-6
7.	Andhra Pradesh	Karnool	Nandyal	APMP-7
8.	Telangana	Ranga Reddy	ICRISAT	TEMP-8
9.	Telangana	Warangal	RARS, Warangal	TEMP-9
10.	Tamil Nadu	Coimbatore	TNAU (Pappanaicken Pudur)	TNMP-10
11.	Tamil Nadu	Coimbatore	Periyanakulampalyam	TNMP-11
12.	Maharastra	Jalna	Badnapur	MHMP-12
13.	Madhya Pradesh	Ujjain	Rajwas	MPMP-13
14.	Punjab	Ludiyana	PAU, Ludiyana	PUMP-14
15.	Uttar Pradesh	Kanpur	IIPR, Kanpur	UPMP-15
16.	Uttar Pradesh	Shahjahanpur	Sujalpur	UPMP-16
17.	Bihar	Patna	Mokama	BHMP-17
18.	Uttarakhand	Udham Singh Nagar	CRC, Pantanagar	UKMP-18
19.	West Bengal	Burdwan	Durgapur	WBMP-19
20.	Himachal Pradesh	Sirmour	Dhaulakaun	HPMP-20

4.2.2 Pathogenicity and virulence test of representative isolates of *M. phaseolina*

Pathogenicity and virulence test was conducted by following artificial inoculation of soil with 20 isolates of *M. phaseolina* separately. The giant culture was prepared by using chickpea seeds as a substrate. Ten seeds of each cultivars viz., GNG 2237, BG 1053, DBGV 151, L-550 and BG-212 were sown in trays filled with soil to which the gaint culture of *M. phaseolina* was added. The dry root rot symptoms was recorded at 60 days after sowing. Symptoms involved viz., seed rot, yellowing, drooping and drying of leaves, the root system of infected plant turned dark and extensive rotting destroyed the lateral roots. Symptoms due to dry root rot disease affected plants in tray inoculated with *M. phaseolina* were found similar to that of the plants drying in the field (Plate 5). Re-isolation of the pathogen from infected root and the cultures obtained were compared with original culture of *M. phaseolina*. The cultures obtained were found to be similar with regards to all morphological characters viz., microsclerotial bodies, right angle branching of mycelia on PDA. Out of twenty isolates, five isolates viz., KAMP-2, KAMP-3, KAMP-4, TNMP-10 and PUMP-14 caused cent per cent dry root rot incidence on all five cultivars; four isolates viz., KAMP-1, MHMP-12, UPMP-16 and BHMP-17 caused 90-92 per cent incidence; six isolates viz., KAMP-5, APMP-6, APMP-7, MPMP-13, UKMP-18 and WBMP-19 caused 80-90 per cent incidence and the remaining five isolates TEMP-8, TEMP-9, TNMP-11, UPMP-15 and HPMP-20 caused 70-80 per cent dry root rot incidence (Table 6 and Plate 6).

These symptoms were in conformity with Kataria *et al.* (2007) who assessed nine isolates of *R. bataticola*, the incitant of dry root rot in chickpea and revealed that all nine isolates were found pathogenic on C-235 chickpea cultivar in both sterilized and unsterilized soil. The root rot incidence varied from 26 to 100 per cent in sterilized soil and 20 to 80 per cent in an unsterilized soil. Similarly Manjunatha *et al.* (2014) who conducted study by seedling inoculation of *M. phaseolina* in chickpea and obtained yellowing, drooping and drying of leaves, the root system of infected plant turned dark and extensive rotting destroyed the lateral roots.

According to Aghakhani and Dubey (2009a) twenty-three isolates of *M. phaseolina* causing dry root rot of chickpea where collected from 10 different major chickpea growing states of India. All the isolates were highly variable in their cultural characters as well as pathogenicity/virulence. The virulence analysis of the isolates on a set of chickpea cultivars viz., ICC 12441, ICC 11332, ICC 11224, ICC 12450, Pusa 362, BGD 112, Pusa 1103, Pusa 212, Pusa 1088 and BCP 17 under sick soil grouped them into 6 pathotypes. The pathotype groups were related to agro-ecological regions of the country. In contrast to the present results, Fernandiz *et al.* (2006) observed that pathogenic specialization of *M. phaseolina* was related to the geographic origin of the isolates.

4.2.3 Identification of *M. phaseolina* strains in chickpea cultivars

All ten host differentials viz., HK 11-104, DBGV 151, BGD 1094, AKG 1108, GJG 1205, H 11-22, DBGV 101, GNG 2226, DKG 964 and L550 were found susceptible for dry root rot at three locations viz., Bengaluru, Coimbatore and Raipur.



Plate 4. *M. phaseolina* isolates collected from different locations of India



Infected plants



Healthy Plants



Infected plants in field

Plate 5. Chickpea plants showing symptoms of dry root rot disease

Table 6. Virulence of *M. phaseolina* isolates on chickpea cultivars

Sl. No.	Cultivars	Isolates																			
		Per cent dry root rot disease incidence and reaction of different isolates*																			
		KAMP-1	KAMP-2	KAMP-3	KAMP-4	KAMP-5	APMP-6	APMP-7	TEMP-8	TEMP-9	TNMP-10	TNMP-11	MHMP-12	MPMP-13	PUMP-14	UPMP-15	UPMP-16	BHMP-17	UKMP-18	WBMP-19	HPMP-20
1.	GNG 2237	80.00 (63.43) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	70.00 (56.79) S	60.00 (50.77) S	80.00 (63.43) S	50.00 (45.00) S	60.00 (50.77) S	100.00 (90.00) S	60.00 (50.77) S	90.00 (71.57) S	80.00 (63.43) S	100.00 (90.00) S	70.00 (56.79) S	80.00 (63.43) S	90.00 (71.57) S	90.00 (71.57) S	80.00 (63.43) S	80.00 (63.43) S
2.	BG 1053	70.00 (56.79) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	60.00 (50.77) S	40.00 (39.29) S	80.00 (63.43) S	30.00 (33.21) S	70.00 (56.79) S	100.00 (90.00) S	40.00 (39.23) S	70.00 (56.79) S	50.00 (45.00) S	100.00 (90.00) S	40.00 (39.23) S	80.00 (63.43) S	90.00 (71.57) S	80.00 (63.43) S	60.00 (50.77) S	60.00 (50.77) S
3.	DBGV 151	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	80.00 (63.43) S	100.00 (90.00) S	70.00 (56.79) S	100.00 (90.00) S	80.00 (63.43) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	90.00 (71.57) S	100.00 (90.00) S	90.00 (71.57) S	90.00 (71.57) S	100.00 (90.00) S	80.00 (63.43) S
4.	L-550	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	80.00 (63.43) S	100.00 (90.00) S	70.00 (56.79) S	100.00 (90.00) S	90.00 (71.57) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	90.00 (71.57) S	100.00 (90.00) S	90.00 (71.57) S	90.00 (71.57) S	100.00 (90.00) S	80.00 (63.43) S
5.	BG-212	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	80.00 (63.43) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	90.00 (71.57) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	90.00 (71.57) S	100.00 (90.00) S	90.00 (71.57) S	100.00 (90.00) S	90.00 (71.57) S
Mean		90.00 (71.57) S	100.00 (90.00) S	100.00 (90.00) S	100.00 (90.00) S	86.00 (68.03) S	80.00 (63.43) S	80.00 (63.43) S	76.00 (90.00) S	74.00 (59.34) S	100.00 (90.00) S	72.00 (58.05) S	92.00 (73.57) S	86.00 (68.03) S	100.00 (90.00) S	78.00 (62.03) S	90.00 (71.57) S	92.00 (73.57) S	88.00 (69.73) S	88.00 (69.73) S	78.00 (62.03) S
S. Em ±		Cultivars				Isolates				Cultivars × Isolates											
		1.38				2.77				6.19											
CD @ 1 %		3.86				7.71				17.25											

*Figures in parenthesis are arc sin transformed values

R=Resistant (1-20 % dry root rot incidence)

S= Susceptible (> 20 % dry root rot incidence)

Host differentials GNG 2226 was found resistant at Durgapur whereas, all other host differentials were found susceptible at Durgapur.

The cultivars viz., GJG 1205, H 11-22, GNG 2226 and DKG 964 were found resistant at Sabour, whereas remaining differentials were found susceptible.

Based on the host differentials reaction against dry root rot at different locations, *M. phaseolina* isolates of Bengaluru, Coimbatore and Raipur were designated as *M. phaseolina* strain I because all host differentials viz., HK 11-104, DBGV 151, BGD 1094, AKG 1108, GJG 1205, H 11-22, DBGV 101, GNG 2226, DKG 964 and L550 showed susceptible reaction. Durgapur isolate designated as strain II because only one host differential viz., GNG 2226 showed resistance reaction, whereas, other host differentials showed susceptible reaction. The Sabour isolate was designated as strain III because four host differentials viz., GJG 1205, H 11-22, GNG 2226 and DKG 964 showed resistant reaction, whereas others showed susceptible reaction.

Based on the host differentials reaction of *M. phaseolina* isolates at five different locations, there may be prevalence of three different strains viz., I, II, III (Table 7).

Similarly Pathogenic variation in *Fusarium oxysporium* f. sp. *cubans* was well demonstrated by Haware and Nene (1982); Cabrera *et al.* (1985). Whereas, Mahesh (2008); Shashi Tiwari and Vishwa Dhar (2011) suggested the possibility of existence of pathogenic strains of the fungus *Fusarium udum* based on host differential reaction.

4.2.4 Molecular identification of *M. phaseolina* isolates

Twenty different isolates of *M. phaseolina* was collected from different locations of India and were used to characterize with respect to molecular identification of isolates by using species specific primers and for studying the molecular variability among the isolates by using ITS sequence.

The total genomic DNA of twenty *M. phaseolina* isolates were amplified using species specific primers MpKFI (5'-CCGCCAGAGGACTATCAAAC-3') and MpKRI (5'-CGTCCGAAGCGAGGTGTATT-3') by using PCR. The primers yielded single amplified product of 350 bp in all the isolates tested. This amplification of specific band confirmed that all the twenty isolates belonged to *M. phaseolina* (Plate 7).

The results are in conformity with Mahdizadeh *et al.* (2011) who identified fifty two *M. phaseolina* isolates that were recovered from 24 host plant species through the 14 Iranian provinces using species-specific primers MpKFI and MpKRI.

Babu *et al.* (2010) designed two species-specific primers MpKFI and MpKRI from the conserved region, adjacent to ribosomal 5.8 S gene. The designed primers yielded single amplified product of 350 bp with all the *M. phaseolina* isolates tested. Similarly twenty isolates of *M. phaseolina* were identified in chickpea by using species specific primers MpKFI and MpKRI (Manjunatha, 2014).



Plate 6. Virulence test of *M. phaseolina* isolates on different cultivars

Table 7. Identification of *M. phaseolina* strains in chickpea cultivars

Sl. No.	Cultivars	Disease incidence and reaction of chickpea cultivars				
		(Anon., 2015)				
		Bangalore (KAMP)	Coimbatore (TNMP)	Raipur (CGMP)	Durgapura (WBMP)	Sabour (BHMP)
1.	HK 11-104	50.00 (S)	100.00 (S)	73.30 (S)	82.50 (S)	26.70 (S)
2.	DBGV 151	75.00 (S)	59.80 (S)	77.80 (S)	81.70 (S)	25.60 (S)
3.	BGD 1094	66.67 (S)	65.70 (S)	91.70 (S)	100.00 (S)	23.00 (S)
4.	AKG 1108	100.00 (S)	42.70 (S)	45.40 (S)	36.50 (S)	22.60 (S)
5.	GJG 1205	30.90 (S)	29.20 (S)	88.70 (S)	37.20 (S)	17.30 (R)
6.	H 11-22	45.00 (S)	36.50 (S)	52.50 (S)	41.70 (S)	16.30 (R)
7.	DBGV 101	29.55 (S)	32.50 (S)	50.00 (S)	66.40 (S)	26.30 (S)
8.	GNG 2226	38.57 (S)	29.40 (S)	28.10 (S)	14.70 (R)	16.60 (R)
9.	DKG 964	32.14 (S)	34.20 (S)	34.70 (S)	36.50 (S)	18.60 (R)
10.	L550 (Check)	74.28 (S)	100.00 (S)	76.00 (S)	82.50 (S)	20.90 (S)
Strain		Strain I	Strain I	Strain I	Strain II	Strain III

R=Resistant (1-20 % dry root rot incidence)

S= Susceptible (> 20 % dry root rot incidence)

4.2.5 Molecular variability among *M. phaseolina* isolates by using Internal Transcribed Spacer (ITS) region

Genomic DNA of twenty isolates of *M. phaseolina* causing dry root rot of chickpea collected from different geographical regions was amplified. Main focus was on the ITS regions of ribosomal genes to identify and to study variability among *M. phaseolina* isolates. ITS region of rDNA was amplified using ITS-1 and ITS-4 primers for sequencing of its 18S rDNA region. The PCR amplified 18S rDNA region using ITS primers produced a band size of about 600bp (Plate 8). The sequence obtained upon sequencing of the ITS region from 20 different isolates were subjected to homology study using Mega 7 software along with sequences downloaded from NCBI database.

The homology studies involving isolates of *M. phaseolina* revealed that the partial sequence of ITS1-5.8S-ITS4 rRNA gene was as variable as rDNA regions. The sequence information was then analyzed through BLASTn program which indicated that the sequences contain the genetic information of internal transcribed spacer region of rDNA gene of *M. phaseolina* with nearly 96 - 100 per cent similarity.

The identified *M. phaseolina* rDNA gene sequences obtained from NCBI GenBank of various host plants were selected for comparison with the rDNA gene sequence of *M. phaseolina* isolate of chickpea. The sequence alignment of the isolate of *M. phaseolina* shows variation in this gene. The available sequences of *M. phaseolina* from NCBI were used in the pair wise and multiple sequence alignment using Bioedit software for determining the conserved regions of rDNA gene. Multiple sequence alignment revealed that there were quite a number of gaps introduced in the alignment within the ITS region which were closely related. Multiple and pair wise sequence alignment were generated and used to calculate evolutionary distances and per cent homology of sequence and the phylogenetic tree was constructed.

The evolutionary history was inferred using the Neighbor-Joining method. The bootstrap consensus tree inferred from 1000 replicates was taken to represent the evolutionary history of the taxa analyzed. Branches corresponding to partitions reproduced in less than 50 per cent bootstrap replicates were collapsed. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (1000 replicates) was shown next to the branches. The evolutionary distances were computed using the Maximum Composite Likelihood method and were in the units of number of base substitutions per site. The analysis involved 35 nucleotide sequences. Codon positions included were 1st+2nd+3rd+Noncoding. All positions containing gaps and missing data were eliminated. There were a total of 509 positions in the final dataset. Evolutionary analyses was conducted in MEGA7.

The phylogenetic tree was constructed by using nucleotide sequences from ITS region of 20 different *M. phaseolina* isolates along with 15 sequences of *M. phaseolina* sequences obtained from NCBI GenBank data base which generated a phylogenetic tree with clusters A and B indicating the variability in *M. phaseolina* and existence of two types of pathotype (Fig. 2).

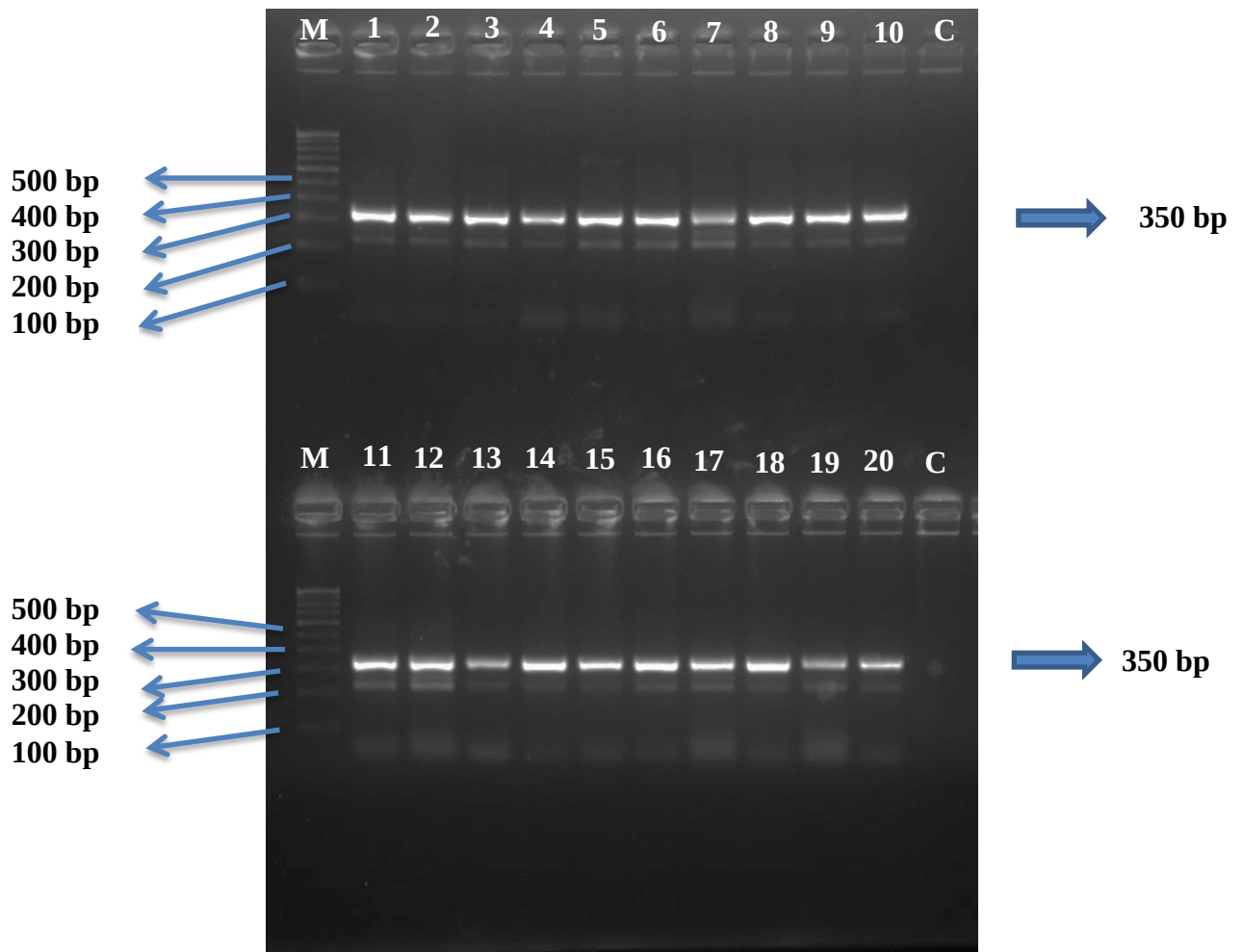


Plate 7. PCR amplification of *M. phaseolina* with species specific primers MpKF1 and MpKR1

M: Marker 1000 bp
1-20: *M. phaseolina* samples
C: Control

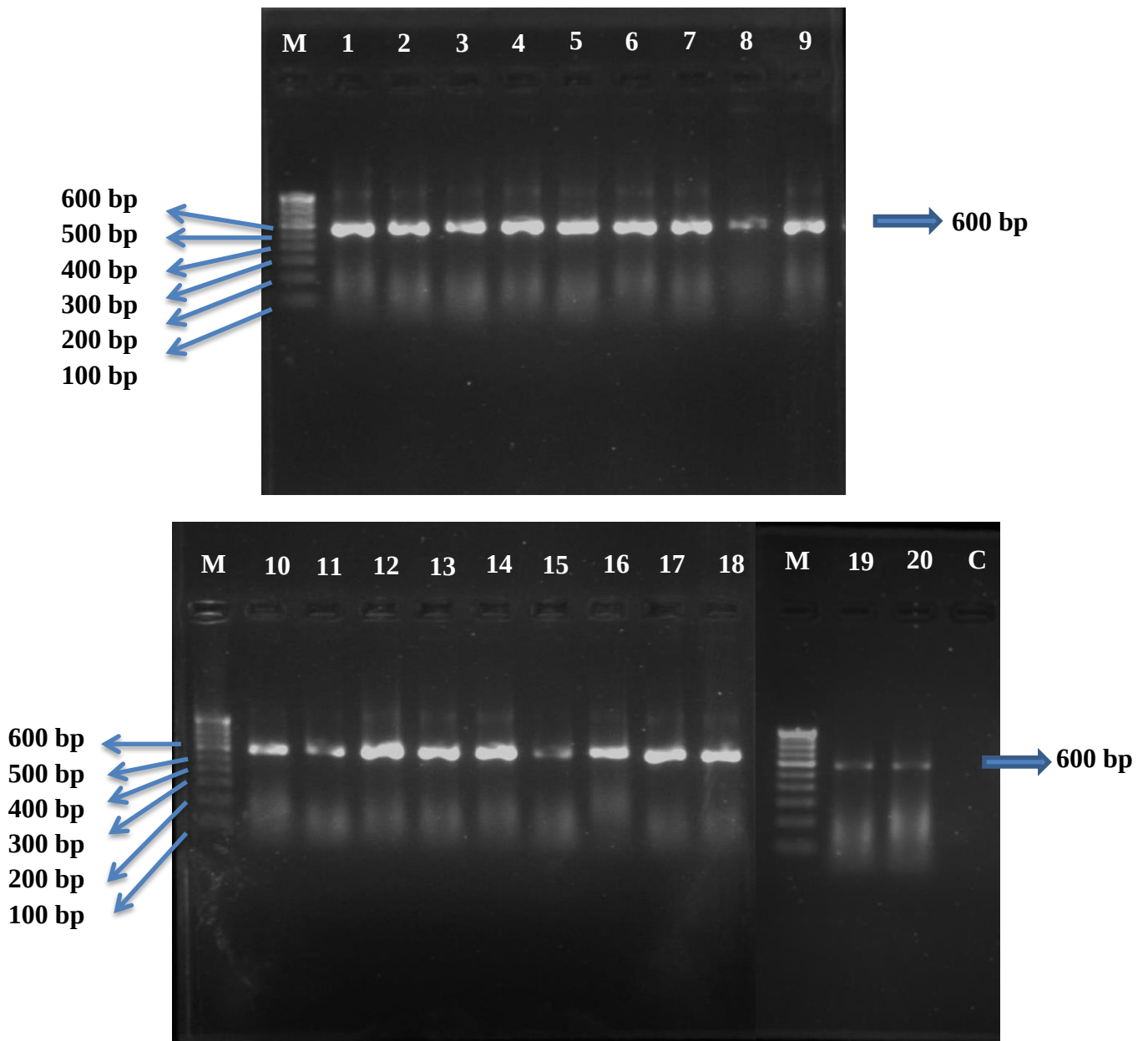


Plate 8. PCR amplification of ITS region of *M. phaseolina* with primers ITS1 and ITS4

M: Marker 1000 bp
 1-20: *M. phaseolina* samples
 C: Control

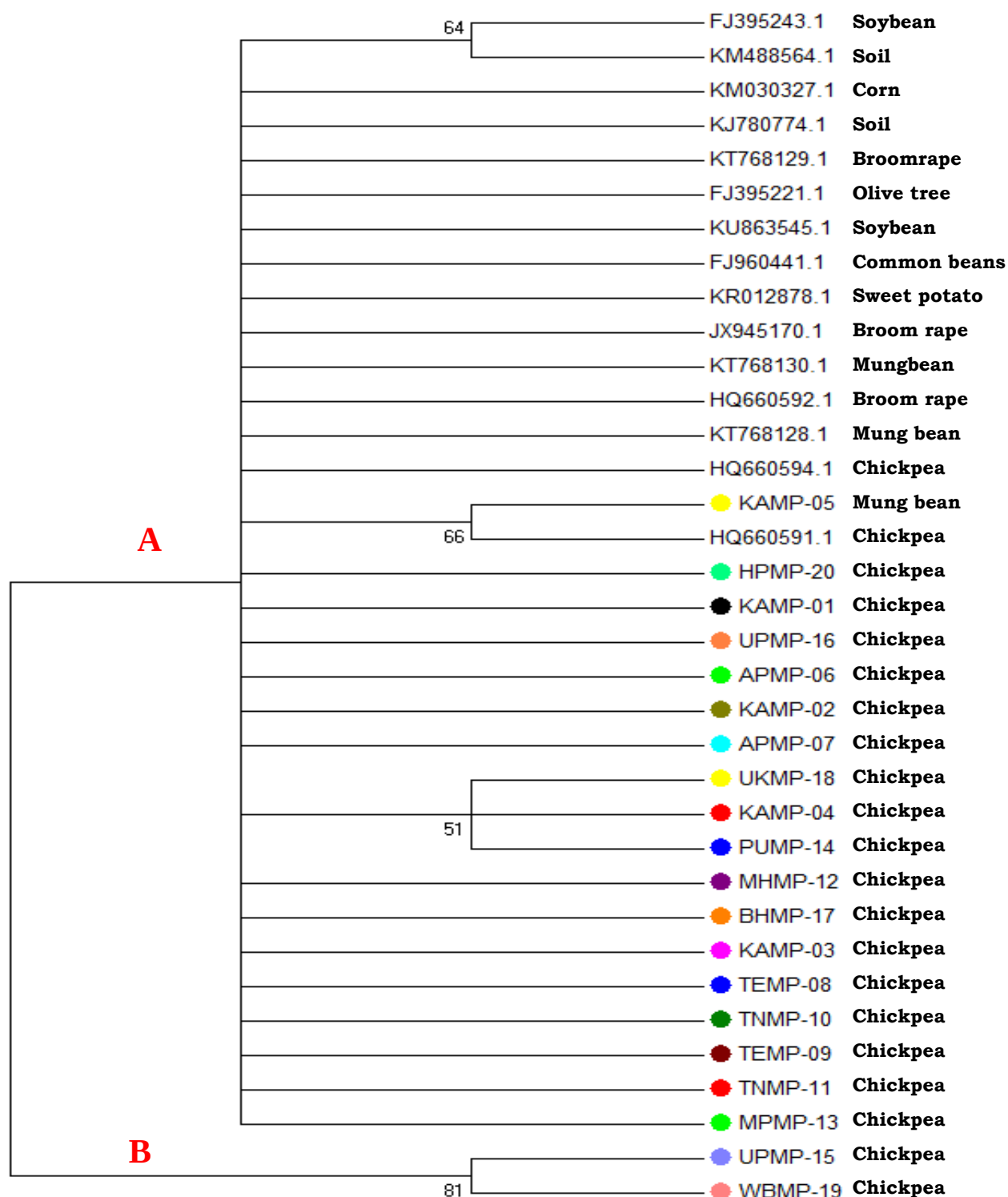


Fig. 2. Phylogenetic tree showing relationship among the *M. phaseolina* isolates based on ITS sequences

All the isolates showed about 65.28 to 100 per cent nucleotide similarity with already deposited *M. phaseolina* sequences in NCBI data base. Among the 20 isolates used in the present study the highest similarity (100 %) was observed between most of the isolates (Table 8).

Similar results were recorded by Manjunatha (2014) using the primers ITS-1 and ITS-4, a single band of 600bp was generated in the amplified pattern of all the isolates. the ITS sequence of all the isolates showed about 97 to 100 per cent similarity with already deposited *M. phaseolina* sequences in NCBI data base, the lowest similarity (32 to 37 %) was observed between the isolates and other genus plant pathogens.

Monga *et al.* (2007) made studies on 25 isolates of *R. bataticola* and obtained similar results. All the isolates belonging to one geographical location had not come under a single group, reflecting the fact that the variation was independent of geographical nearness of agroclimatic zone. These findings indicated a clear variation among the isolates and strongly support the possibility of existence of pathotypes in *M. phaseolina*.

Taxon-selective amplification of ITS regions was a common approach in molecular identification strategies. ITS regions was used successfully to generate specific primers capable of differentiating closely related fungal species (Bryan *et al.*, 1995). These rDNA was highly stable and exhibit a mosaic of conserved and diverse regions within the genome (Hibbett, 1992).

4.3 Screening of chickpea germplasm against dry root rot and biochemical analysis of cultivars

Chemical control of dry root rot is not effective as *M. phaseolina* has a broad host range and survives in soil for longer periods in the form of microsclerotia. The microsclerotial bodies can survive up to 10 months even in the absence of the host plants and under prevailing dry soil conditions. Use of host plant resistance is the most economical approach for the management of dry root in chickpea. A few chickpea lines with field tolerance to dry root rot has identified, but high levels of resistance are scarce in cultivated genotypes (Reddy *et al.*, 1991). Keeping this in view the present investigations were undertaken to evaluate performance of chickpea genotypes against dry root rot disease under sick plot condition (Plate 9) and blotter paper technique (Plate 10).

4.3.1 Screening of chickpea germplasm against dry root rot

Different chickpea germplasm were screened for dry root rot disease in sick plot maintained at All India Co-ordinated Research Project on chickpea, ZARS, GKVK, Bengaluru during 2014-15 and *in-vitro* screening was done by blotter paper method. The observations were recorded around flowering to maturity stage. The germplasm were categorized into disease reactions based on the disease incidence.

Table 8. Per cent similarity matrix for ITS region sequences of *M. phaseolina* with other reference sequences from GeneBank

Sequence	UPMP-15	WBMP-19	KM030327.1	FJ395221.1	FJ395243.1	KAMP-03	HPMP-20	KAMP-01	TNMP-11	JX945170.1	KU863545.1	KT768129.1	KT768128.1	KT768130.1	MPMP-13
UPMP-15	ID														
WBMP-19	91.17	ID													
KM030327.1	91.49	94.50	ID												
FJ395221.1	92.88	94.79	99.35	ID											
FJ395243.1	92.71	94.62	99.25	99.93	ID										
KAMP-03	89.41	90.42	93.92	95.27	95.10	ID									
HPMP-20	90.67	93.33	97.87	97.40	97.22	92.94	ID								
KAMP-01	91.60	93.61	97.50	98.42	98.25	94.00	96.30	ID							
TNMP-11	93.17	94.67	98.60	99.69	99.54	95.13	97.17	98.15	ID						
JX945170.1	93.17	95.00	98.92	100.00	99.85	95.50	97.50	98.50	99.71	ID					
KU863545.1	93.17	95.00	98.91	100.00	99.85	95.46	97.50	98.49	99.71	100.00	ID				
KT768129.1	93.17	95.00	98.91	100.00	99.85	95.46	97.50	98.49	99.71	100.00	100.00	ID			
KT768128.1	93.17	95.00	98.91	100.00	99.85	95.46	97.50	98.49	99.71	100.00	100.00	100.00	ID		
KT768130.1	93.17	95.00	98.91	100.00	99.85	95.46	97.50	98.49	99.71	100.00	100.00	100.00	100.00	ID	
MPMP-13	81.33	82.83	69.63	69.37	69.29	81.80	86.00	84.64	81.89	81.20	81.89	81.89	81.89	81.89	ID
UPMP-16	85.12	88.80	81.71	81.97	81.83	87.46	90.47	91.14	85.99	85.61	86.30	86.30	86.30	86.30	68.66
KAMP-02	86.59	90.07	92.69	91.14	90.97	89.63	94.43	93.32	91.14	91.14	91.14	91.14	91.14	91.14	87.59
BHMP-17	88.95	90.94	97.67	98.48	98.30	91.70	93.66	94.40	95.65	96.03	96.01	96.01	96.01	96.01	69.54
PUMP-14	86.50	88.50	67.26	65.43	65.28	88.74	91.83	91.93	87.12	87.35	87.42	87.42	87.42	87.42	80.43
MHMP-12	88.40	90.92	69.96	67.76	67.69	91.02	93.61	93.90	88.13	88.16	88.13	88.13	88.13	88.13	80.02
APMP-07	85.12	87.46	72.93	72.84	72.64	86.96	90.97	90.30	87.00	86.43	87.31	87.31	87.31	87.31	71.27
UKMP-18	89.11	91.96	70.82	68.21	68.06	91.60	94.64	94.79	90.91	90.98	91.22	91.22	91.22	91.22	81.11
KAMP-04	87.12	89.80	69.45	67.86	67.70	90.07	91.97	93.10	88.28	88.08	88.58	88.58	88.58	88.58	77.55
TNMP-10	90.47	91.30	69.48	69.12	68.96	92.41	94.31	95.28	90.56	90.43	90.56	90.56	90.56	90.56	77.31
TEMP-08	91.18	91.52	97.23	97.83	97.65	92.75	94.46	94.82	96.19	96.37	96.54	96.54	96.54	96.54	86.40
APMP-06	89.63	91.14	70.03	68.71	68.55	92.24	94.15	95.28	90.21	90.14	90.52	90.52	90.52	90.52	78.12
KJ780774.1	89.83	92.24	98.71	98.20	98.02	92.08	95.00	95.18	96.55	96.73	96.90	96.90	96.90	96.90	86.45
TEMP-09	90.30	92.14	73.27	74.16	73.97	93.25	95.15	96.29	90.43	90.06	90.43	90.43	90.43	90.43	77.11
KAMP-05	90.78	92.73	98.48	99.44	99.26	93.10	95.92	96.28	97.70	97.88	98.05	98.05	98.05	98.05	86.44
KM488564.1	90.16	92.06	97.61	98.56	98.74	92.41	95.16	95.52	96.89	97.07	97.24	97.24	97.24	97.24	86.36
KR012878.1	91.35	93.25	98.89	99.82	99.64	93.61	96.37	96.72	98.10	98.27	98.44	98.44	98.44	98.44	86.00
FJ960441.1	91.18	93.08	98.71	99.64	99.46	93.44	96.19	96.55	97.92	98.10	98.27	98.27	98.27	98.27	87.44
HQ660594.1	91.18	92.91	98.71	99.64	99.46	93.26	96.02	96.37	97.75	97.93	98.10	98.10	98.10	98.10	87.61
HQ660592.1	91.00	92.91	98.71	99.64	99.46	93.26	96.02	96.37	97.75	97.93	98.10	98.10	98.10	98.10	87.61
HQ660591.1	91.00	92.91	98.71	99.64	99.46	93.26	96.02	96.37	97.75	97.93	98.10	98.10	98.10	98.10	87.61

Contd...

Sequences	UPMP-16	KAMP-02	BHMP-17	PUMP-14	MHMP-12	APMP-07	UKMP-18	KAMP-04	TNMP-10	TEMP-08	APMP-06	KJ780774.1	TEMP-09	KAMP-05
UPMP-16	ID													
KAMP-02	87.48	ID												
UPMP-17	70.75	95.82	ID											
PUMP-14	80.70	89.27	92.93	ID										
MHMP-12	79.77	92.49	93.60	77.59	ID									
APMP-07	71.29	91.77	73.28	76.26	77.53	ID								
UKMP-18	81.55	92.21	93.45	79.80	82.52	78.02	ID							
KAMP-04	78.80	91.14	91.88	77.03	80.19	77.42	80.67	ID						
TNMP-10	81.13	90.78	94.91	75.92	82.12	78.72	80.54	83.04	ID					
TEMP-08	90.02	94.95	94.39	92.21	93.74	92.25	93.46	95.88	95.52	ID				
APMP-06	82.83	91.62	95.64	76.07	78.14	80.07	81.16	78.62	82.31	96.03	ID			
KJ780774.1	90.43	95.14	94.41	92.41	93.93	91.97	93.85	96.07	95.17	96.73	95.52	ID		
TEMP-09	80.97	90.64	95.64	79.00	80.71	78.03	84.47	79.68	82.73	96.21	86.83	96.38	ID	
KAMP-05	89.75	95.93	92.08	94.68	94.47	91.76	94.98	96.97	97.17	97.72	97.70	97.55	98.06	ID
KM488564.1	89.27	95.32	92.55	93.44	93.75	90.76	93.76	95.88	96.04	97.08	96.73	96.92	97.07	98.82
KR012878.1	89.59	96.39	92.72	93.77	94.96	91.74	94.09	96.70	97.07	98.11	97.59	98.11	98.10	99.66
FJ960441.1	91.74	96.21	95.48	93.60	94.78	93.63	94.84	97.25	96.90	97.94	97.41	98.11	97.93	99.47
HQ660594.1	91.57	96.21	95.66	93.43	94.61	93.46	94.66	96.91	96.72	97.60	97.24	97.76	97.76	99.12
HQ660592.1	91.57	96.21	95.66	93.43	94.61	93.46	94.66	96.91	96.72	97.60	97.24	97.76	97.76	99.12
HQ660591.1	91.57	96.21	95.66	93.43	94.61	93.46	94.66	97.25	96.72	97.60	97.24	97.76	97.76	99.47

Contd...

Sequences	KM488564.1	KR012878.1	FJ960441.1	HQ660594.1	HQ660592.1	HQ660591.1
KM488564.1	ID					
KR012878.1	98.35	ID				
FJ960441.1	98.80	99.83	ID			
HQ660594.1	98.46	99.49	99.31	ID		
HQ660592.1	98.46	99.49	99.31	99.66	ID	
HQ660591.1	98.46	99.49	99.31	99.66	99.66	ID

4.3.1.1 Screening of desi germplasm for dry root rot under field condition

Among one hundred and twenty two chickpea germplasm screened for host plant resistance against dry root rot incidence in field condition, none of the germplasm showed resistant reaction, while fifteen germplasm viz., Phule G 0405, Phule G 12107, GJG 1202, BG 372, BG 3054, IPC 2010-134, NDG 13-21, Phule G 0611, GJG 1205, IPC 2010-112, BG 3055, GJG 1207, DIBG 201, GNG 2258 and IC 83325 showed moderately resistant reaction with 11-20 per cent disease, thirty seven were found moderately susceptible, thirty four were susceptible and thirty six were highly susceptible (Table 9 and 10).

4.3.1.2 Screening of desi germplasm for dry root rot under blotter paper method

Out of fifty nine chickpea germplasm screened for resistance against dry root rot incidence in blotter paper method, none of the germplasm showed resistant reaction, while thirteen germplasms viz., Phule G 12107, GJG 1202, BG 372, BG 3054, IPC 2010-134, NDG 13-21, Phule G 0611, GJG 1205, IPC 2010-112, BG 3055, GJG 1207, DIBG 201 and GNG 2258 showed moderately resistant reaction with 11-20 per cent disease incidence, thirty one were found moderately susceptible, nine were susceptible and six were highly susceptible (Table 11 and 12).

4.3.1.3 Screening of Kabuli germplasm for dry root rot under field condition

Among forty five germplasm screened in field, none of the germplasm showed resistant reactions, seven germplasm viz., GNG 1969, HK 4, JGK 29, JGK 31, BG 3059, JGK 30 and Phule G 12404 showed moderately resistant reactions with 11-20 per cent disease incidence, nine germplasm showed moderately susceptible reaction, seventeen germplasm showed susceptible reaction and twelve germplasm showed highly susceptible reaction (Table 13 and 14).

4.3.1.4 Screening of Kabuli germplasm for dry root rot under blotter paper method

Out of thirty four chickpea germplasm screened for resistance against dry root rot incidence in blotter paper method, none of the germplasm showed resistant reaction, while six germplasm viz., GNG 1969, HK 4, JGK 29, JGK 31, JGK 30, and Phule G 12404 showed moderately resistant reaction with 11-20 per cent disease incidence, ten were found moderately susceptible, fifteen were susceptible and three were highly susceptible (Table 15 and 16).

4.3.1.5 Screening of ICRISAT germplasm for dry root rot under field condition

Among thirty five germplasm screened in field, none of the germplasm showed resistant reactions, three germplasm viz., ICCV 08412, ICCV08114 and ICCV05534 showed moderately resistant reactions with 11-20 per cent disease incidence, two germplasm showed moderately susceptible reaction, seven germplasm showed susceptible reaction and twenty three germplasm showed highly susceptible reaction (Table 17 and 18).



Plate 9. General view of dry root rot screening plot



Resistant (1)



Moderately Resistant (3)



Moderately Susceptible (5)



Susceptible (7)



Highly Susceptible (9)

Plate 10. Disease reaction of chickpea germplasm for dry root rot under blotter paper technique

Table 9. Screening of desi germplasm for dry root rot under field condition

Sl. No.	Germplasm	DRR incidence	Disease reaction
1.	GNG 469	22.50	MS
2.	GNG 2171	28.00	MS
3.	GNG 1581	28.93	MS
4.	GNG 1958	45.00	S
5.	Phule G 0405	18.10	MR
6.	DKG 876	66.67	HS
7.	NBeG 452	70.00	HS
8.	AKG 1108	100.00	HS
9.	GNG 2207	45.80	S
10.	GL 10023	100.00	HS
11.	BDNG 804	83.30	S
12.	DKG 964	32.14	S
13.	GAG 1107	35.00	S
14.	Phule G 0408	66.67	HS
15.	BG 3044	45.00	S
16.	PBG 5	62.50	HS
17.	IPC 2010-14	45.00	S
18.	BG 3043	66.70	HS
19.	GL 10006	100.00	HS
20.	BAUG 15	47.90	S
21.	BDNG 2013-1	57.50	HS
22.	DBGV 102	50.00	S
23.	Phule G 12110	39.29	S
24.	GNG 2264	54.17	HS
25.	DCP 92-3	32.50	HS
26.	BG 3051	79.17	HS
27.	GJG 1209	68.33	HS
28.	Phule G 12107	15.00	MR
29.	GJG 1202	14.44	MR
30.	AKG 1106	23.00	MS
31.	NBeG 453	29.21	MS
32.	H 11-58	23.93	MS
33.	JG 41	30.00	MS
34.	KGD 2011-1	29.17	MS
35.	DIBG 202	25.00	MS
36.	IPC 2011-138	25.00	MS
37.	GNG 2249	20.77	MS
38.	BGD 1088	23.33	MS
39.	Phule G 08108 (Mech.)	21.67	MS
40.	H 12-63 (Mech.)	33.33	S

Sl. No.	Germplasm	DRR incidence	Disease reaction
41.	HC 5 (Mech.)	24.29	MS
42.	NBeG 780 (Mech.)	25.00	MS
43.	BGD 1082	25.00	MS
44.	GNG 2215	28.57	MS
45.	Pusa 547	23.33	MS
46.	GJG 0831	29.17	MS
47.	GNG 2219	28.33	MS
48.	H 10-22	28.33	MS
49.	PG 0109	50.00	S
50.	IPC 2010-62	25.00	MS
51.	BG 3046	57.14	HS
52.	CSJ 769	29.17	MS
53.	IPC 2006-77	70.00	HS
54.	GNG 2127	35.00	S
55.	JG 14	45.00	S
56.	GNG 2144	68.75	HS
57.	BG 372	16.67	MR
58.	GNG 2259	28.33	MS
59.	GJG 1208	24.17	MS
60.	IPC 2007-28	39.29	S
61.	GNG 2261	28.10	MS
62.	BG 3054	15.83	MR
63.	IPC 2010-134	11.00	MR
64.	H 11-41	21.43	MS
65.	NBeG 454	23.00	MS
66.	Phule G 0609-15	24.29	MS
67.	NDG 13-21	15.00	MR
68.	BGD 1091	25.00	MS
69.	IPC 2010-72	55.28	HS
70.	H 10-41	47.50	S
71.	GAG 1107	61.25	HS
72.	GNG 2226	78.57	HS
73.	IPC 2009-191	83.33	HS
74.	Phule G 10114	50.00	S
75.	JG 16	32.92	MS
76.	GNG 2263	30.00	MS
77.	DBGV 101	49.55	S
78.	Phule G 0611	14.09	MR
79.	GJG 1205	10.95	MR
80.	IPC 2010-112	15.48	MR
81.	BG 3055	12.50	MR

Sl. No.	Germplasm	DRR incidence	Disease reaction
82.	GJG 1207	16.25	MR
83.	NBeG 47	28.00	MS
84.	Phule G 0302	46.67	S
85.	BG 3056	38.75	S
86.	H 11-22	45.00	S
87.	DIBG 201	12.50	MR
88.	GNG 2258	18.75	MR
89.	JG 62	57.94	HS
90.	JG 11	55.11	HS
91.	WR 315	83.33	HS
92.	Vishal	66.67	HS
93.	K850	50.00	S
94.	BBG1	70.00	HS
95.	GJG1003	80.00	HS
96.	AT-4-4	50.00	S
97.	BDNG-2003-1	57.14	HS
98.	JAKI 9218	20.83	MS
99.	IPC04-52	70.00	HS
100.	BDG-1061	42.86	S
101.	ICCV1116	68.89	HS
102.	HC-5	45.00	S
103.	GJG0907	22.50	MS
104.	NBeG165	65.71	HS
105.	Phule G 0204	51.43	HS
106.	ICC37	56.25	HS
107.	ST 10-4	20.56	MS
108.	BCP 60	42.86	S
109.	IC 83325	16.25	MR
110.	JG 27	33.93	S
111.	JG 40	41.67	S
112.	PBC 101	45.83	S
113.	JG 16	50.00	S
114.	WR 315	35.00	S
115.	IPCK 2004-29	100.00	HS
116.	C-104	42.50	S
117.	JG 74	91.67	HS
118.	GPF 2	33.76	S
119.	ICCV 07104	70.00	HS
120.	ICC 4951	46.97	S
121.	ICCV 08119	34.29	S
122.	L550 (check)	74.28	HS

Table 10. Disease reaction of desi germplasm for dry root rot under field condition

Sl. No.	Reaction	Germplasm	No. of germplasm
1.	Resistant (0-10 %)	-----	
2.	Moderately Resistant (11-20 %)	Phule G 0405, Phule G 12107, GJG 1202, BG 372, BG 3054, IPC 2010-134, NDG 13-21, Phule G 0611, GJG 1205, IPC 2010-112, BG 3055, GJG 1207, DIBG 201, GNG 2258 and IC 83325.	15
3.	Moderately Susceptible (21-30 %)	GNG 469, GNG 2171, GNG 1581, GNG 1958, AKG 1106, NBeG 453, H 11-58, JG 41, KGD 2011-1, DIBG 202, IPC 2011-138, GNG 2249, BGD 1088, Phule G 08108 (Mech.), HC 5 (Mech.), NBeG 780 (Mech.), BGD 1082, GNG 2215, Pusa 547, GJG 0831, GNG 2219, H 10-22, IPC 2010-62, CSJ 769, GNG 2259, GJG 1208, GNG 2261, H 11-41, NBeG 454, Phule G 0609-15, BGD 1091, JG 16, GNG 2263, NBeG 47, JAKI 9218, GJG0907 and ST 10-4.	37
4.	Susceptible (31-50 %)	GNG 2207, BDNG 804, DKG 964, GAG 1107, BG 3044, IPC 2010-14, BAUG 15, DBGV 102, Phule G 12110, H 12-63 (Mech.), PG 0109, GNG 2127, JG 14, IPC 2007-28, H 10-41, Phule G 10114, DBGV 101, Phule G 0302, BG 3056, H 11-22, K850, AT-4-4, BDG-1061, HC-5, BCP 60, JG 27, JG 40, PBC 101, JG 16, WR 315, C-104, GPF 2, ICC 4951 and ICCV 08119.	34
5.	Highly Susceptible (51-100 %)	DKG 876, NBeG 452, AKG 1108, GL 10023, Phule G 0408, PBG 5, BG 3043, GL 10006, BDNG 2013-1, GNG 2264, DCP 92-3, BG 3051, GJG 1209, BG 3046, IPC 2006-77, GNG 2144, IPC 2010-72, GAG 1107, GNG 2226, IPC 2009-191, JG 62, JG 11, WR 315, Vishal, BBG1, GJG1003, BDNG-2003-1, IPC04-52, ICCV1116, NBeG165, Phule G 0204, ICC37, IPCK 2004-29, JG 74, ICCV 07104 and L550 (check).	36
Total No. of germplasm			122

Table 11. Screening of desi germplasm for dry root rot under blotter paper method

Sl. No.	Germplasm	Dry root rot	
		Disease score	Disease reaction
1.	GNG 469	5	MS
2.	GNG 2171	5	MS
3.	GNG 1581	5	MS
4.	GNG 1958	7	S
5.	Phule G 12107	3	MR
6.	GJG 1202	3	MR
7.	AKG 1106	5	MS
8.	NBeG 453	5	MS
9.	H 11-58	5	MS
10.	JG 41	5	MS
11.	KGD 2011-1	5	MS
12.	DIBG 202	5	MS
13.	IPC 2011-138	5	MS
14.	GNG 2249	5	MS
15.	BGD 1088	5	MS
16.	Phule G 08108 (Mech.)	5	MS
17.	H 12-63 (Mech.)	7	S
18.	HC 5 (Mech.)	5	MS
19.	NBeG 780 (Mech.)	5	MS
20.	BGD 1082	5	MS
21.	GNG 2215	5	MS
22.	Pusa 547	5	MS
23.	GJG 0831	5	MS
24.	GNG 2219	5	MS
25.	H 10-22	5	MS
26.	PG 0109	7	S
27.	IPC 2010-62	5	MS
28.	BG 372	3	MR
29.	GNG 2259	5	MS
30.	GJG 1208	5	MS

Sl. No.	Germplasm	Dry root rot	
		Disease score	Disease reaction
31.	IPC 2007-28	7	S
32.	GNG 2261	5	MS
33.	BG 3054	3	MR
34.	IPC 2010-134	3	MR
35.	H 11-41	5	MS
36.	NBeG 454	5	MS
37.	Phule G 0609-15	5	MS
38.	NDG 13-21	3	MR
39.	BGD 1091	5	MS
40.	IPC 2010-72	9	HS
41.	H 10-41	7	S
42.	GAG 1107	9	HS
43.	GNG 2226	9	HS
44.	IPC 2009-191	9	HS
45.	JG 16	7	S
46.	GNG 2263	5	MS
47.	DBGV 101	7	S
48.	Phule G 0611	3	MR
49.	GJG 1205	3	MR
50.	IPC 2010-112	3	MR
51.	BG 3055	3	MR
52.	GJG 1207	3	MR
53.	NBeG 47	3	MS
54.	Phule G 0302	7	S
55.	BG 3056	7	S
56.	H 11-22	7	S
57.	DIBG 201	3	MR
58.	GNG 2258	3	MR
59.	L550 (check)	9	HS

Table 12. Disease reaction of desi germplasm for dry root rot under blotter paper method

Sl. No.	Reaction	Germplasm	No. of germplasm
1.	Resistant	-----	0
2.	Moderately Resistant	Phule G 12107, GJG 1202, BG 372, BG 3054, IPC 2010-134, NDG 13-21, Phule G 0611, GJG 1205, IPC 2010-112, BG 3055, GJG 1207, DIBG 201 and GNG 2258	13
3.	Moderately Susceptible	GNG 469, GNG 2171, GNG 1581, AKG 1106, NBeG 453, H 11-58, JG 41, KGD 2011-1, DIBG 202, IPC 2011-138, GNG 2249, BGD 1088, Phule G 08108 (Mech.), HC 5 (Mech.), NBeG 780 (Mech.), BGD 1082, GNG 2215, Pusa 547, GJG 0831, GNG 2219, H 10-22, IPC 2010-62, GNG 2259, GJG 1208, GNG 2261, H 11-41, NBeG 454, Phule G 0609-15, BGD 1091, GNG 2263 and NBeG 47	31
4.	Susceptible	GNG 1958, H 12-63 (Mech.), PG 0109, IPC 2007-28, H 10-41, JG 16, DBGV 101, Phule G 0302 and BG 3056	09
5.	Highly Susceptible	IPC 2010-72, GAG 1107, GNG 2226, IPC 2009-191, H 11-22 and L550 (check)	06
Total No. of germplasm			59

Table 13. Screening of Kabuli germplasm for dry root rot under field condition

Sl. No.	Germplasm	DRR incidence	Disease reaction	Sl. No.	Germplasm	DRR incidence	Disease reaction
1.	GNG 2237	32.00	S	24.	Phule G 12310	34.76	S
2.	BG 1003	62.50	HS	25.	HK 11-104	50.00	S
3.	HK 09-211	26.00	MS	26.	JGK 29	15.00	MR
4.	PG 071	22.50	MS	27.	HK 4	25.00	MS
5.	GNG 2228	39.29	S	28.	NBeG 460	26.66	MS
6.	HK 10-103	35.00	S	29.	DBGV 151	49.00	S
7.	HK 2	47.00	S	30.	BG 3057	24.50	MS
8.	GNG 1969	17.00	MR	31.	Phule G 12302	55.00	HS
9.	HK 4	13.68	MR	32.	HK 10-124	66.67	HS
10.	BG 1053	48.00	S	33.	IPCK 2009-79	33.33	S
11.	PKV Kabuli-2	37.50	S	34.	JSC 26	70.00	HS
12.	IPCK 2010-92	45.50	S	35.	NBeG 177	58.33	HS
13.	CSJK 77	100.00	HS	36.	JGK 31	12.50	MR
14.	PKV Kabuli-4	35.00	S	37.	NBeG 179	26.25	MS
15.	CSJK 4	43.75	S	38.	IPCK 2010-124	30.00	MS
16.	JGK 1	62.50	HS	39.	NBeG 399	36.67	S
17.	GNG 2285	50.00	S	40.	BG 3059	12.50	MR
18.	IPCK 2009-165	38.00	S	41.	JGK 30	16.67	MR
19.	NBeG 439	29.29	MS	42.	Phule G 12404	19.17	MR
20.	GNG 2281	46.67	S	43.	BGD 1097	87.50	HS
21.	IPCK 2002-29	70.00	HS	44.	KAK2	68.33	HS
22.	BGD 128	100.00	HS	45.	L550 (check)	80.44	HS
23.	JGK 28	28.33	MS				

Table 14. Disease reaction of Kabuli germplasm for dry root rot under field condition

Sl. No.	Reaction	Germplasm	No. of germplasm
1.	Resistant (0-10 %)	-----	0
2.	Moderately Resistant (11-20 %)	GNG 1969, HK 4, JGK 29, JGK 31, BG 3059, JGK 30 and Phule G 12404	07
3.	Moderately Susceptible (21-30 %)	HK 09-211, PG 071, NBeG 439, JGK 28, HK 4, NBeG 460, BG 3057, NBeG 179 and IPCK 2010-124	09
4.	Susceptible (31-50 %)	GNG 2237, GNG 2228, HK 10-103, HK 2, BG 1053, PKV Kabuli-2, IPCK 2010-92, PKV Kabuli-4, CSJK 4, GNG 2285, IPCK 2009-165, GNG 2281, Phule G 12310, HK 11-104, DBGV 151, IPCK 2009-79 and NBeG 399	17
5.	Highly Susceptible (51-100 %)	BG 1003, CSJK 77, JGK 1, IPCK 2002-29, BGD 128, Phule G 12302, HK 10-124, JSC 26, NBeG 177, KAK2, BGD 1094, and L550 (check)	12
Total No. of germplasm			45

Table 15. Screening of Kabuli germplasm for dry root rot under blotter paper method

Sl. No.	Germplasm	Dry root rot	
		Disease score	Disease reaction
1.	GNG 2237	7	S
2.	BG 1003	9	HS
3.	HK 09-211	5	MS
4.	PG 071	5	MS
5.	GNG 2228	7	S
6.	HK 10-103	7	S
7.	HK 2	7	S
8.	GNG 1969	3	MR
9.	HK 4	3	MR
10.	BG 1053	7	S
11.	PKV Kabuli-2	7	S
12.	IPCK 2010-92	7	S
13.	PKV Kabuli-4	7	S
14.	JGK 1	9	HS
15.	GNG 2285	7	S
16.	IPCK 2009-165	7	S
17.	NBeG 439	5	MS
18.	GNG 2281	7	S
19.	JGK 28	5	MS
20.	Phule G 12310	7	S
21.	JGK 29	3	MR
22.	HK 4	5	MS
23.	NBeG 460	5	MS
24.	DBGV 151	7	S
25.	BG 3057	5	MS
26.	JGK 31	3	MR
27.	NBeG 179	5	MS
28.	IPCK 2010-124	5	MS
29.	NBeG 399	7	S
30.	JGK 30	3	MR
31.	Phule G 12404	3	MR
32.	BGD 1097	5	MS
33.	Phule G 12407	5	MS
34.	L550 (check)	9	HS

Table 16. Disease reaction of Kabuli germplasm for dry root rot under blotter paper method

Sl. No.	Reaction	Germplasm	No. of germplasm
1.	Resistant	-----	0
2.	Moderately Resistant	GNG 1969, HK 4, JGK 29, JGK 31, JGK 30, and Phule G 12404	06
3.	Moderately Susceptible	HK 09-211, PG 071, NBeG 439, JGK 28, HK 4, NBeG 460, BG 3057, NBeG 179, IPCK 2010-124, and BGD 1097	10
4.	Susceptible	GNG 2237, GNG 2228, HK 10-103, HK 2, BG 1053, PKV Kabuli-2, IPCK 2010-92, PKV Kabuli-4, GNG 2285, IPCK 2009-165, GNG 2281, Phule G 12310, DBGV 151, NBeG 399 and Phule G 12407	15
5.	Highly Susceptible	BG 1003, JGK 1 and L550 (check)	03
Total No. of germplasm			34

Table 17. Screening of ICRISAT germplasm for dry root rot under field condition

Sl. No.	Germplasm	DRR incidence	Disease reaction
1.	ICC 1449	67.86	HS
2.	ICCV 055321	33.33	S
3.	ICCV 06302	50.00	S
4.	ICC 4951	58.93	HS
5.	ICCV 08412	20.00	MR
6.	ICC 11322	66.43	HS
7.	ICC 4951	61.11	HS
8.	ICC 07307	62.50	HS
9.	ICCV 0818	62.50	HS
10.	ICCV 08101	37.50	S
11.	ICC 5003	46.67	S
12.	ICC 08115	100.00	HS
13.	ICCV08101	80.00	HS
14.	ICC08102	45.00	S
15.	ICC14199	100.00	HS
16.	ICCV07312	100.00	HS
17.	ICCV07117	60.00	HS
18.	ICC4951	51.43	HS
19.	ICC04608	20.20	MS
20.	ICCV08114	16.67	MR
21.	ICC14402	87.50	HS
22.	ICCV08312	42.86	S
23.	ICCV05534	12.88	MR
24.	ICCV07112	71.67	HS
25.	ICCV08324	72.22	HS
26.	ICCV08112	83.33	HS
27.	ICCV05530	66.67	HS
28.	ICC4951	44.44	S
29.	ICCV96854	71.43	HS
30.	JAKI92418	75.00	HS
31.	GNG2182	90.00	HS
32.	ICC4951	56.25	HS
33.	PSG93	62.50	HS
34.	Phule G 10404	26.67	MS
35.	L550 (check)	77.98	HS

Table 18. Disease reaction of ICRISAT germplasm for dry root rot under field condition

Sl. No.	Reaction	Germplasm	No. of germplasm
1.	Resistant (0-10 %)	-----	0
2.	Moderately Resistant (11-20 %)	ICCV 08412, ICCV08114 and ICCV05534	3
3.	Moderately Susceptible (21-30 %)	ICC04608 and Phule G 10404	2
4.	Susceptible (31-50 %)	ICCV 055321, ICCV 06302, ICCV 08101, ICC 5003, ICC08102, ICCV08312 and ICC4951	7
5.	Highly Susceptible (51-100 %)	ICC 1449, ICC 4951, ICC 11322, ICC 4951, ICC 07307, ICCV 0818, ICC 08115, ICCV08101, ICC14199, ICCV07312, ICCV07117, ICC4951, ICC14402, ICCV07112, ICCV08324, ICCV08112, ICCV05530, ICCV96854, JAKI92418, GNG2182, ICC4951, PSG93 and L550 (check)	23
Total No. of germplasm			35

Since *M. phaseolina* is having wide host range, prolonged saprophytic survival ability and the sclerotia can survive upto 10 months even in the absence of the host can make any resistant cultivar susceptible with in a less time, so the development of resistant cultivars for dry root rot disease is becoming challenge for both breeders and pathologists. The present study was well supported by Saifulla *et al.* (2011b) who screened 16 chickpea genotypes for collar rot, dry root rot, wilt and rust diseases at GKVK, Bengaluru during the year 2010-11. For dry root rot HIR 60, BGD 256, A-1, JG 11, BBG 1, BBG 2, and JG 62 genotypes were found resistant to moderately resistant reaction, whereas HIR 60, BGD 256, A-1, BBG 1 and BBG 2 showed multiple disease resistance to collar rot, dry root rot, wilt and rust diseases.

Om Gupta *et al.* (2012) screened 170 chickpea genotypes for dry root rot at JNKVV, Jabalpur during the year 2007-10. Sixty eight genotypes exhibited resistant reaction (< 10 % mortality), out of which 26 were the promising lines viz., JG 1-14, 2-125, 2-4-110, 14-11, 14-10, 2001-13, 2001-13, 2001-18, 2001-80, 2001-115, 2002-20, 2003-95, 2003-14-16, 2004-110, 210, 9605, 1-9, 99-115, 2001-04, 2003-14-2, JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas. These lines were further evaluated for their performance in sick field for three consecutive years and six lines viz., JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and Rajas were found to be resistant exhibiting < 10 per cent mortality, however 14 lines showed moderately resistance reaction.

Nagamma and Saifulla (2012) screened sixty four Kabuli genotypes against dry root rot, out of which six genotypes viz., Phule G 04305, IPCK 07-62, RVSSG 12, HK 08-212, Phule G 09305 and AKG 2002-1K showed resistant reaction, nine genotypes viz., HK 94-134, GNG 1888, HK 06-152, GNG 2112, HK 06-171, Phule G 09316, CSJK 74, JGK 13 and CSJK 70 showed moderately resistant reaction. Thirty-five chickpea cultivars were screened for resistance to dry root rot in a field experiment, only 10 cultivars viz., ICC 2644, 10384, 10630, 112244, 11332, ICCL 81002, 810810, ICC 12263, 12441 and ICCV 90254 were found resistant. Whereas, five cultivars viz., Phule G 9504, 96020, 96105, 96313 and GL 91059 were moderately resistant to dry root rot (Gangwar *et al.*, 2002).

4.3.2 Biochemical analysis of chickpea cultivars

4.3.2.1 Peroxidase activity

There was an increase in the activity of peroxidase in all the genotypes when inoculated with *M. phaseolina*. Highest enzymatic activity was recorded at 48 hpi (hours after post inoculation) in moderately resistant genotypes, Phule G 12107 (3.80 min/g), NDG 13-21 (3.08 min/g) and IPC 2010-112 (2.93 min/g) compared to susceptible genotype GNG 2228 (2.64 min/g) and highly susceptible genotype L550 (2.56 min/g). However peroxidase activity started declining after 48 hpi (Table 19 and Fig 3). The mean total peroxidase activity was found to be more in Phule G 12107 (2.73 min/g) followed by NDG 13-21 (2.33 min/g), IPC 2010-112 (2.12 min/g), GNG 2228 (1.78 min/g) and L550 (1.67 min/g) at all stages of sampling.

Table 19. Peroxidase activity in chickpea genotypes

Peroxidase activity (min/g)							
Sl. No.	Genotypes	Duration of inoculation					
		Healthy	24 hpi	48 hpi	96 hpi	144 hpi	Mean
1.	IPC 2010-112 (MR)	2.52	2.54	2.93	2.14	0.45	2.12
2.	Phule G 12107 (MR)	3.24	3.54	3.80	2.57	0.48	2.73
3.	NDG 13-21 (MR)	2.82	2.91	3.08	2.36	0.46	2.33
4.	GNG 2228 (S)	1.98	2.14	2.64	1.87	0.26	1.78
5.	L550 (HS)	1.75	1.98	2.56	1.80	0.23	1.67
S. Em ±		Genotypes		Hours		Genotypes × Hours	
		0.01		0.014		0.031	
CD at 1 %		0.053		0.053		0.118	

hpi – Hours after post inoculation

MR – Moderately resistant

S – Susceptible

HS – Highly susceptible

During oxidative stress, the plant protects itself against reactive oxygen species by antioxidant enzymes as well as a wide array of non-enzymatic antioxidants (Das and Roychoudhury, 2014). Superoxide dismutase was considered to be the first line of defense against reactive oxygen species (ROS) and was the major O_2^- scavenger. Its enzymatic action resulted in H_2O_2 and O_2 formation. The H_2O_2 produced was then scavenged by several classes of peroxidases. Peroxidases decomposed H_2O_2 by the oxidation of phenolic compounds. Increased peroxidase activity had often been studied in connection with the oxidation of phenolic substances in the diseased plants and resistance in host attributed to toxicity of these oxidation products (Fric, 1976). Moreover, enhanced peroxidase activity was linked with synthesis of lignin (Ride, 1975).

Similarly there was an increase in the activity of peroxidase in all the genotypes when inoculated with pigeonpea sterility mosaic virus. Highest enzymatic activity after 28 DAI (Days after inoculation) was recorded in resistant/moderately resistant genotypes, ICP 7035 (37.40 %) and BRG 1 (34.69 %) at all stages of sampling compared to susceptible genotypes maruthi (19.44 %) and TTB 7 (18.91 %). However in susceptible genotypes peroxidase activity slightly increase over healthy and was observed upon 21 DAI later on it was decreased (Manjunatha, 2012). The peroxidase was increased in resistant cultivars than susceptible ones in *Fusarium oxysporum* f. sp. *ciceri* inoculated seedlings over control. In Vijay (wilt resistant), peroxidase activity was increased from 2.06 to 8.54 units (un-inoculated over inoculated) and in JG-62 (susceptible) activity was increased from 1.68 to 1.96 units (un-inoculated over inoculated) in seedlings (Dalvi *et al.*, 2011).

4.3.2.2 Catalase activity

There was an increase in the activity of catalase in all the genotypes when inoculated with *M. phaseolina*. Highest enzymatic activity was recorded at 144 hpi (hours after post inoculation) in moderately resistant genotypes, Phule G 12107 (5.42 $\mu M/min$), NDG 13-21 (4.66 $\mu M/min$) and IPC 2010-112 (4.01 $\mu M/min$) compared to susceptible genotype GNG 2228 (1.30 $\mu M/min$) and highly susceptible genotype L550 (1.25 $\mu M/min$) which recorded highest activity at 96 hpi. Catalase activity started declining after 96 hpi in susceptible and highly susceptible genotypes whereas, in moderately resistant genotypes the catalase activity prolonged upto 144 hpi (Table 20 and Fig 4). The mean catalase activity was found to be more in Phule G 12107 (2.04 $\mu M/min$) followed by NDG 13-21 (1.62 $\mu M/min$), IPC 2010-112 (1.45 $\mu M/min$), GNG 2228 (0.60 $\mu M/min$) and L550 (0.49 $\mu M/min$).

Superoxide dismutase was considered as first line of defense against reactive oxygen species (ROS) and was the major O_2^- scavenger. Its enzymatic action resulted in H_2O_2 and O_2 formation. The H_2O_2 produced was then scavenged by catalase and several classes of peroxidases. The catalase was a tetrameric heme protein found in peroxisomes, cytosol and mitochondria (Krych *et al.*, 2014). This enzyme had hyperoxidase activity which catalyzed the dismutation of hydrogen peroxide into water and oxygen. Catalase activity increased during infection as a mechanism to scavenge fungitoxic H_2O_2 . The major function of catalase within cells was to prevent the accumulation of toxic levels of

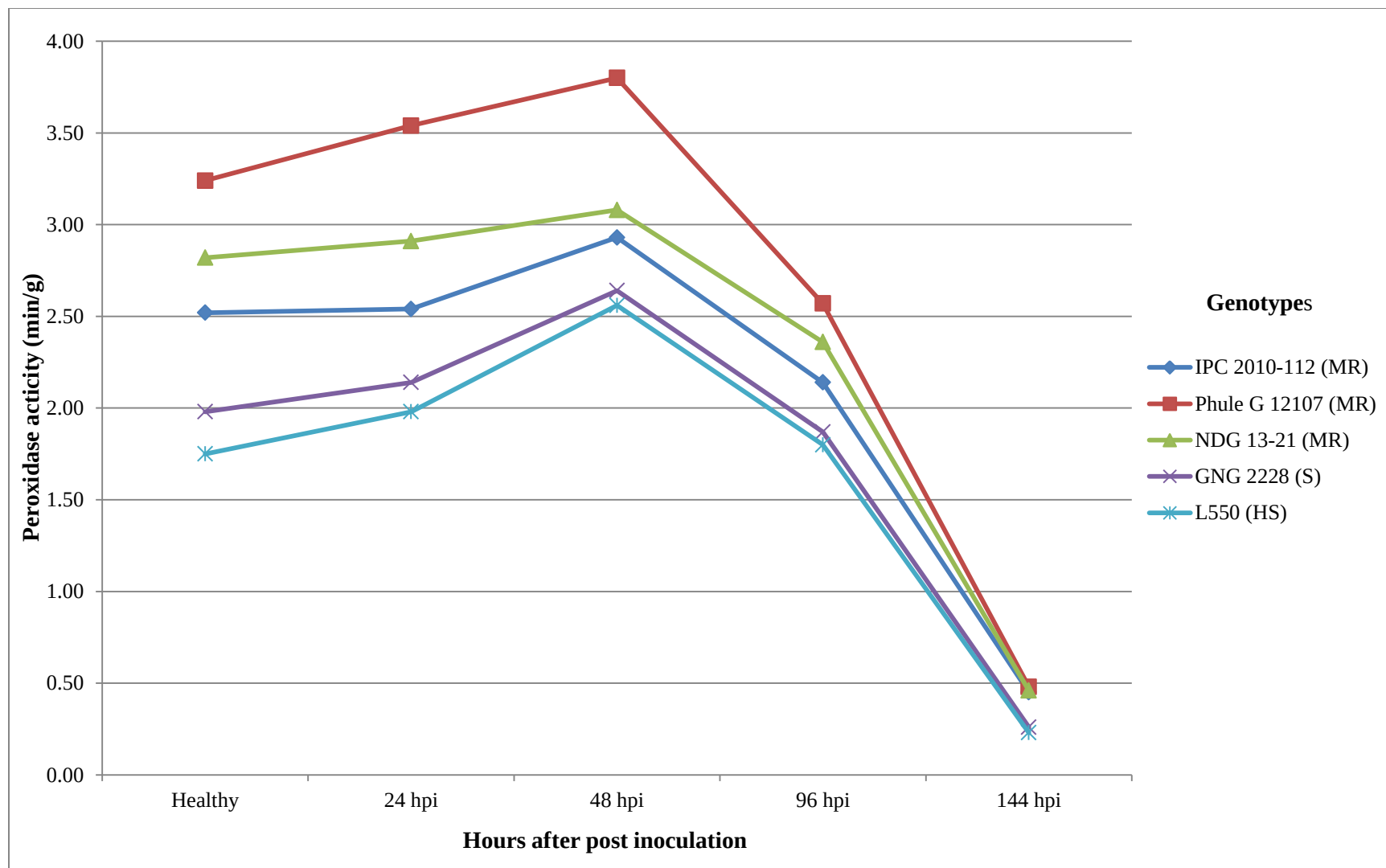


Fig. 3. Peroxidase activity in chickpea genotypes

Table 20. Catalase activity in chickpea genotypes

Catalase activity (µM/min)							
Sl. No.	Genotypes	Duration of inoculation					
		Healthy	24 hpi	48 hpi	96 hpi	144 hpi	Mean
1.	IPC 2010-112 (MR)	0.22	0.33	0.54	2.16	4.01	1.45
2.	Phule G 12107 (MR)	0.30	0.64	0.69	3.18	5.42	2.04
3.	NDG 13-21 (MR)	0.26	0.38	0.54	2.29	4.66	1.62
4.	GNG 2228 (S)	0.10	0.28	0.49	1.30	0.83	0.60
5.	L550 (HS)	0.08	0.16	0.36	1.25	0.62	0.49
S. Em ±		Genotypes	Hours		Genotypes × Hours		
		0.018	0.018		0.039		
CD at 1 %		0.067	0.067		0.149		

hpi – Hours after post inoculation

MR – Moderately resistant

S – Susceptible

HS – Highly susceptible

hydrogen peroxide formed as a by-product of metabolic processes primarily that of the electron transport pathway (Montalbini, 1991).

Similar results were obtained by Amoako (2015) where he observed that infected leaf extract had 141.02 ± 3.536 mg/mL protein, apparent Michealis constant (K_{mapp}) of 26.7 μ M and maximum rate of reaction (V_{max}) of 54.50 μ M/min compared to the uninfected leaf extract with 75.04 ± 0.560 mg/mL protein, K_{mapp} of 39.61 μ M and V_{max} of 143.06 μ M/min. The activation energy of the infected extract was 0.1578 J/mol compared to 0.2181 J/mol obtained for the uninfected extract. Activity of the crude catalase in the viral infected leaf extract was higher than that in the uninfected one.

4.3.2.3 Phenylalanine ammonia lyase (PAL) activity

The increase in the activity of PAL in all the genotypes was recorded when chickpea roots were inoculated with *M. phaseolina*. Highest enzymatic activity was recorded at 144 hpi (hours after post inoculation) in moderately resistant genotypes, Phule G 12107 (8.01 μ l/ml), NDG 13-21 (7.47 μ l/ml) and IPC 2010-112 (7.24 μ l/ml) compared to susceptible genotype GNG 2228 (1.09 μ l/ml) and highly susceptible genotype L550 (1.06 μ l/ml) which recorded highest activity at 96 hpi. PAL activity started declining after 96 hpi in susceptible and highly susceptible genotypes whereas in moderately resistant genotypes the PAL activity prolonged upto 144 hpi (Table 21 and Fig 5). The mean PAL activity was found to be more in Phule G 12107 (3.37 μ l/ml) followed by NDG 13-21 (3.07 μ l/ml), IPC 2010-112 (2.88 μ l/ml), GNG 2228 (0.67 μ l/ml) and L550 (0.52 μ l/ml). Therefore, one can conclude that the plant defense mechanism started after the pathogen attack with higher synthesis of PAL compounds and defense related enzymes.

Phenylalanine ammonia-lyase (PAL) was the main enzyme of plant phenolics production including those under stress exposure. It was the key enzyme that catalyzed core reaction in phenylpropanoid metabolism leading to functionally diverse defense related products such as lignin, suberin, wall-bound phenolics, flavanoids etc (Wen *et al.*, 2005). There was an increase in total polyphenols and activity of PAL in infected plants of banana against *Fusarium oxysporum*. Though the plant pathogen interaction had triggered the activities of defense enzymes initially but later the activities drastically declined when pathogen colonized the root tissues (Thakkar *et al.*, 2007). Biochemical response of guar against *M. phaseolina* infection revealed that there was maximum accumulation of phenolic acids in infected plants *i.e.*, 23 per cent higher than in control after 120 hours of infection in all the four cultivars. Similarly, PAL activity was also significantly increased by 37 per cent after 96 or 120 hours of infection depending upon the cultivar in comparison to control. Phenolic acid accumulation and enhanced PAL activity in the compatible host-pathogen combination presumes that both participated actively in the guar resistance to root rot (Sharma *et al.*, 2011).

Similarly biochemical changes in banded leaf and sheath blight affected maize plants caused by *Rhizoctonia solani* f. sp. *sasakii* recorded increased phenylalanine ammonia lyase (PAL) activities in leaf sheaths when inoculated with the pathogen (Sivakumar and Sharma, 2003). Fungal elicitors 34 are known to induce the production

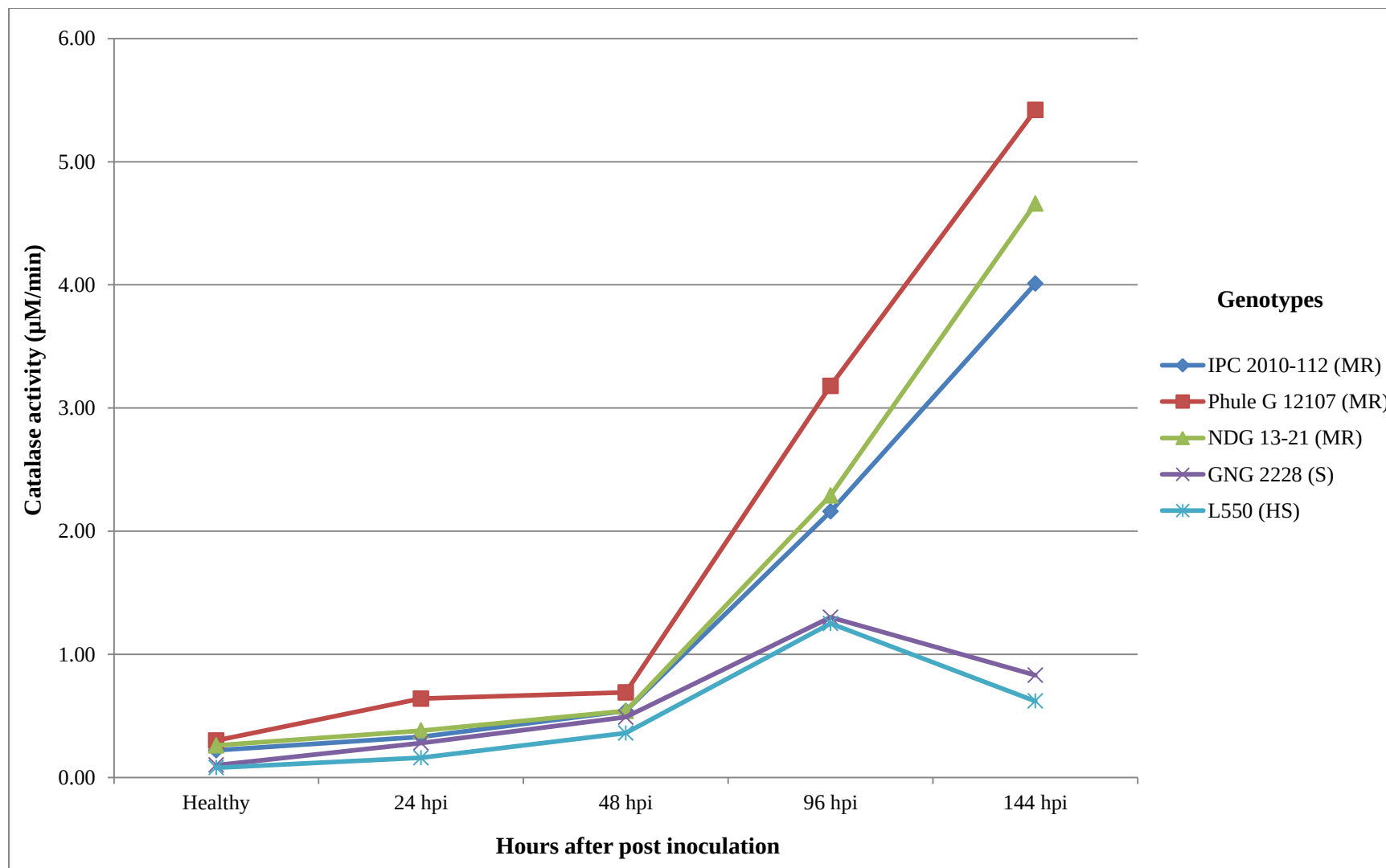


Fig. 4. Catalase activity in chickpea genotypes

Table 21. Phenylalanine ammonia lyase activity in chickpea genotypes

Phenylalanine ammonia lyase activity (µl/ml)							
Sl. No.	Genotypes	Duration of inoculation					
		Healthy	24 hpi	48 hpi	96 hpi	144 hpi	Mean
1.	IPC 2010-112 (MR)	0.39	0.64	1.14	5.01	7.24	2.88
2.	Phule G 12107 (MR)	0.66	0.84	1.36	6.00	8.01	3.37
3.	NDG 13-21 (MR)	0.41	0.64	1.23	5.63	7.47	3.07
4.	GNG 2228 (S)	0.08	0.63	0.92	1.09	0.64	0.67
5.	L550 (HS)	0.02	0.57	0.63	1.06	0.32	0.52
S. Em ±		Genotypes	Hours		Genotypes × Hours		
		0.016	0.016		0.037		
CD at 1 %		0.062	0.062		0.138		

hpi – Hours after post inoculation

MR – Moderately resistant

S – Susceptible

HS – Highly susceptible

of phenyl ammonia lyase (PAL) and peroxidase which were involved in the synthesis and depolymerization of lignin precursors. The rapid increase and higher levels of PAL and peroxidases activity was found in resistant cultivars as compared to the susceptible cultivars of chickpea against *Fusarium* wilt (Aguilar *et al.*, 2000).

4.3.2.4 Total phenol activity

The increase in the activity of phenol was recorded when chickpea roots were inoculated with *M. phaseolina*. Highest phenol activity was recorded at 96 hpi (hours after post inoculation) in moderately resistant genotypes, Phule G 12107 (1.43 µl/ml), NDG 13-21 (1.35 µl/ml) and IPC 2010-112 (1.03 µl/ml) compared to susceptible genotype GNG 2228 (0.86 µl/ml) and highly susceptible genotype L550 (0.74 µl/ml) which recorded highest activity at 48 hpi. Phenol activity started declining after 48 hpi in susceptible and highly susceptible genotypes whereas in moderately resistant genotypes the phenol activity prolonged upto 96 hpi (Table 22 and Fig 6). The mean phenol activity was found to be more in Phule G 12107 (0.97 µl/ml) followed by NDG 13-21 (0.93 µl/ml), IPC 2010-112 (0.83 µl/ml), GNG 2228 (0.67 µl/ml) and L550 (0.54 µl/ml).

Phenolic compounds were found to be most important group of secondary plant products that played an important role in reducing the susceptibility of a plant to pathogen (Jyosthna *et al.*, 2004). During pathogen infection, deposition of phenolics into the cell wall was an important defense mechanism, either because of a hypersensitive reaction (HR) of entire cells or due to local wall reinforcement (Conceicao *et al.*, 2006). A distinct correlation between the degree of plant resistance and phenolics was reported in plant tissues. phenolic compounds such as cinnamic and ferulic acids released into plant cell walls was a common and early response to fungal attack, which resulted in cell strengthening and then enhanced resistance to pathogen penetration (Stadnik and Buchenauer, 2000). This showed that when plant cells were recruited into infection, it switches from normal primary metabolism to a multitude of secondary metabolism defense pathway and activation of novel defense enzymes and genes takes place (Tan *et al.*, 2004).

The results obtained are in conformity with Shreenivas (2000) reported that resistant varieties (ICPL 8863 and ICPL 87119) showed higher levels of phenolics contents for *Fusarium* in both the control and inoculated pigeonpea seedlings over the susceptible (TTB 7 and AKT 9221) at all the growth stage. Similarly Mandavia *et al.* (2002) collected root exudates from ten day old seedlings and were analyzed for total phenol content. A significant inverse relationship was found between wilt susceptibility and total phenol content in root exudates of chickpea seedlings. Resistant genotypes (0-30 % wilt) had the highest amount of phenol, followed by moderately resistant (31-70 % wilt) and susceptible (71-100 % wilt) genotypes.

Manjunatha (2012) reported that total phenol content of healthy leaves and resistant/moderately susceptible varieties, ICP 7035 (4.416 mg) and BRG 1 (2.981 mg), exhibited highly increase in phenolic content than susceptible variety maruthi (2.241 mg) and TTB 7 (2.251 mg). However phenol content was increased upon inoculation with

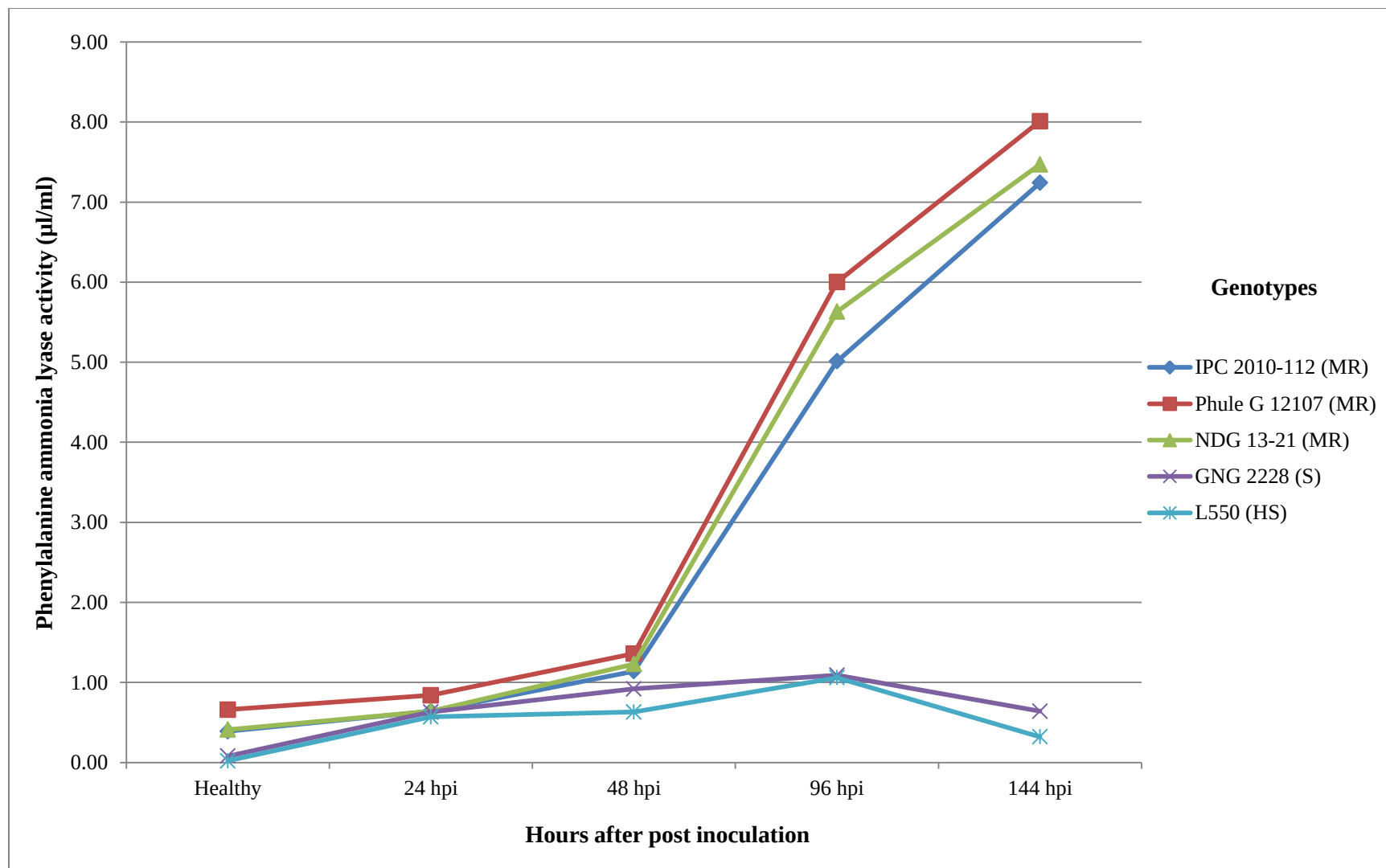


Fig. 5. Phenylalanine ammonia lyase activity in chickpea genotypes

Table 22. Total phenol activity in chickpea genotypes

Total phenol activity(µl/ml)							
Sl. No.	Genotypes	Duration of inoculation					
		Healthy	24 hpi	48 hpi	96 hpi	144 hpi	Mean
1.	IPC 2010-112 (MR)	0.63	0.71	0.93	1.03	0.86	0.83
2.	Phule G 12107 (MR)	0.69	0.79	1.00	1.43	0.94	0.97
3.	NDG 13-21 (MR)	0.68	0.76	0.95	1.35	0.92	0.93
4.	GNG 2228 (S)	0.63	0.84	0.86	0.68	0.32	0.67
5.	L550 (HS)	0.59	0.63	0.74	0.52	0.20	0.54
S. Em ±		Genotypes	Hours		Genotypes × Hours		
		0.018	0.018		0.040		
CD at 1 %		0.067	0.067		0.150		

hpi – Hours after post inoculation

MR – Moderately resistant

S – Susceptible

HS – Highly susceptible

pigeonpea sterility mosaic virus upto 28 days in resistant genotypes, whereas it started declining after 21 days in susceptible genotypes.

Variety ICP 8863 recorded total phenol content of 0.49, 1.03, 2.36 and 3.45 mg/g of fresh roots at 30, 60, 90 and 120 days respectively upon inoculation with pigeon pea sterility mosaic virus. WRP 1 had total phenol content of 0.32, 0.79, 1.56 and 2.02 mg/g of fresh roots at 30, 60, 90 and 120 days respectively. Similarly, TTB 7 registered total phenol content of 0.20, 0.57, 1.02 and 1.98 mg/g of fresh roots at 30, 60, 90 and 120 days respectively. ICP 8863 a wilt resistant variety had highest total phenol content compared to WRP 1, whereas, TTB 7 wilt susceptible variety had lowest total phenol content. As the age of the crop progressed, consequently there was increase in the total phenol content (Asha, 2012). In the present study also phenol content increased after infection and it was high in resistant genotypes than susceptible genotypes.

4.3.2.5 Total free amino acid activity

There was an increase in the activity of total free amino acids in all the genotypes when chickpea roots were inoculated with *M. phaseolina*. Highest amino acid activity was recorded at 48 hours after inoculation (hpi) and continued to be constant upto 96 hpi and started declining after 96 hpi. At 48 hpi enzymatic activity was highest in resistant genotypes Phule G 12107 (2.45 mg/ml), NDG 13-21 (2.26 mg/ml) and IPC 2010-112 (2.18 mg/ml) compared to susceptible genotype GNG 2228 (1.17 mg/ml) and highly susceptible genotype L550 (1.09 mg/ml). The mean amino acid activity was found to be more in Phule G 12107 (2.06 mg/ml), followed by NDG 13-21 (1.88 mg/ml), IPC 2010-112 (1.79 mg/ml), GNG 2228 (0.93 mg/ml) and L550 (0.81 mg/ml) data presented in Table 23 and Fig 7.

Involvement of amino acids in the relationship between host and parasite was established (Omokolo *et al.*, 2002; Omokolo and Boudjeko, 2005), indicating an accumulation during the development of infections. Amino acids might act directly to inhibit fungal development, or indirectly by their implication in the metabolic ways associated with resistance to diseases (Graham *et al.*, 1990). After infection of resistant genotypes of some host species, accumulation of certain specific amino acids such as glutamine, histidine, glycine and arginine were observed in tomatoes (Hassan *et al.*, 1994; Starrat and Lazarovits, 1996), tyrosine and alanine in wheat (Tyuterev and Tarlakovskii, 1994) and asparagin, glutamic acid, proline, glycine and arginine in citrus (Nemec, 1995). Free amino acids were important indicators of the plant conditions, arising as a consequence of protein degradation in tissues under programmed cell death (Scarpari, *et al.* 2005). The increase in amino acids under infection conditions may also indicate pathogen effects on molecular transportation via phloem tissues. Indeed, inhibition of metabolite translocation due to the presence of pathogen in infected plants was described by Guthrie *et al.* (2001) in papaya and by Maust *et al.* (2003) in coconut.

These results are in accordance with Gupta and Khare (1992) who carried out biochemical analysis of chickpea resistant and susceptible cultivars to vascular wilt, caused by *F. oxysporum* f. sp. *ciceris*. They established that the quantities of total polysaccharides, total carbohydrates, total amino acids, total phenols and mineral

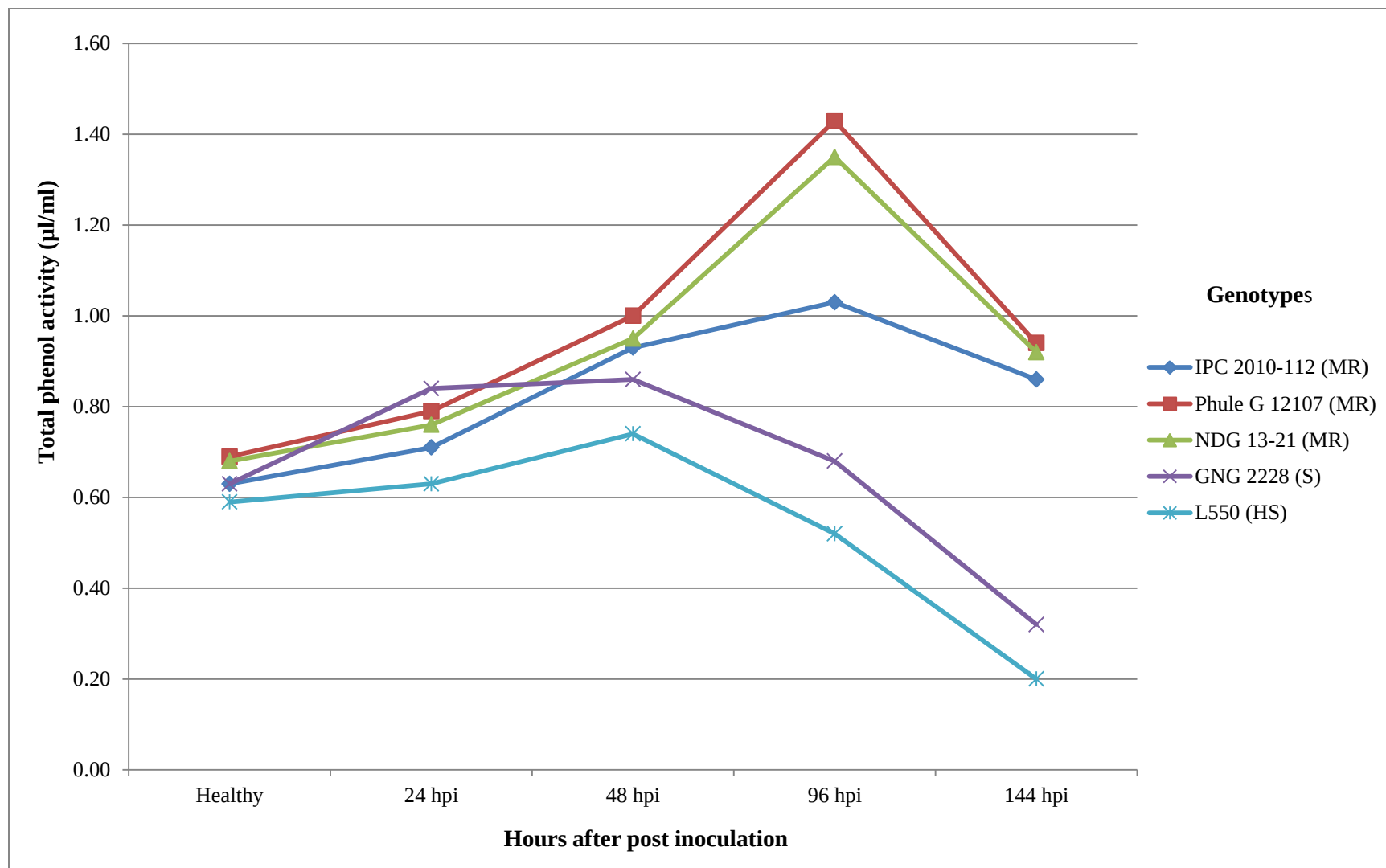


Fig. 6. Total phenol activity in chickpea genotypes

Table 23. Total free amino acids content in chickpea genotypes

Total free amino acids content (mg/ml)							
Sl. No.	Genotypes	Duration of inoculation					
		Healthy	24 hpi	48 hpi	96 hpi	144 hpi	Mean
1.	IPC 2010-112 (MR)	1.15	1.69	2.18	2.18	1.75	1.79
2.	Phule G 12107 (MR)	1.40	2.00	2.45	2.45	2.01	2.06
3.	NDG 13-21 (MR)	1.20	1.80	2.26	2.26	1.86	1.88
4.	GNG 2228 (S)	0.40	0.92	1.17	1.17	1.01	0.93
5.	L550 (HS)	0.32	0.73	1.09	1.08	0.83	0.81
S. Em $\pm$		Genotypes	Hours		Genotypes $\times$ Hours		
		0.007	0.007		0.016		
CD at 1 %		0.027	0.027		0.061		

hpi – Hours after post inoculation

MR – Moderately resistant

S – Susceptible

HS – Highly susceptible

contents were greater in the roots of resistant cultivars. Similarly amino acids content increased in 70 per cent of genotypes after injury or infection by *Phytophthora megakarya* in cocoa. A significant positive relationship was observed between amino acid contents and the severity of necrosis (Pierr *et al.*, 2011). In the present study also amino acid activity was recorded high in resistant genotypes compare to susceptible genotypes and the activity increased after infection.

4.3.2.6 Total sugar content

There was a decrease in the activity of total sugar content in all the genotypes when chickpea roots were inoculated with *M. phaseolina*. The mean total sugar content was found to be more in resistant genotypes Phule G 12107 (3.37 mg/ml), NDG 13-21 (3.04 mg/ml) and IPC 2010-112 (2.50 mg/ml) compared to susceptible genotype GNG 2228 (1.74 mg/ml) and highly susceptible genotype L550 (1.64 mg/ml) data presented in Table 24 and Fig 8.

4.3.2.7 Reducing sugar content

The decrease in the activity of reducing sugar content was observed in all the genotypes when chickpea roots were inoculated with *M. phaseolina*. The mean reducing sugar content was found to be more in resistant genotypes Phule G 12107 (1.71 mg/ml), NDG 13-21 (1.57 mg/ml) and IPC 2010-112 (1.46 mg/ml) compared to susceptible genotype GNG 2228 (0.54 mg/ml) and highly susceptible genotype L550 (0.47 mg/ml) data presented in Table 25 and Fig 9.

4.3.2.8 Non-Reducing sugar content

The mean non-reducing sugar content was found to be more in resistant genotypes Phule G 12107 (1.66 mg/ml), NDG 13-21 (1.48 mg/ml) and IPC 2010-112 (1.04 mg/ml) compared to susceptible genotype GNG 2228 (1.21 mg/ml) and highly susceptible genotype L550 (1.17 mg/ml) data presented in Table 26 and Fig 10.

A similar trend of decreased sugar levels in diseased plants was observed by Prasad *et al.* (1976). The depletion of sugars during host-parasite interaction might be due to increased respiration or utilization of sugars by the fungi which depends on the capability of fungi to secrete carbohydrate degrading enzyme. Similarly Nema (1989) suggested that reduction in sugars during disease development might be due to utilization of sugars probably for energy and synthetic reactions involved in multiplication of the pathogen.

These results are in agreement with Kaur and Dhillon (1989) they also noticed susceptible cultivars of groundnut showed a rapid decrease of sugars but the decrease was relatively slow in resistant cultivars. Similarly, Sindhan *et al.* (1999) screened two hundred and sixty genotypes of mungbean against *Cercospora* leaf spot reported that total sugar, reducing sugar and non reducing sugar content was decreased in diseased leaves of susceptible varieties than resistant varieties. The *Fusarium* infected plants resistant varieties ICP 8863 had highest total sugar content compared to BDN 2010 (moderately resistant), whereas, TTB 7 (susceptible) has lowest total sugar content

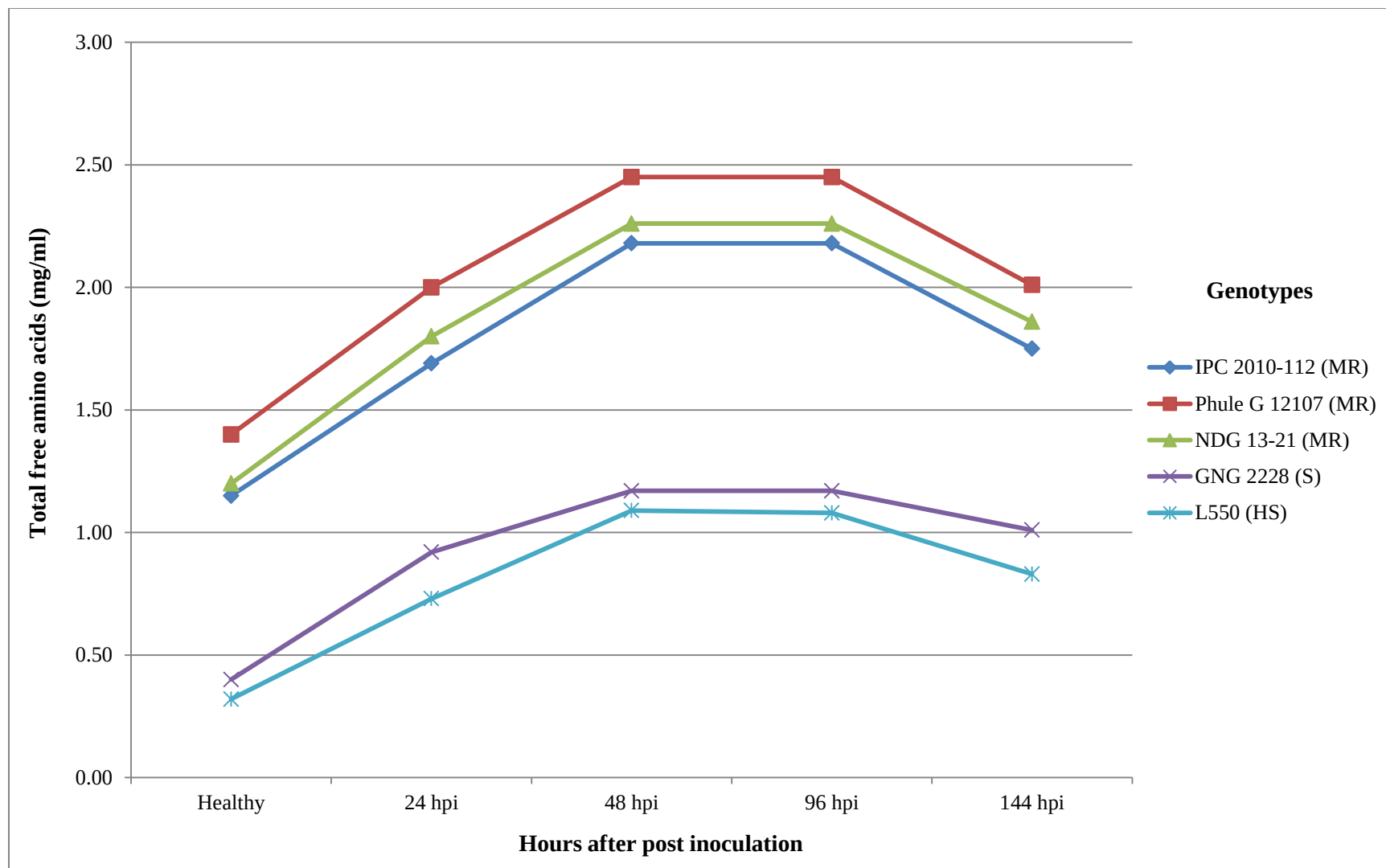


Fig. 7. Total free amino acid content in chickpea genotypes

Table 24. Total sugar content in chickpea genotypes

Total sugar content (mg/ml)							
Sl. No.	Genotypes	Duration of inoculation					
		Healthy	24 hpi	48 hpi	96 hpi	144 hpi	Mean
1.	IPC 2010-112 (MR)	5.09	3.12	2.55	0.99	0.73	2.50
2.	Phule G 12107 (MR)	5.96	4.48	3.64	1.95	0.82	3.37
3.	NDG 13-21 (MR)	5.63	4.21	2.71	1.89	0.78	3.04
4.	GNG 2228 (S)	3.05	2.37	1.92	0.90	0.48	1.74
5.	L550 (HS)	3.01	2.01	1.89	0.82	0.46	1.64
S. Em ±		Genotypes		Hours		Genotypes × Hours	
		0.079		0.079		0.177	
CD at 1 %		0.299		0.299		0.669	

hpi – Hours after post inoculation

MR – Moderately resistant

S – Susceptible

HS – Highly susceptible

Table 25. Reducing sugar content in chickpea genotypes

Reducing sugar (mg/ml)							
Sl. No.	Genotypes	Duration of inoculation					
		Healthy	24 hpi	48 hpi	96 hpi	144 hpi	Mean
1.	IPC 2010-112 (MR)	2.55	2.1	1.53	0.81	0.29	1.46
2.	Phule G 12107 (MR)	2.70	2.52	1.87	0.94	0.51	1.71
3.	NDG 13-21 (MR)	2.56	2.28	1.67	0.9	0.42	1.57
4.	GNG 2228 (S)	1.21	0.99	0.3	0.10	0.08	0.54
5.	L550 (HS)	1.05	0.81	0.22	0.19	0.06	0.47
S. Em ±		Genotypes	Hours		Genotypes × Hours		
		0.005	0.005		0.010		
CD at 1 %		0.017	0.017		0.039		

hpi – Hours after post inoculation

MR – Moderately resistant

S – Susceptible

HS – Highly susceptible

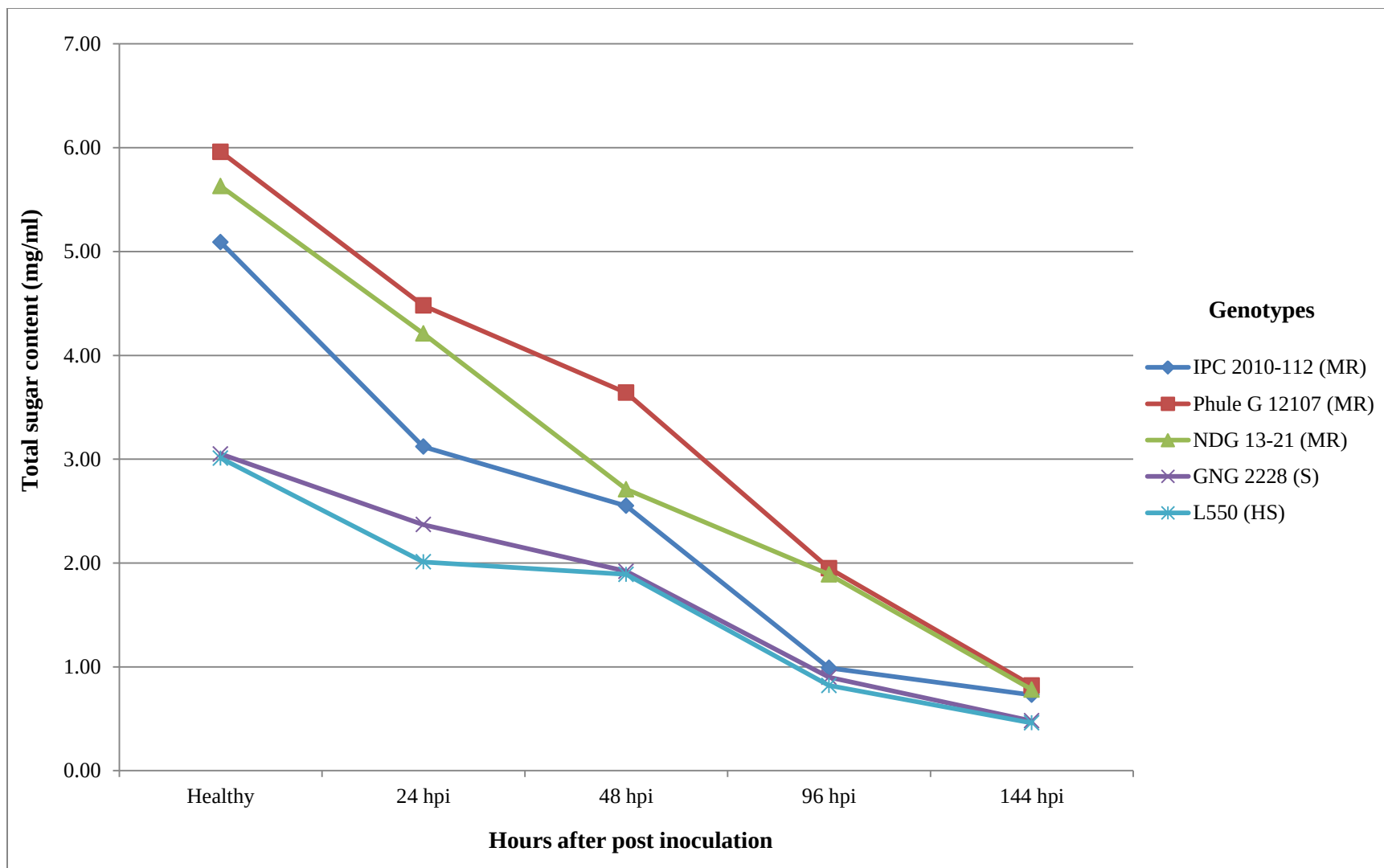


Fig. 8. Total sugar content in chickpea genotypes

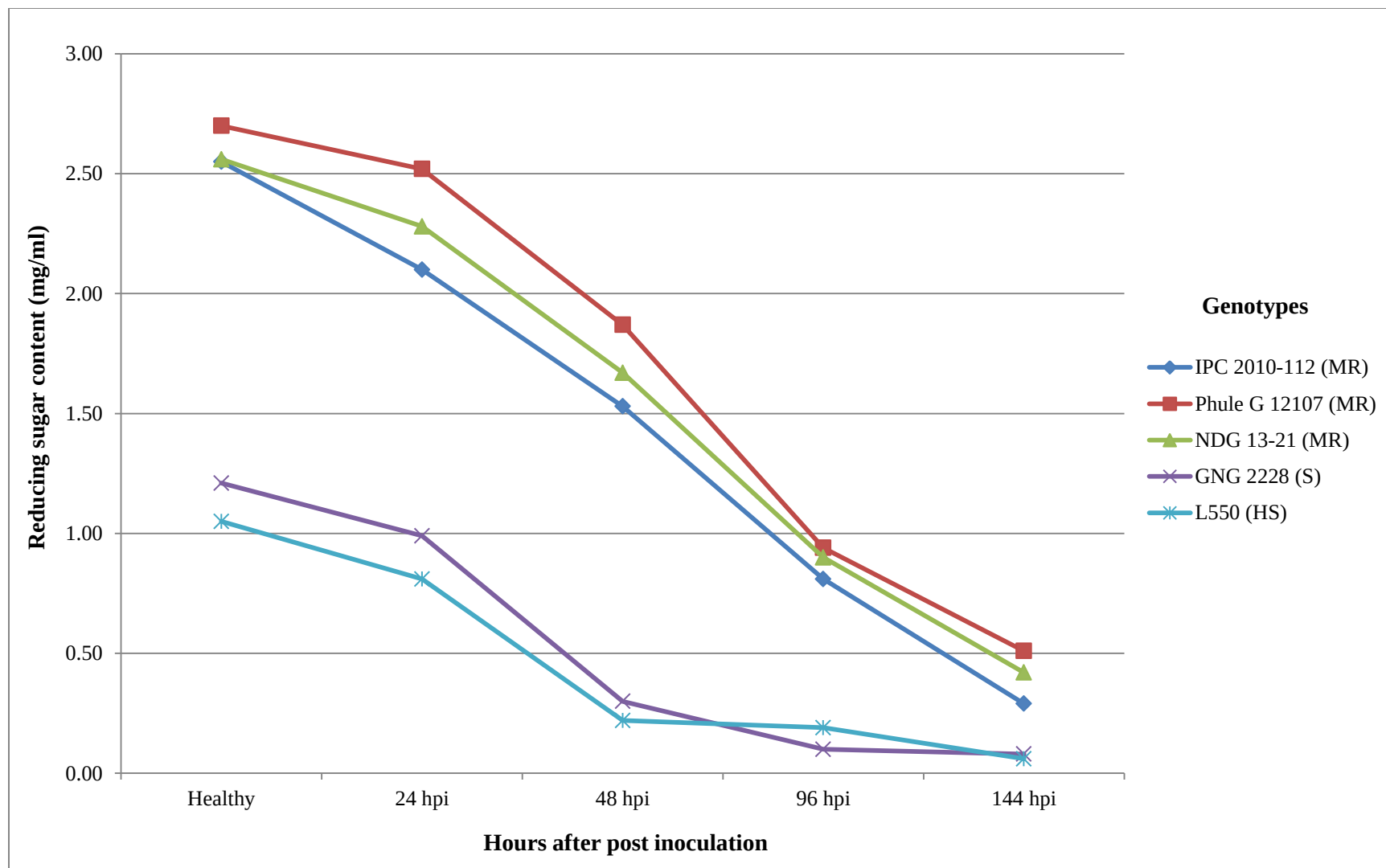


Fig. 9. Reducing sugar content in chickpea genotypes

Table 26. Non-reducing sugar content in chickpea genotypes

Sl. No.	Genotypes	Total sugars (mg/ml)	Reducing sugars (mg/ml)	Non-reducing sugars (mg/ml)
1.	IPC 2010-112 (MR)	2.50	1.46	1.04
2.	Phule G 12107 (MR)	3.37	1.71	1.66
3.	NDG 13-21 (MR)	3.04	1.57	1.48
4.	GNG 2228 (S)	1.74	0.54	1.21
5.	L550 (HS)	1.64	0.47	1.17

MR – Moderately resistant

S – Susceptible

HS – Highly susceptible

(Prasad, 2011). The present study also correlates with decrease in total sugars, reducing sugars and non-reducing sugars after infection with *M. phaseolina*. The sugar content was found to be more in resistant genotypes compare to susceptible genotypes.

4.4 Management of *Macrophomina phaseolina* causing dry root rot in chickpea

4.4.1 *In-vitro* evaluation of bioagents against *M. phaseolina* by dual culture technique

Efficacy of fungal and bacterial bio control agents was studied in *in-vitro* condition by following dual culture method against *M. phaseolina* isolate of chickpea and the results are presented here under.

The results of the study indicated that all the antagonists significantly inhibited mycelial growth of *M. phaseolina* and per cent inhibition ranged from 62.96 to 81.44. Among fungal and bacterial antagonists, the maximum inhibition of mycelial growth was observed in *T. harzianum* (Th-55) isolate with 81.44 per cent followed by *B. subtilis* (75.56 %), *T. viride* (Tv-27) with 74.44 per cent, *P. fluorescens* (70.37 %), *T. harzianum* (IIHR) with 70.04 per cent, *T. viride* (B) with 64.44 per cent and minimum inhibition was observed in *T. harzianum* (GKVK) with 62.96 per cent (Table 27, Fig 11 and Plate 11). In the present study also *Trichoderma* was found effective in inhibition of mycelial growth of the fungus.

The suggested mechanisms for biocontrol of plant pathogens by *Trichoderma* were antibiosis, lysis, competition and mycoparasitism (Cook and Baker, 1983). *T. harzianum* and *T. viride* both suppressed the growth of *M. phaseolina* and this may be due to coiling and disintegration of hyphae of the test fungus resulting in the loss of competitive saprophytic ability. The results of the present study was supported by the previous reports Manjunatha (2009), Ammajamma *et al.* (2009), Indra *et al.* (2003) and Sachidananda (2005). Similarly Kaswate *et al.* (2003) evaluated the antagonism of *T. viride*, *P. fluorescens* and *B. subtilis* isolates against *R. bataticola* (*M. phaseolina*) collected from pigeonpea, chickpea, green gram, cluster bean, field pea, cotton, okra and safflower in *in-vitro*. *T. viride* was most effective in the inhibition of the various isolates of *R. bataticola* (100 %), followed by *B. subtilis* (87.41 - 92.89 %) and *P. fluorescens* (73.98 - 78.94 %).

4.4.2 *In-vitro* evaluation of fungicides by poison food technique

4.4.2.1 *In-vitro* evaluation of systemic fungicides against *M. phaseolina*

Five systemic fungicides viz., thiophanate methyl, tebuconazole, carbendazim, difenoconazole and propiconazole were evaluated in *in-vitro* condition against *M. phaseolina* at different concentrations (50, 100, 250 and 500 ppm) as described in material and methods. The data on the per cent inhibition of radial growth of the fungus is presented here under.

Among five systemic fungicides evaluated, tebuconazole, carbendazim, difenoconazole and propiconazole inhibited cent per cent mycelial growth at all concentrations. Whereas, the least inhibition of 53.33, 89.26 and 93.33 per cent was

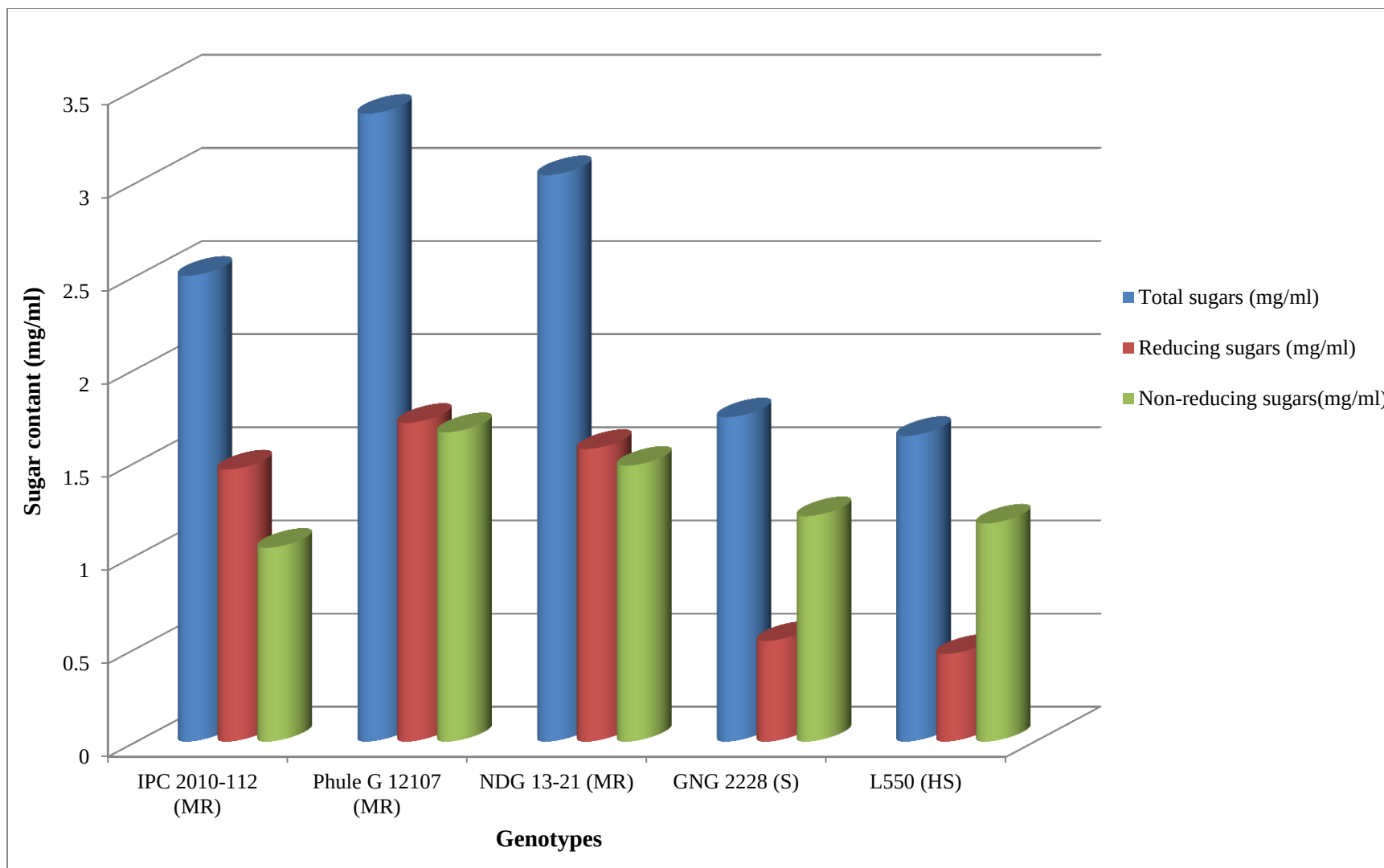


Fig. 10. Total sugar, reducing sugar and non-reducing sugar content in chickpea genotypes

Table 27. *In-vitro* evaluation of bioagents against *M. phaseolina* by dual culture technique

Sl. No.	Bio agents	Per cent inhibition over control*
1.	<i>Trichoderma viride</i> (B)	64.44 (53.40)
2.	<i>Trichoderma viride</i> (Tv-27)	74.44 (59.63)
3.	<i>Trichoderma harzianum</i> (GKVK)	62.96 (52.51)
4.	<i>Trichoderma harzianum</i> (IIHR)	70.04 (56.81)
5.	<i>Trichoderma harzianum</i> (Th-55)	81.44 (64.48)
6.	<i>Pseudomonas fluorescens</i>	70.37 (57.02)
7.	<i>Bacillus subtilis</i>	75.56 (60.37)
8.	Control	0.00
S. Em $\pm$		0.15
CD at 1%		0.63

* Figures in parentheses are arc sin angular transformed values

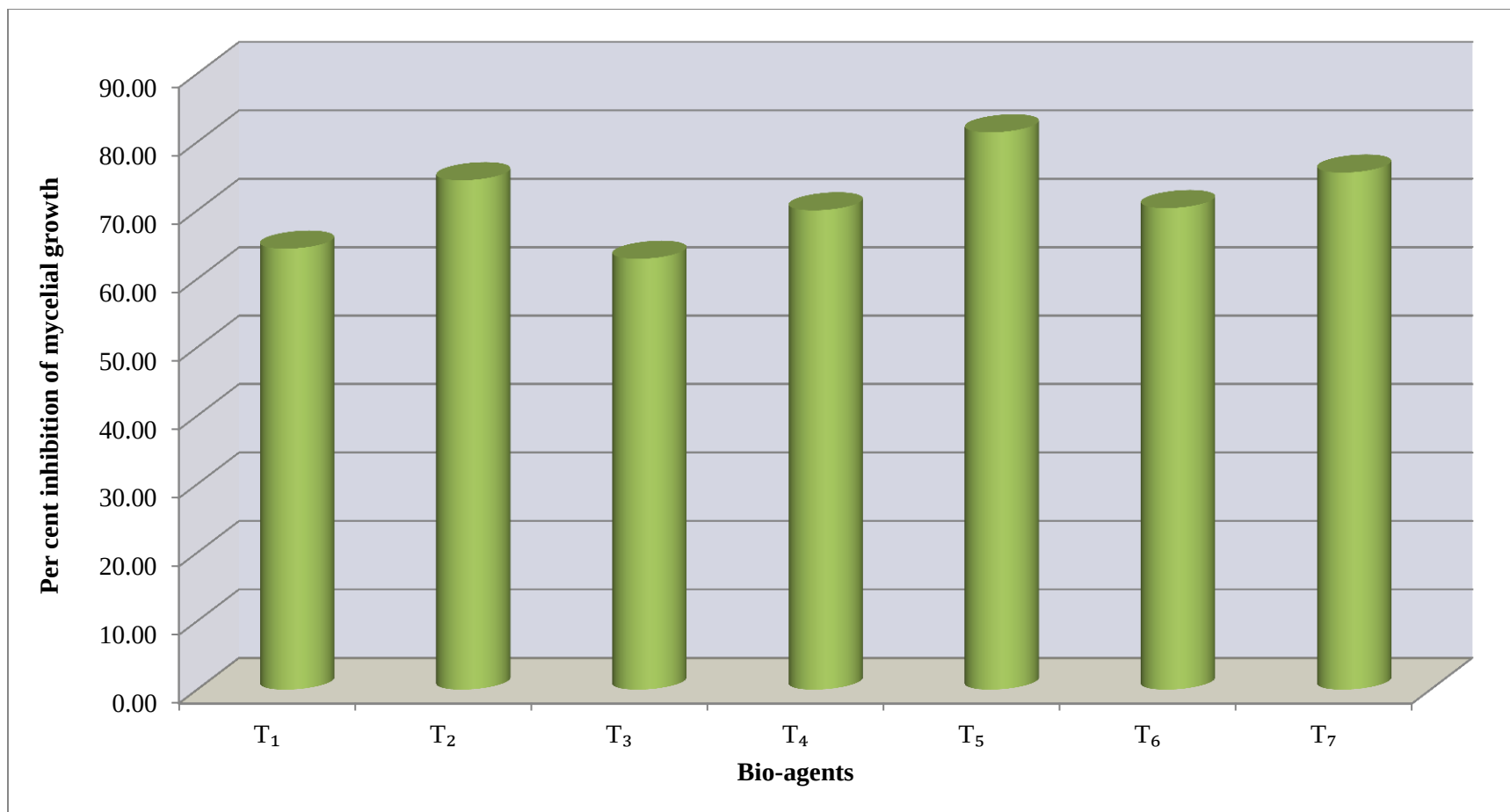


Fig. 11. In-vitro evaluation of bioagents against *M. phaseolina* by dual culture technique

T₁: *Trichoderma viride* (B)
 T₂: *Trichoderma viride* (Tv-27)
 T₃: *Trichoderma harzianum*

T₄: *Trichoderma harzianum* (IIHR)
 T₅: *Trichoderma harzianum* (Th-55)
 T₆: *Pseudomonas fluorescens*

T₇: *Bacillus subtilis*

observed in case of thiophanate methyl at 50, 100 and 250 ppm concentration respectively. However, cent per cent inhibition was observed at 500 ppm concentration (Table 28, Fig 12 and Plate 12).

4.4.2.2 In-vitro evaluation of contact and combi - products fungicides against *M. phaseolina*

Among the contact fungicides viz., mancozeb, thiram, chlorothalonil and captan evaluated against *M. phaseolina*, mancozeb recorded cent per cent inhibition of mycelial growth at all the concentrations (250, 500, 750 and 1000 ppm), followed by thiram which recorded the inhibition of mycelial growth of 65.56, 78.52, 85.19 and 91.11 per cent at 250, 500, 750 and 1000 ppm concentration respectively, followed by captan 48.15, 68.89, 77.41 and 84.44 per cent at 250, 500, 750 and 1000 ppm concentration respectively. However, chlorothalonil recorded the least inhibition of 46.30, 54.81, 68.15, 79.63 per cent mycelial growth at 250, 500, 750 and 1000 ppm concentration respectively (Plate 13).

Among the combi - products viz., tricyclozole 4 % + mancozeb 62 % WP, hexaconazole 4 % + zineb 68 % WP, carboxin 37.5 % + thiram 37.5 % WP and carbendazim 12 % + mancozeb 63 % WP evaluated against the *M. phaseolina*, carboxin 37.5 % + thiram 37.5 % WP and carbendazim 12 % + mancozeb 63 % WP showed cent per cent inhibition of mycelia growth at all concentrations (250, 500, 750 and 1000 ppm) respectively followed by tricyclozole 4 % + mancozeb 62 % WP with 79.63, 89.26, 100.00 and 100.00 per cent at 250, 500, 750 and 1000 ppm concentration respectively. The least inhibition was observed in hexaconazole 4 % + zineb 68 % WP with 62.22, 75.93, 82.59 and 97.41 per cent at 250, 500, 750 and 1000 ppm concentration respectively (Table 29, Fig. 13 and Plate 14).

Similarly carbendazim being benzimidazole group of fungicide, it interferes with energy production and cell wall synthesis of fungi (Nene and Thapliyal, 1973). Further, they also observed the effectiveness of triazoles, which inhibited sterol biosynthetic pathway in fungi. According to Davidse (1986) carbendazim induced nuclear instability by disturbing the mitosis and meiosis. Similar results were obtained by Waterfield and Sisler (1985), Virupaksha Prabhu (2009), Prashanthi *et al.* (2000), Prajapati *et al.* (2002) Lokesh (2003), Konde *et al.* (2008) and Salunke *et al.* (2008).

4.4.3 Management of dry root rot under greenhouse condition

4.4.3.1 Per cent seed germination

Among the different treatments, the seed germination was maximum in seed treatment with tebuconazole at 2 g/kg of seed (98.01 %) which was on par with seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed (98.00 %) followed by seed treatment with carbendazim 2 g + captan 1 g/kg of seed (97.03 %) which was on par with seed treatment with carbendazim 2 g + thiram 1 g/kg of seed (97.05 %) over the control which recorded least per cent seed germination (90.05 %) presented in Table 30 and Plate 15.

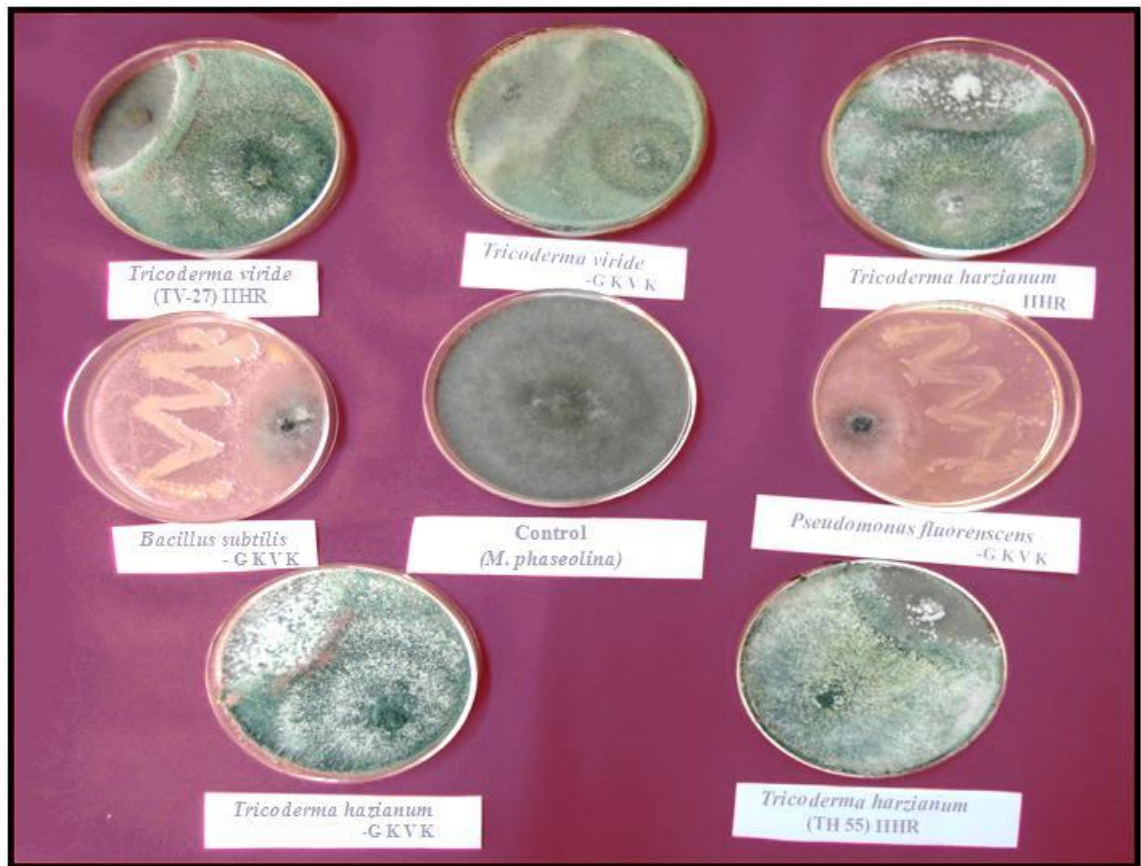


Plate 11. In-vitro evaluation of bioagents against *M. phaseolina*

Table 28. *In-vitro* evaluation of systemic fungicides against *M. phaseolina*

Sl. No.	Fungicides	Per cent inhibition of mycelial growth*				
		Concentrations				
		50 ppm	100 ppm	250 ppm	500 ppm	Mean
1.	Thiophanate methyl	53.33 (46.91)	89.26 (70.87)	93.33 (75.03)	100.00 (90.00)	83.98 (66.41)
2.	Tebuconazole	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
3.	Carbendazim	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
4.	Difenoconazole	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
5.	Propiconazole	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
S. Em $\pm$		Fungicides	Concentration		Fungicide $\times$ concentration	
		0.06	0.05		0.11	
CD at 1 %		0.22	0.19		0.43	

* Figures in parentheses are arc sin angular transformed values

Table 29. *In-vitro* evaluation of contact and combi- products fungicides against *M. phaseolina*

Sl. No.	Fungicides	Per cent inhibition of mycelial growth*				
		Concentrations				
		250 ppm	500 ppm	750 ppm	1000 ppm	Mean
Contact fungicides						
1.	Mancozeb	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
2.	Thiram	65.56 (54.07)	78.52 (62.39)	85.19 (67.37)	91.11 (72.65)	80.10 (63.50)
3.	Chlorothalonil	46.30 (42.88)	54.81 (47.76)	68.15 (55.64)	79.63 (63.17)	62.22 (52.07)
4.	Captan	48.15 (43.94)	68.89 (56.10)	77.41 (61.62)	84.44 (66.77)	69.72 (56.62)
Combi-products						
5.	Tricyclazole 4 %+ Mancozeb 62 % WP	79.63 (63.17)	89.26 (70.87)	100.00 (90.00)	100.00 (90.00)	92.22 (73.81)
6.	Hexaconazole 4 % + Zineb 68 % WP	62.22 (52.07)	75.93 (60.62)	82.59 (65.34)	97.41 (80.74)	79.54 (63.11)
7.	Carboxin 37.5 %+ Thiram 37.5 % WP	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
8.	Carbendazim 12 % + Mancozeb 63 % WP	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)	100.00 (90.00)
S. Em ±		Fungicides	Concentration		Fungicides × concentration	
		0.11	0.08		0.22	
CD at 1 %		0.41	0.29		0.82	

* Figures in parentheses are arc sin angular transformed

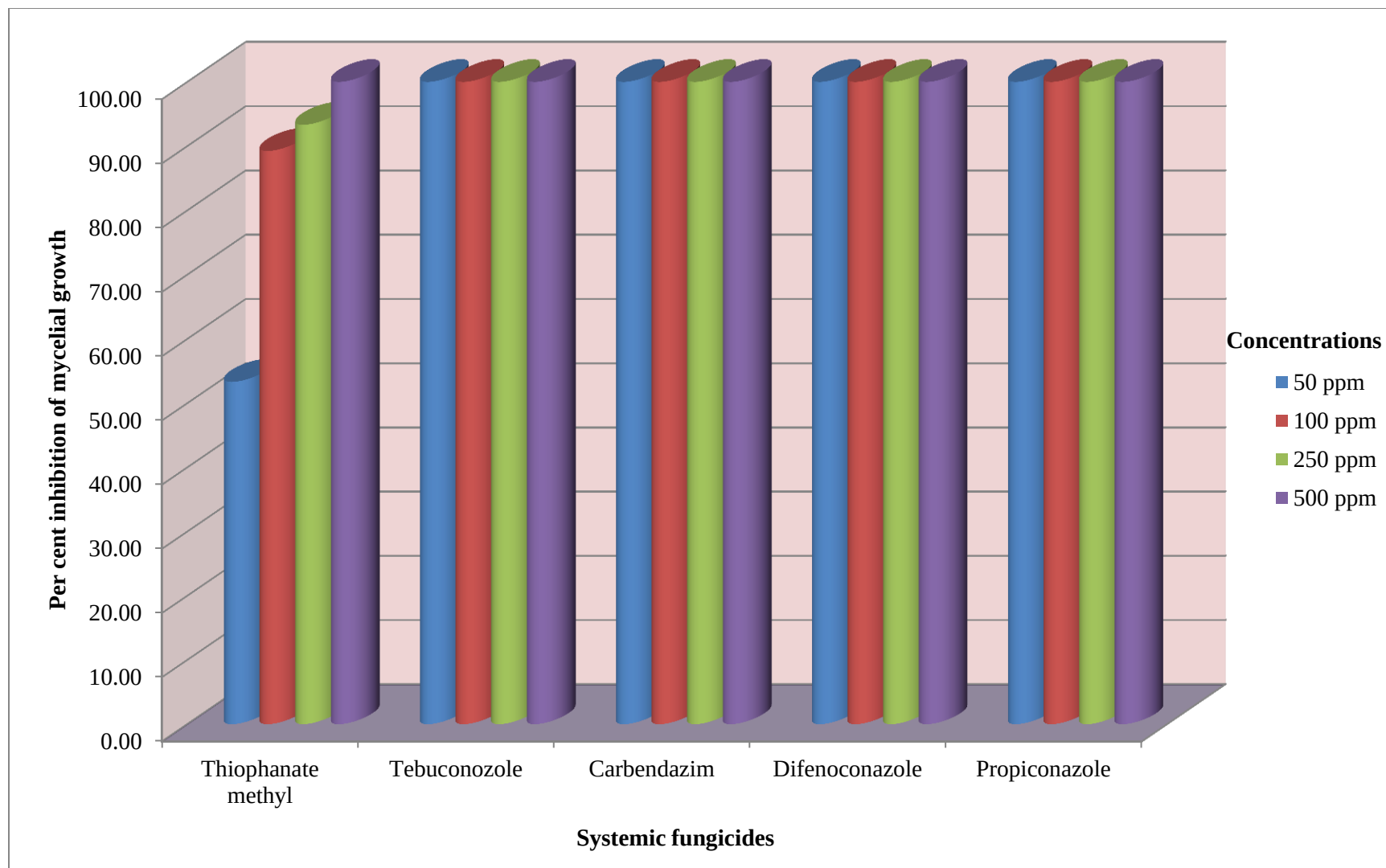


Fig. 12. *In-vitro* evaluation of systemic fungicides against *M. phaseolina*

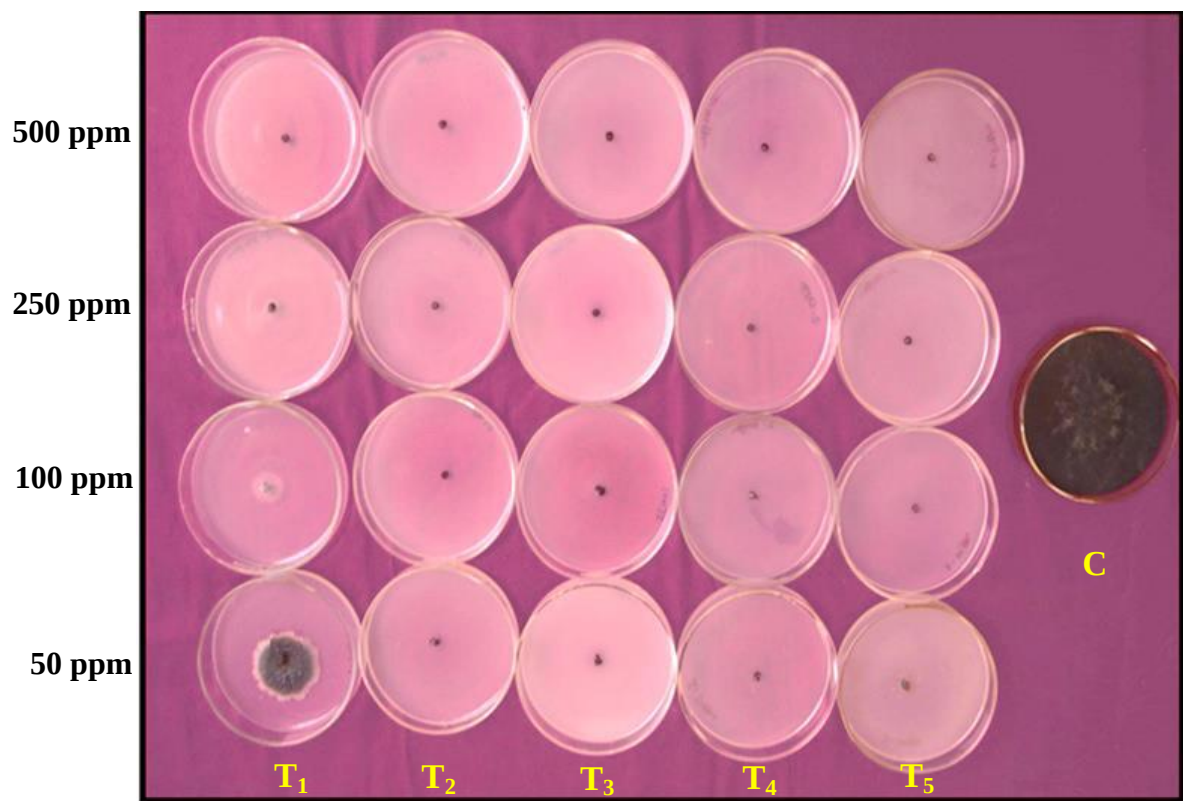


Plate 12. *In-vitro* evaluation of systemic fungicides against *M. phaseolina*

T₁: Thiophanate methyl

T₂: Tebuconazole

T₃: Carbendazim

T₄: Difenoconazole

T₅: Propiconazole

C: Control

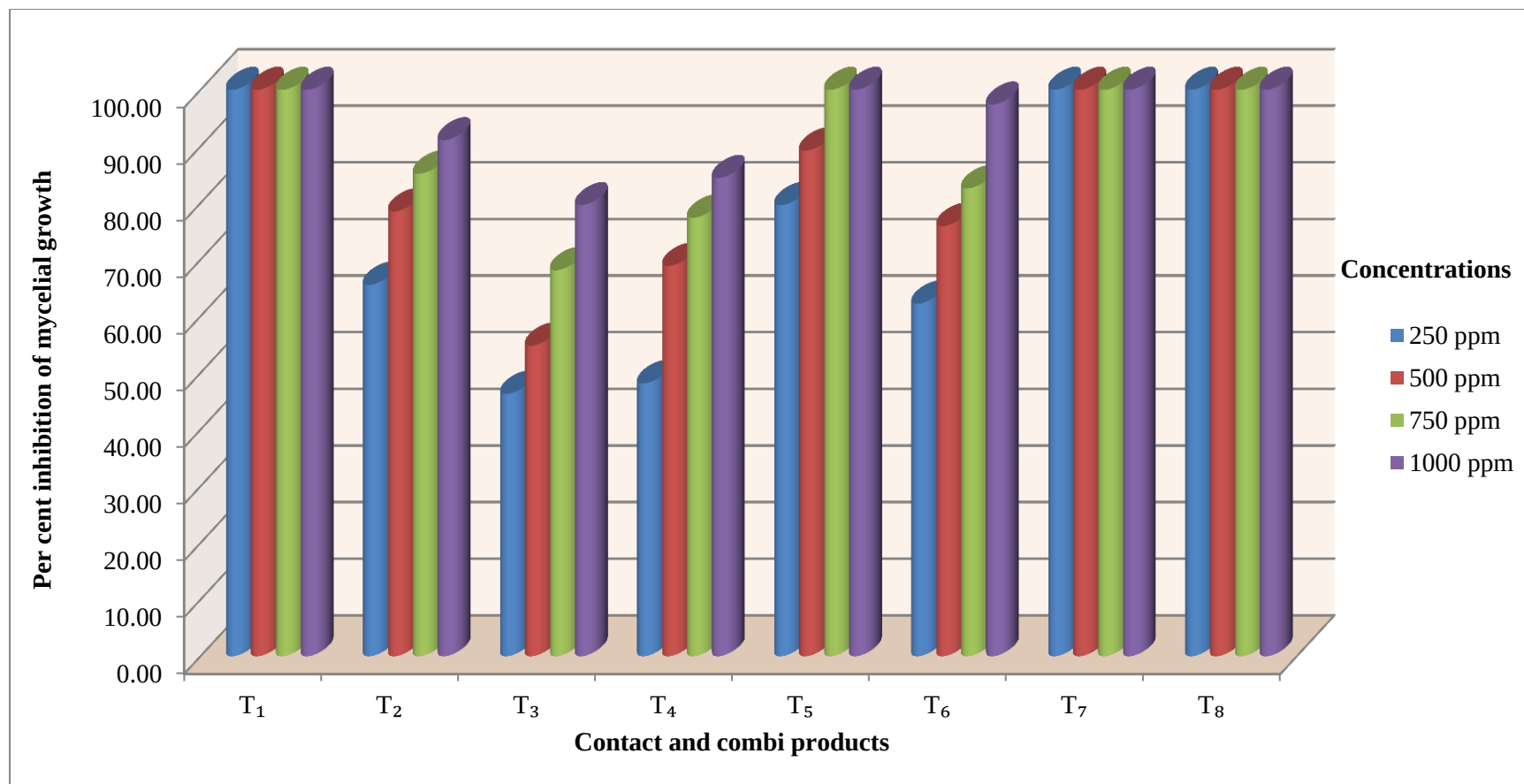


Fig. 13. In-vitro evaluation of contact and combi- products fungicides against *M. phaseolina*

T₁: Mancozeb

T₂: Thiram

T₃: Chlorothalonil

T₄: Captan

T₅: Tricyclazole 4 % + Mancozeb 62 % WP

T₆: Hexaconazole 4 % + Zineb 68% WP

T₇: Carboxin 37.5 % + Thiram 37.5 % WP

T₈: Carbendazim 12 % + Mancozeb 63 % WP

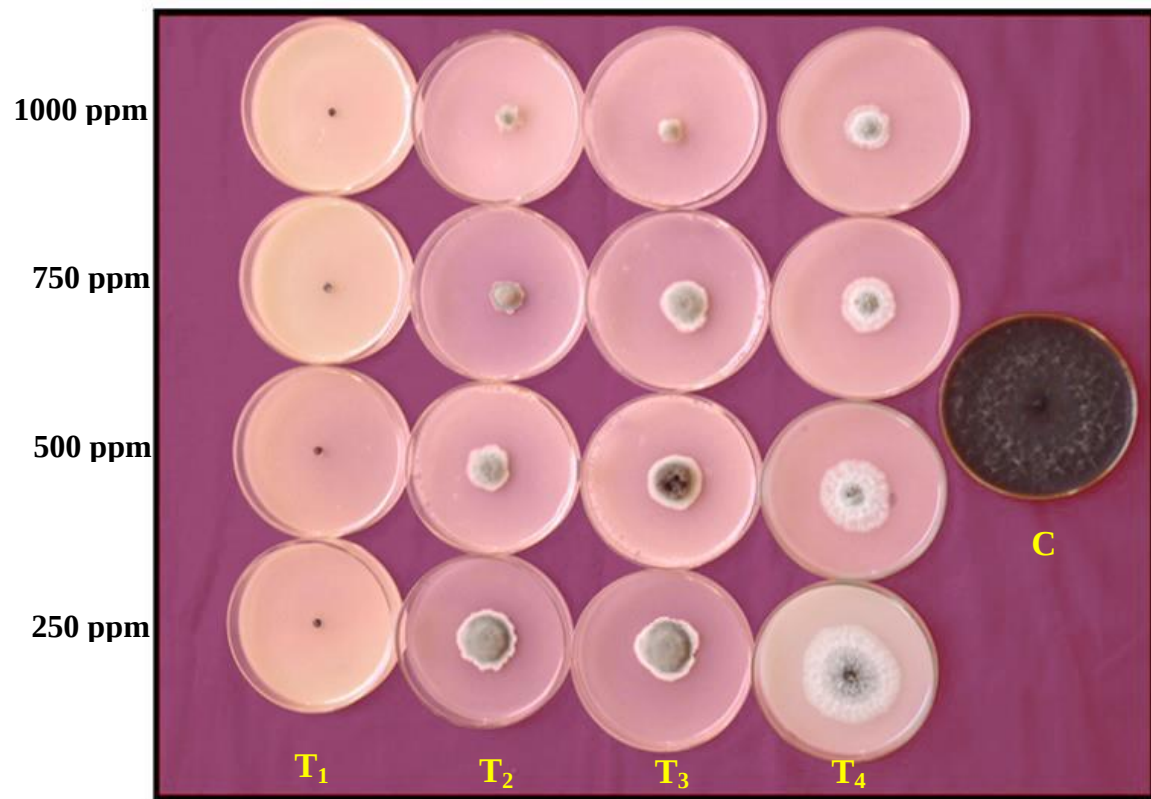


Plate 13. *In-vitro* evaluation of contact fungicides against *M. phaseolina*

T₁: Mancozeb
T₂: Thiram
T₃: Chlorothalonil
T₄: Captan
C : Control

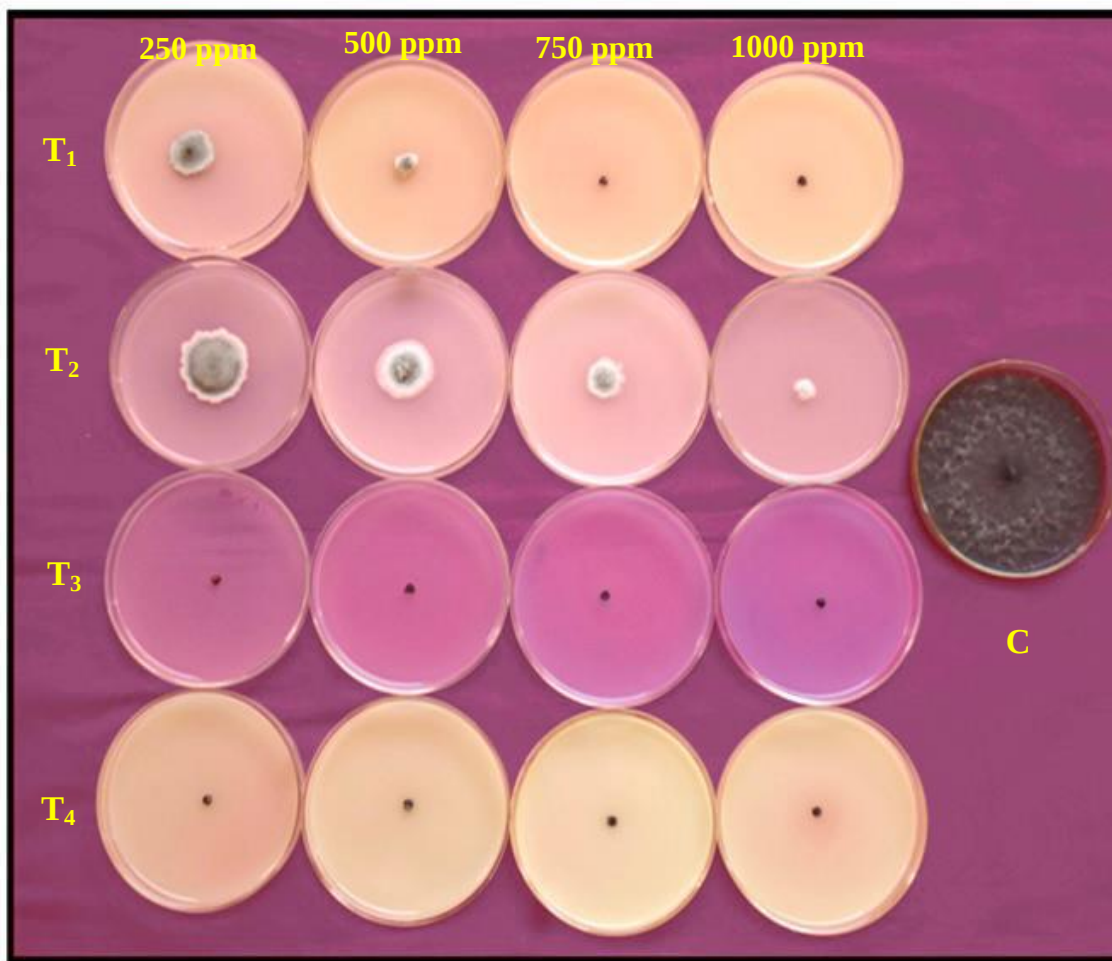


Plate 14. *In-vitro* evaluation of combi-product fungicides against *M. phaseolina*

T₁: Tricyclazole 4 % + Mancozeb 62 % WP (Merger)

T₂: Hexaconazole 4 % + Zineb 68 % WP (Avtar)

T₃: Carboxin 37.5 + Thiram 37.5 % WP (Vitavax power)

T₄: Carbendazim 12 % + Mancozeb 63 % WP (Companion)

C : Control

4.4.3.2 Shoot length

The shoot length was found to be maximum in seed treatment with tebuconazole at 2 g/kg of seed (20.51 cm) which was on par with seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed (20.50 cm) followed by seed treatment with carbendazim 2 g + thiram 1 g/kg of seed (19.40 cm) and seed treatment with carbendazim 2 g + captan 1 g/kg of seed (19.29 cm) over the control which recorded least shoot length of 11.01 cm (Table 30 and Plate 15).

4.4.3.3 Root length

The highest root length was observed when the seed were treated with tebuconazole at 2 g/kg of seed (17.22 cm) which was on par with seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed (17.20 cm) followed by seed treatment with carbendazim 2 g + thiram 1 g/kg of seed (16.01 cm) which was on par with seed treatment with carbendazim 2 g + captan 1 g/kg of seed 15.99 over the control (8.38 cm) presented in Table 30 and Plate 15.

4.4.3.4 Number of branches

The number of branches ranged from 3 to 5 in different treatments, number of branches were found to more (5) in seed treatment with tebuconazole at 2 g/kg of seed and seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed and least (3) were recorded in control (Table 30 and Plate 15). However, there was no significant difference in number of branches between the treatments.

4.4.3.5 Disease incidence

The disease incidence was found to be less in seed treatment with tebuconazole at 2 g/kg of seed (12.44 %) which was followed by seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed (20.19 %) and the highest per cent disease incidence was recorded in control with 50.50 per cent (Table 30 and Plate 15).

4.4.3.6 Shoot weight

The shoot weight was found to be maximum in seed treatment with tebuconazole at 2 g/kg of seed (1.72 g) which was on par with seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed (1.70 g) followed by seed treatment with carbendazim 2 g + thiram 1 g/kg of seed (1.05 g) which was on par with seed treatment with carbendazim 2 g + captan 1 g/kg of seed (1.03 g) over the control which recorded least shoot weight of 0.29 g (Table 30 and Plate 15).

4.4.3.7 Root weight

The highest root weight was recorded when the seed were treated with tebuconazole at 2 g/kg of seed (0.99 g) which was on par with seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed (0.98 g) followed by seed treatment with carbendazim 2 g + thiram 1 g/kg of seed (0.75 g) which was on par with seed treatment with carbendazim 2 g + captan 1 g/kg of seed (0.75 g) over the control (0.09 g) presented in Table 30 and Plate 15.

Table 30. Effect of treatments on germination, shoot length, root length, no. of branches, disease incidence, shoot weight and root weight under greenhouse condition

Treatments	Germination (%)	Shoot length (cm)	Root length (cm)	No. of branches	Disease incidence (%)	Shoot weight (g)	Root weight (g)
T ₁ : ST with <i>T. viridae</i> @ 10 g/kg of seed	96.30 ^e (78.91)	19.01 ^d (25.85)	14.51 ^d (22.39)	4.00 (12.00)	34.75 ^h (36.12)	0.87 ^d (5.35)	0.67 ^d (4.70)
T ₂ : Application of <i>T. viridae</i> 5 kg in 50 kg of FYM/ha	96.31 ^e (78.93)	19.00 ^e (25.84)	14.50 ^d (22.38)	4.00 (12.00)	34.91 ^g (36.22)	0.86 ^d (5.32)	0.67 ^d (4.70)
T ₃ : Application of <i>T. viridae</i> 10 kg in 50 kg of FYM/ha	96.31 ^e (78.93)	19.03 ^d (25.86)	14.52 ^d (22.40)	4.00 (12.00)	33.17 ⁱ (35.17)	0.86 ^d (5.32)	0.66 ^d (4.66)
T ₄ : ST with carbendazim 2 g + thiram 1 g/kg of seed	97.05 ^b (80.11)	19.40 ^b (26.13)	16.01 ^b (23.59)	4.00 (11.54)	23.43 ^m (28.95)	1.05 ^b (5.88)	0.75 ^b (4.97)
T ₅ : ST with carbendazim 2 g + captan 1 g/kg of seed	97.03 ^b (80.08)	19.29 ^c (26.05)	15.99 ^b (23.57)	4.00 (11.48)	25.63 ^l (30.42)	1.03 ^b (5.82)	0.75 ^b (4.97)
T ₆ : ST with tebuconazole 2 g/kg of seed	98.01 ^a (81.89)	20.51 ^a (26.93)	17.22 ^a (24.52)	5.00 (12.00)	12.44 ^o (20.65)	1.72 ^a (7.54)	0.99 ^a (5.71)
T ₇ : ST with tebuconazole 2 g + carbendazim 1g/kg of seed	98.00 ^a (81.87)	20.50 ^a (26.92)	17.20 ^a (24.50)	5.00 (11.94)	20.19 ⁿ (26.70)	1.70 ^a (7.49)	0.98 ^a (5.68)
T ₈ : ST with vitavax power 2 g/kg of seed	96.55 ^c (79.30)	18.50 ^f (25.47)	15.09 ^c (22.86)	5.00 (11.48)	29.50 ^j (32.90)	0.92 ^c (5.50)	0.70 ^c (4.80)
T ₉ : ST with vitavax power 2 g + <i>T. viridae</i> 5g/kg of seed	96.30 ^e (78.91)	19.00 ^e (25.84)	14.50 ^d (22.38)	4.00 (11.54)	26.24 ^k (30.81)	0.87 ^d (5.35)	0.67 ^d (4.70)
T ₁₀ : Neem cake 250 kg/ha	95.58 ⁱ (77.86)	14.53 ^h (22.41)	11.62 ⁱ (19.93)	4.00 (11.54)	37.74 ^d (37.90)	0.48 ^f (3.97)	0.35 ⁱ (3.39)
T ₁₁ : Pongamia cake 250 kg/ha	95.54 ^g (77.81)	14.48 ⁱ (22.37)	11.60 ⁱ (19.91)	4.00 (11.54)	39.60 ^c (39.00)	0.47 ^f (3.93)	0.37 ⁱ (3.49)
T ₁₂ : Neem cake 250 kg/ha + <i>T. viridae</i> 5 g/kg of seed	96.40 ^d (79.06)	15.51 ^g (23.19)	13.74 ^e (21.76)	4.00 (11.54)	35.92 ^f (36.82)	0.52 ^e (4.14)	0.44 ^e (3.80)
T ₁₃ : Pongamia cake 250 kg/ha + <i>T. viridae</i> 5 g/kg of seed	96.42 ^d (79.09)	15.50 ^g (23.18)	13.72 ^e (21.74)	4.00 (12.00)	37.43 ^e (37.72)	0.52 ^e (4.14)	0.44 ^e (3.80)
T ₁₄ : Sorghum 1:3 ratio	94.27 ^h (76.15)	13.16 ⁱ (21.27)	11.00 ^g (19.37)	4.00 (11.54)	41.82 ^b (40.29)	0.41 ^g (3.67)	0.28 ^g (3.03)
T ₁₅ : Control	90.05 ⁱ (71.61)	11.01 ^k (19.38)	8.38 ^h (16.83)	3.00 (11.94)	50.50 ^a (45.29)	0.29 ^h (3.09)	0.09 ^h (1.72)
S. Em ±	0.009	0.004	0.010	NS	0.011	0.020	0.026
CD at 1 %	0.035	0.017	0.038	NS	0.043	0.079	0.101

* Figures in parentheses are arc sin angular transformed values



Plate 15. Management of dry root rot under greenhouse condition

Similarly in safflower, anucop, captan and mancozeb M-45 were found extremely effective in reducing *Fusarium oxysporum* f. sp. *carthami* wilt. The seed treatments improved seed germination, seedling vigour and plant stand. Due to these treatments many of the seed-borne fungi failed to express. Bioagents formulations viz., *Trichoderma harzianum*, *Bacillus subtilis* and *Pseudomonas fluorescens* reduced the wilt incidence both under greenhouse and field conditions, thereby enhancing the growth of the seedlings. These antagonists significantly reduced the population of *F. oxysporum* f.sp. *carthami*, increased the seed germination and seedling vigour. Nine fungicide enhanced seed germination, root-shoot length and seedling vigour. Among them captan (0.2 %) and mancozeb M-45 (0.3 %) followed by vitavax, bavistin, captafol, milzim (0.2 %) and volzim (0.3 %) were more effective (Govindappa *et al.*, 2011).

Studies on chickpea revealed that benomyl was found effective followed by *P. fluorescens* in the inhibition of mycelial growth of *Fusarium solani*. Maximum germination percentage was recorded with benomyl (Benomyl 500 WP) at 2 g/kg seed (treated check) and Tebuconazole at 1 ml/kg seed (95 per cent) followed by treatment with *P. fluorescens* at 10 g/kg seed along with soil application of *P. fluorescens* at 3 kg/acre (94 per cent). Maximization of growth parameters like root length, shoot length and total seedling length were observed highest with benomyl at 2 g/kg seed as 17.0 cm, 10.3 cm and 27.3 cm, respectively (Bharathi *et al.*, 2014).

The germination of chickpea in pot was found to be increased by treating seeds with Secure 600 WG (51.03 %) and Folicure 200 W (51.03 %) followed by BAU-Biofungicide (48.98 %) over soil inoculated with *M. phaseolina*. Disease incidence decreased by treating seeds with Secure 600 WG (74.04 %) which was statistically similar to the treating seeds with Heprodin 50 WP (73.94 %), BAU-Biofungicide (73.64 %), Provex 200 WP (73.32 %), Gaivet (72.17 %) and Du-du 70 SL (71.75 %). Shoot length (38.41 %), root length (66.83 %), vigour index (122.44 %), shoot dry weight (117.78 %) and root dry weight (158.33 %) were increased when seeds were treated with BAU-Biofungicide over soil inoculated with pathogens (Golam Morshed, 2013).

4.4.4 Management of dry root rot in field condition

Management of soil borne diseases with host plant resistance and cultural methods have limited value in chickpea. Thus there is an urgent need to formulate effective disease management module by using chemicals as a first barrier for plant disease management by treating the seed with chemicals while sowing. Therefore, experiments were conducted to find out the best chemical and biocontrol agents for effective management of dry root rot of chickpea caused by *M. phaseolina* (Plate 16).

4.4.4.1 Management of dry root rot disease (DRR) during *rabi* season (2014-15)

Among the treatments employed in management of DRR during the *rabi* season of 2014-15, seed treatment with tebuconazole (2 g/kg of seed) recorded significantly lowest dry root rot incidence of 7.12 per cent with highest yield of 754.74 kg/ha, followed by seed treatment with tebuconazole (2 g/kg) + carbendazim (1 g/kg) recorded 14.87 per cent disease incidence with a yield of 721.05 kg/ha and seed treatment with carbendazim 2 g + thiram 1 g/kg of seed recorded 18.11 per cent disease incidence with

yield of 699 kg/ha compared to control which recorded 42.18 per cent diseases incidence with yield of 309.86 kg/ha (Table 31).

The per cent disease reduction over control was superior in seed treatment with tebuconazole at 2 g/kg of seed (83.12 %) followed by seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed (64.75%) and seed treatment with carbendazim 2 g + thiram 1 g/kg of seed (57.06 %).

The per cent yield increase over the control was highest in seed treatment with tebuconazole at 2 g/kg of seed (58.95 %) followed by seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed (57.03 %) and seed treatment with carbendazim 2 g + thiram 1 g/kg of seed (55.70 %).

4.4.4.2 Management of dry root rot disease during *kharif* season (2014-15)

During *kharif* 2014-15, seed treatment with tebuconazole (2 g/kg of seed) recorded lowest dry root rot incidence (10.28 %) with highest yield of 697.36 kg/ha, followed by seed treatment with tebuconazole (2 g/kg) + carbendazim (1 g/kg) which recorded 16.44 per cent disease incidence with yield of 637.86 kg/ha and seed treatment with carbendazim 2 g + thiram 1 g/kg of seed recorded 19.97 per cent disease incidence with yield of 600.45 kg/ha compared to control which recorded highest disease incidence of 45.64 per cent with yield of 259.55 kg/ha (Table 32).

The per cent disease reduction over control was higher in seed treatment with tebuconazole at 2 g/kg of seed (77.48 %) followed by seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed (63.98 %) and seed treatment with carbendazim 2 g + thiram 1 g/kg of seed (56.24 %).

The per cent yield increase over the control was highest in seed treatment with tebuconazole at 2 g/kg of seed (62.78 %) followed by seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed (59.31 %) and seed treatment with carbendazim 2 g + thiram 1 g/kg of seed (56.77 %).

4.4.4.3 Pooled data of two seasons employed to manage dry root rot of chickpea

The data obtained from two seasons (*rabi* and *kharif*) was pooled and analyzed statistically to know the best treatments in both the seasons and the results are presented here.

Mean data of both *rabi* and *kharif* indicated that seed treatment with tebuconazole (2 g/kg of seed) recorded least disease incidence of 8.70 per cent with highest yield of 726.05 kg/ha and C: B (cost benefit) ratio (1:2.37), followed by seed treatment with tebuconazole (2 g/kg) + carbendazim (1 g/kg) which recorded diseases incidence of 15.66 per cent with yield of 679.45 kg/ha and C:B ratio (1: 2.20) which was followed by with carbendazim 2 g + thiram 1 g/kg of seed which recorded disease incidence of 19.04 per cent with yield of 649.93 kg/ha and C: B ratio (1:2.12) compared to control which

Table 31: Management of dry root rot disease of chickpea during *rabi* 2014-15

Sl. No.	Treatments	Per cent disease incidence*	Per cent disease reduction over control	Yield kg/ha	Per cent yield increase over control
1.	T ₁ : ST with <i>T. viridae</i> @ 10 g/kg of seed	29.43 (32.85) ^{fg}	30.23	510.70 ^h	39.33
2.	T ₂ : Application of <i>T. viridae</i> 5 kg in 50 kg of FYM/ha	29.59 (32.95) ^{fg}	29.85	507.93 ⁱ	39.00
3.	T ₃ : Application of <i>T. viridae</i> 10 kg in 50 kg of FYM/ha	27.85 (31.85) ^g	33.97	550.17 ^g	43.68
4.	T ₄ : ST with carbendazim 2 g + thiram 1 g/kg of seed	18.11 (25.19) ^j	57.06	699.40 ^c	55.70
5.	T ₅ : ST with carbendazim 2 g + captan 1 g/kg of seed	20.31 (26.79) ⁱ	51.85	671.88 ^d	53.88
6.	T ₆ : ST with tebuconazole 2 g/kg of seed	7.12 (15.48) ^l	83.12	754.74 ^a	58.95
7.	T ₇ : ST with tebuconazole 2 g + carbendazim 1g/kg of seed	14.87 (22.68) ^k	64.75	721.05 ^b	57.03
8.	T ₈ : ST with vitavax power 2 g/kg of seed	24.18 (29.45) ^h	42.67	600.36 ^f	48.39
9.	T ₉ : ST with vitavax power 2 g + <i>T. viridae</i> 5 g/kg of seed	20.92 (27.22) ⁱ	50.40	641.03 ^e	51.66
10.	T ₁₀ : Neem cake 250 kg/ha	32.42 (34.71) ^d	23.14	480.24 ^l	35.48
11.	T ₁₁ : Pongamia cake 250 kg/ha	34.28 (35.84) ^c	18.73	461.03 ^m	32.79
12.	T ₁₂ : Neem cake 250 kg/ha + <i>T. viridae</i> 5 g/kg of seed	30.60 (33.58) ^{ef}	27.45	490.82 ^j	36.87
13.	T ₁₃ : Pongamia cake 250 kg/ha + <i>T. viridae</i> 5 g/kg of seed	32.11 (34.52) ^{de}	23.87	485.91 ^k	36.23
14.	T ₁₄ : Sorghum 1:3 ratio	36.50 (37.17) ^b	13.47	440.86 ⁿ	29.72
15.	T ₁₅ : Control	42.18 (40.50) ^a	0.00	309.86 ^o	0.00
S. Em ±		0.37	----	0.45	----
CD at 5 %		1.08	----	1.29	----

*Figures in the parenthesis are arcsine transformed values

ST - Seed Treatment

Table 32: Management of dry root rot disease of chickpea during *kharif* 2014-15

Sl. No.	Treatments	Per cent disease incidence*	Per cent disease reduction over control	Yield kg/ha	Per cent yield increase over control
1.	T ₁ : ST with <i>T. viridae</i> @ 10 g/kg of seed	30.51 (33.53) ^e	33.15	484.00 ^h	46.37
2.	T ₂ : Application of <i>T. viridae</i> 5 kg in 50 kg of FYM/ha	30.69 (33.64) ^e	32.76	481.39 ^h	46.08
3.	T ₃ : Application of <i>T. viridae</i> 10 kg in 50 kg of FYM/ha	29.22 (32.72) ^f	35.98	500.28 ^g	48.12
4.	T ₄ : ST with carbendazim 2 g + thiram 1 g/kg of seed	19.97 (26.54) ⁱ	56.24	600.45 ^c	56.77
5.	T ₅ : ST with carbendazim 2 g + captan 1 g/kg of seed	23.08 (28.71) ^h	49.43	565.57 ^d	54.11
6.	T ₆ : ST with tebuconazole 2 g/kg of seed	10.28 (18.70) ^k	77.48	697.36 ^a	62.78
7.	T ₇ : ST with tebuconazole 2 g + carbendazim 1g/kg of seed	16.44 (23.92) ^j	63.98	637.86 ^b	59.31
8.	T ₈ : ST with vitavax power 2 g/kg of seed	26.08 (30.71) ^g	42.86	521.63 ^f	50.24
9.	T ₉ : ST with vitavax power 2 g + <i>T. viridae</i> 5 g/kg of seed	23.33 (28.88) ^h	48.88	550.54 ^e	52.86
10.	T ₁₀ : Neem cake 250 kg/ha	32.83 (34.96) ^d	28.07	433.77 ^j	40.16
11.	T ₁₁ : Pongamia cake 250 kg/ha	35.41 (36.52) ^c	22.41	422.16 ^k	38.52
12.	T ₁₂ : Neem cake 250 kg/ha+ <i>T. viridae</i> 5 g/kg of seed	31.70 (34.27) ^{de}	30.54	475.61 ^h	45.43
13.	T ₁₃ : Pongamia cake 250 kg/ha + <i>T. viridae</i> 5 g/kg of seed	32.62 (34.83) ^d	28.53	461.28 ⁱ	43.73
14.	T ₁₄ : Sorghum 1:3 ratio	38.04 (38.08) ^b	16.65	400.50 ^l	35.19
15.	T ₁₅ : Control	45.64 (42.50) ^a	0.00	259.55 ^m	0.00
S. Em ±		0.26	----	3.54	----
CD at 5 %		0.77	----	10.24	----

*Figures in the parenthesis are arcsine transformed values

ST - Seed Treatment

recorded 43.91 per cent disease incidence with yield of 284.70 kg/ha and C:B ratio of 1:0.95 (Table 33 and Fig 14, 15).

These findings are in confirmation with Vijay *et al.* (2006) they reported that carbendozin (0.2 %) and Etaconazole (0.1 %) used as seed treatment, soil drenching and seed treatment plus soil drenching recorded lowest disease incidence of 15.60 per cent and 18.20 per cent and highest grain yield of 19.20 and 18.90 q/ha respectively, during *rabi* 2001-2002 and 2002-2003 crop seasons. The above treatment recorded 57.20 and 52.40 per cent higher yield over control with per rupee return of 12.78 and 11.80 respectively in fungicidal management of dry root rot of chickpea. Similarly Manjunatha, (2014) reported that seed treatment with tebuconazole 2 g/kg recorded lowest per cent disease incidence of 26.61, with a highest yield of 722.81 kg/ha, which is on par with seed treatment with tebuconazole 2 g + carbendazim 1 g/kg which recorded 38.59 per cent disease incidence with a yield of 596.74 kg/ha followed by seed treatment with carbendazim 2 g + captan 1 g /kg of seed recorded 46.27 per cent disease incidence with a mean yield of 519.92 kg/ha. They are significantly superior to control which recorded 95.77 per cent disease incidence with a lowest mean yield of 362.02 kg/ha.

The fungicide tebuconazole was highly effective against *Fusarium graminearum* (Spolti *et al.*, 2012). Tebuconazole was a systemic fungicide of the triazole group, and the primary mode of action was the inhibition of ergosterol biosynthesis in fungi (Hewitt, 1998). Even though different triazole fungicides had a similar mechanism of action, they may show marked differences in their activity against different fungal pathogens (Buchenauer, 1987; Scheinpflug and Kuck, 1987). This broad-spectrum, relatively new triazole fungicide was being used for its effectiveness against soil-borne and foliar fungal diseases in nut, fruit, cereal and vegetable crops worldwide (Munoz-Leoz *et al.*, 2011). Results of the present study are in agreement with Brenneman *et al.* (1991) who reported the antifungal activity against *S. rolfii* and *R. solani* *in-vitro*. Kanwal *et al.* (2012) evaluated the potential of tebuconazole against three problematic soil-borne fungal phytopathogens viz., *M. phaseolina*, *F. oxysporum* f. sp. *lycopersici* and *S. rolfii*. Different concentrations of this fungicide viz. 35, 70, 105 and 140 ppm were evaluated against the target fungal pathogen using food poisoning technique. All the concentrations of tebuconazole completely arrested the growth of all the three fungal species.

The development of biological techniques using plant growth promoting rhizobacteria (PGPR) amended with suitable bioformulations was an emerging trend in plant protection to reduce the plant diseases. Production of antibiotics viz., siderophore, HCN, pyrrolnitrin, phenazine and 2,4-diacetyl phloroglucinol and lytic enzymes viz., chitinase, β -1,3-glucanase which degraded the fungal cell wall and restricted the growth of the fungus were reported by many workers (Vivekananthan *et al.*, 2004; Sendhilvel *et al.*, 2005 and Saravanakumar *et al.*, 2007).

Out of 8 treatments tested against *Rhizoctonia bataticola*, the lowest per cent disease incidence (4.33 %) was recorded in seed treatment with carbendazim @ 2 g/kg of seed + seed treatment with *T. viride* @ 4 g/kg of seed + soil application of FYM fortified with *T. viride* and highest yield of 893 kg/ha compared to control with highest per cent

Table 33. Management of dry root rot disease of chickpea 2014-15 (Pooled analysis)

Sl. No.	Treatments	Per cent disease incidence*			Yield kg/ha			C:B Ratio
		2014-15 (rabi)	2014-15 (kharif)	Mean	2014-15 (rabi)	2014-15 (kharif)	Mean	
1.	T ₁ : ST with <i>T. viridae</i> @ 10 g/kg of seed	29.43 (32.85)	30.51 (33.53)	29.97 (33.19) ^{ef}	510.70	484.00	497.35 ^{gh}	1:1.64
2.	T ₂ : Application of <i>T. viridae</i> 5 kg in 50 kg of FYM/ha	29.59 (32.95)	30.69 (33.64)	30.14 (33.30) ^{ef}	507.93	481.39	494.66 ^{gh}	1:1.53
3.	T ₃ : Application of <i>T. viridae</i> 10 kg in 50 kg of FYM/ha	27.85 (31.85)	29.22 (32.72)	28.54 (32.29) ^f	550.17	500.28	525.23 ^{fg}	1:1.62
4.	T ₄ : ST with carbendazim 2 g + thiram 1 g/kg of seed	18.11 (25.19)	19.97 (26.54)	19.04 (25.87) ⁱ	699.40	600.45	649.93 ^{bc}	1:2.12
5.	T ₅ : ST with carbendazim 2 g + captan 1 g/kg of seed	20.31 (26.79)	23.08 (28.71)	21.70 (27.76) ^h	671.88	565.57	618.73 ^{cd}	1:2.00
6.	T ₆ : ST with tebuconazole 2 g/kg of seed	7.12 (15.48)	10.28 (18.70)	8.70 (17.15) ^k	754.74	697.36	726.05 ^a	1:2.37
7.	T ₇ : ST with tebuconazole 2 g + carbendazim 1 g/kg of seed	14.87 (22.68)	16.44 (23.92)	15.66 (23.31) ^j	721.05	637.86	679.45 ^b	1:2.20
8.	T ₈ : ST with vitavax power 2 g/kg of seed	24.18 (29.45)	26.08 (30.71)	25.13 (30.09) ^g	600.36	521.63	561.00 ^{ef}	1:1.82
9.	T ₉ : ST with vitavax power 2 g + <i>T. viridae</i> 5 g/kg of seed	20.92 (27.22)	23.33 (28.88)	22.13 (28.06) ^h	641.03	550.54	595.78 ^{de}	1:1.93
10.	T ₁₀ : Neem cake 250 kg/ha	32.42 (34.71)	32.83 (34.96)	32.63 (34.84) ^d	480.24	433.77	457.01 ^{hij}	1:0.39
11.	T ₁₁ : Pongamia cake 250 kg/ha	34.28 (35.84)	35.41 (36.52)	34.85 (36.18) ^c	461.03	422.16	441.60 ^{ij}	1:0.36
12.	T ₁₂ : Neem cake 250 kg/ha + <i>T. viridae</i> 5 g/kg of seed	30.60 (33.58)	31.70 (34.27)	31.15 (33.93) ^{de}	490.82	475.61	483.21 ^{ghi}	1:0.41
13.	T ₁₃ : Pongamia cake 250 kg/ha + <i>T. viridae</i> 5 g/kg of seed	32.11 (34.52)	32.62 (34.83)	32.37 (34.68) ^d	485.91	461.28	473.59 ^{hi}	1:0.38
14.	T ₁₄ : Sorghum 1:3 ratio	36.50 (37.17)	38.04 (38.08)	37.27 (37.63) ^b	440.86	400.50	420.68 ^j	1:1.35
15.	T ₁₅ : Control	42.18 (40.50)	45.64 (42.50)	43.91 (41.50) ^a	309.86	259.55	284.70 ^k	1:0.95
S. Em ±		0.39			14.57			----
CD at 5 %		1.18			44.21			----

*Figures in the parenthesis are arcsine transformed values

ST - Seed Treatment

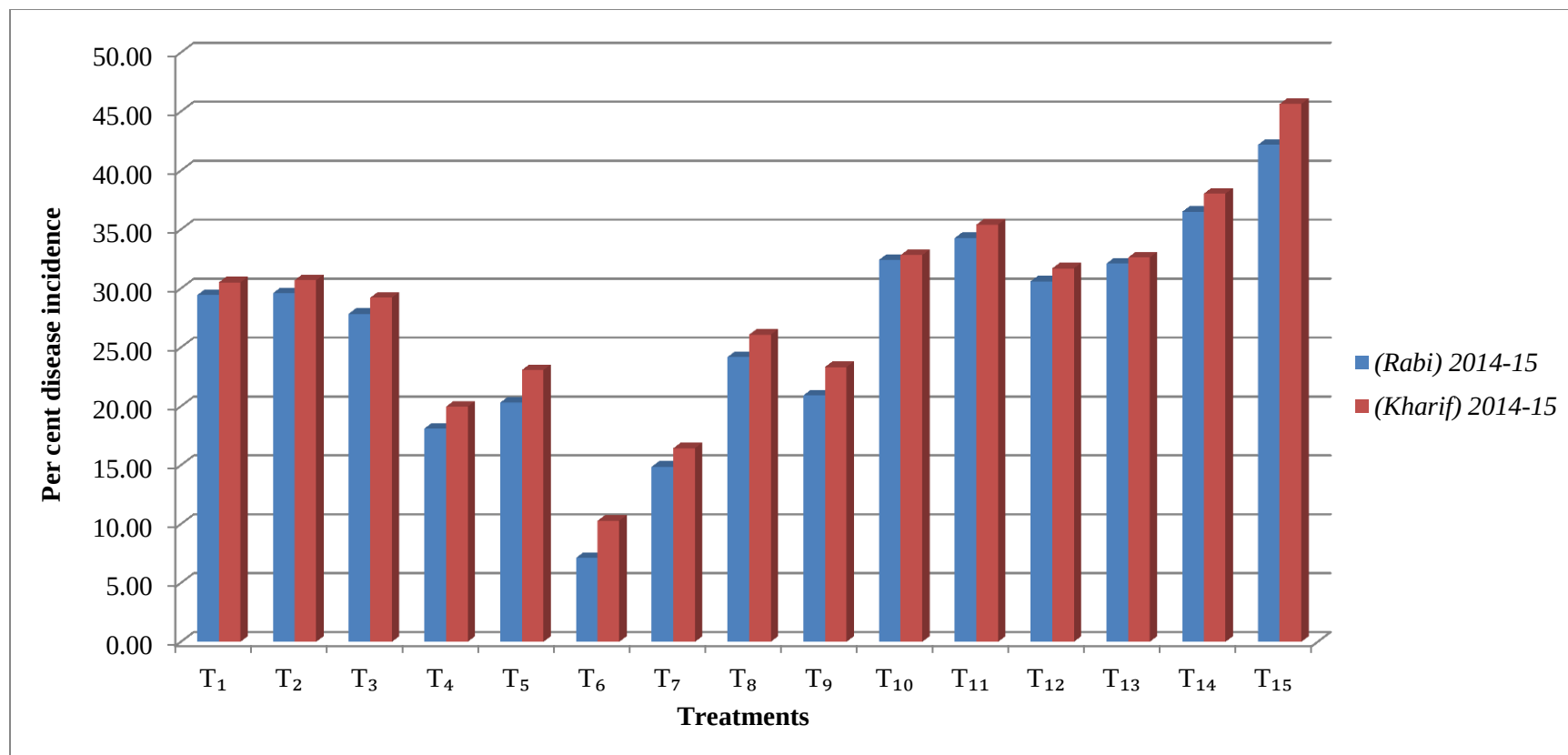


Fig. 14. Per cent dry root rot disease incidence of chickpea during rabi and kharif 2014-15

T₁: ST with *T. viridae* @ 10 g/kg of seed

T₂: Application of *T. viridae* 5 kg in 50 kg of FYM/ha

T₃: Application of *T. viridae* 10 kg in 50 kg of FYM/ha

T₄: ST with carbendazim 2 g + thiram 1 g/kg of seed

T₅: ST with carbendazim 2 g + captan 1 g/kg of seed

T₆: ST with tebuconazole 2 g/kg of seed

T₇: ST with tebuconazole 2 g + carbendazim 1g/kg of seed

T₈: ST with vitavax power 2 g/kg of seed

T₉: ST with vitavax power 2 g + *T. viridae* 5 g/kg of seed

T₁₀: Neem cake 250 kg/ha

T₁₁: Pongamia cake 250 kg/ha

T₁₂: Neem cake 250 kg/ha + *T. viridae* 5 g/kg of seed

T₁₃: Pongamia cake 250 kg/ha + *T. viridae* 5 g/kg of seed

T₁₄: Sorghum 1:3 ratio

T₁₅: Control

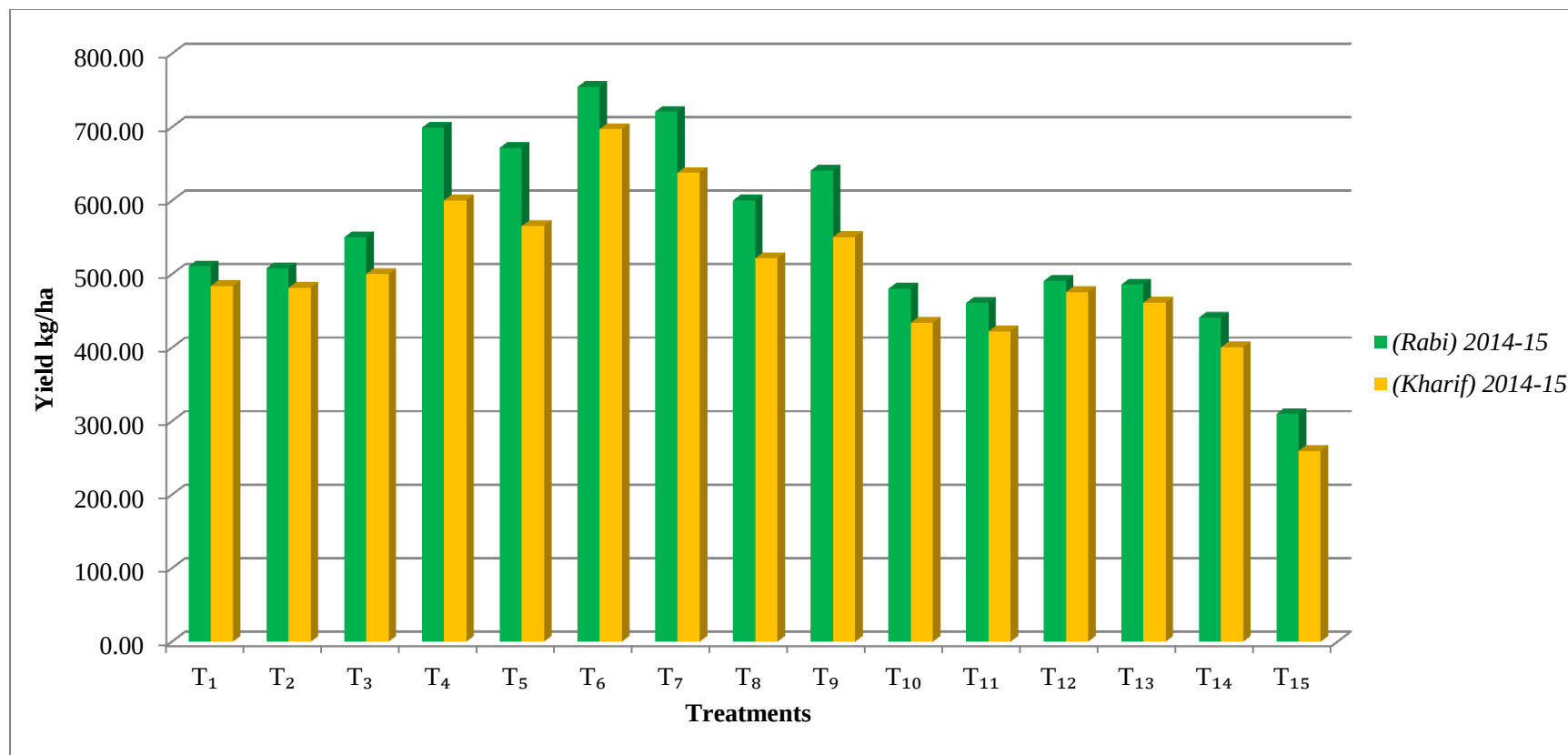


Fig. 15. Yield of chickpea during *rabi* and *kharif* 2014-15

T₁: ST with *T. viridae* @ 10 g/kg of seed

T₂: Application of *T. viridae* 5 kg in 50 kg of FYM/ha

T₃: Application of *T. viridae* 10 kg in 50 kg of FYM/ha

T₄: ST with carbendazim 2 g + thiram 1 g/kg of seed

T₅: ST with carbendazim 2 g + captan 1 g/kg of seed

T₆: ST with tebuconazole 2 g/kg of seed

T₇: ST with tebuconazole 2 g + carbendazim 1g/kg of seed

T₈: ST with vitavax power 2 g/kg of seed

T₉: ST with vitavax power 2 g + *T. viridae* 5g/kg of seed

T₁₀: Neem cake 250 kg/ha

T₁₁: Pongamia cake 250 kg/ha

T₁₂: Neem cake 250 kg/ha + *T. viridae* 5 g/kg of seed

T₁₃: Pongamia cake 250 kg/ha + *T. viridae* 5 g/kg of seed

T₁₄: Sorghum 1:3 ratio

T₁₅: Control



Plate 16. General view of dry root rot management experimental plot

disease incidence (35.03 %) and least yield of 660 kg/ha. The next best treatment was seed treatment with *T. viride* @ 4 g/kg of seed + soil application FYM fortified with *T. viride* with diseases incidence of 8.21 per cent and yield of 850 kg/ha followed by seed treatment with *T. viride* @ 4 g/kg of seed which recorded diseases incidence of 12.45 per cent and yield of 813 kg/ha (Nagamani *et al.*, 2011).

Future line of work

1. Technique for quick detection of dry root rot disease in field
2. Identification of host differentials and prevailing races in *M. phaseolina*
3. Effect of climate change on disease development

V SUMMARY

Dry root rot caused by *Macrophomina phaseolina* (Tassi) Goid., is considered as the important and destructive disease of pulses. Considering the importance of the disease, the present investigation was conducted and the results of the research work carried out with respect to survey for dry root rot of chickpea in southern parts of Karnataka, assessment of molecular variability in *M. phaseolina* isolates from chickpea, screening of chickpea germplasm against dry root rot and biochemical analysis of cultivars and management of dry root rot in chickpea in both *in-vitro* and *in-vivo* condition are summarized below.

A survey conducted for dry root rot in chickpea growing areas of southern parts of Karnataka showed that the disease incidence varied from 8.65 to 69.30 per cent in different villages. Among the districts, the highest mean disease incidence was recorded in Chitradurga (56.10 %) followed by Tumakuru (52.57 %), Bengaluru urban (51.13 %), Bengaluru rural (40.53 %), Chikkaballapur (36.41 %), Mysuru (32.09 %), Hassan (29.97 %) and Chikkamagaluru (24.73 %).

Out of twenty isolates tested for pathogenicity and virulence, five isolates viz., KAMP-2, KAMP-3, KAMP-4, TNMP-10 and PUMP-14 caused cent per cent dry root rot incidence on all five cultivars; four isolates viz., KAMP-1, MHMP-12, UPMP-16 and BHMP-17 caused 90-92 per cent incidence; six isolates viz., KAMP-5, APMP-6, APMP-7, MPMP-13, UKMP-18 and WBMP-19 caused 80-90 per cent incidence and the remaining five isolates TEMP-8, TEMP-9, TNMP-11, UPMP-15 and HPMP-20 caused 70-80 per cent dry root rot disease incidence.

Based on the host differentials reaction against *M. phaseolina* isolates screened at five different locations viz., Bengaluru, Coimbatore, Raipur, Durgapur and Sabour, there may be prevalence of three different strains viz., I (Bengaluru, Coimbatore and Raipur), II (Durgapur), III (Sabour).

Molecular characterization of twenty isolates includes identification of *M. phaseolina* isolates with species specific primers MpKFI (5'-CCGCCAGAGGACTATCAAAC-3') and MpKRI (5' CGTCCGAAGCGAGGTGTATT-3') yielded single amplified product of 350 bp in all the isolates tested and the PCR amplified 18S rDNA region using ITS primers produced a band size of about 600bp. The phylogenetic tree obtained by comparing the nucleotide sequences of ITS region resulted in clusters A and B indicating the variability in *M. phaseolina* and existence of two types of pathotype.

Two hundred and two chickpea germplasm screened for host plant resistance against dry root rot incidence in field condition, none of the germplasm showed resistant reaction, while twenty five germplasm viz., Phule G 0405, Phule G 12107, GJG 1202, BG 372, BG 3054, IPC 2010-134, NDG 13-21, Phule G 0611, GJG 1205, IPC 2010-112, BG 3055, GJG 1207, DIBG 201, GNG 2258, IC 83325, GNG 1969, HK 4, JGK

29, JGK 31, BG 3059, JGK 30, Phule G 12404, ICCV 08412, ICCV08114 and ICCV05534 showed moderately resistant reaction with 11-20 per cent disease.

Out of ninety three chickpea germplasm lines screened for resistance against dry root rot incidence in blotter paper method, none of the germplasm showed resistant reaction, while nineteen germplasm viz., Phule G 12107, GJG 1202, BG 372, BG 3054, IPC 2010-134, NDG 13-21, Phule G 0611, GJG 1205, IPC 2010-112, BG 3055, GJG 1207, DIBG 20, GNG 2258, GNG 1969, HK 4, JGK 29, JGK 31, JGK 30 and Phule G 12404 showed moderately resistant reaction.

There was an increase in the activity of peroxidase, catalase, total phenols, Phenylalanine ammonia lyase and total free amino acids in all the genotypes when inoculated with *M. phaseolina*. Whereas, there was decrease in the activity of total sugar, reducing sugar and non-reducing sugar in all the chickpea genotypes upon inoculation with *M. phaseolina*. Highest enzymatic activity and sugar content was recorded in moderately resistant genotypes viz., Phule G 12107, NDG 13-21 and IPC 2010-112 compared to susceptible genotype GNG 2228 and highly susceptible genotype L550.

All the antagonists significantly inhibited mycelial growth of *M. phaseolina* and per cent inhibition ranged from 62.96 to 81.44 per cent. Among fungal and bacterial antagonists, the maximum inhibition of mycelial growth was observed in *Trichoderma harzianum* (Th-55) isolate with 81.44 per cent and minimum inhibition was observed in *T. harzianum* (GKVK) with 62.96 per cent.

Among five systemic fungicides evaluated, tebuconazole, carbendazim, difenoconazole and propiconazole recorded maximum mycelial inhibition of cent per cent at all concentrations. Whereas, thiophanate methyl recorded cent per cent inhibition at 500 ppm concentration. The contact fungicides viz., mancozeb, recorded cent per cent inhibition of mycelial growth at all the concentrations (250, 500, 750 and 1000 ppm) followed by thiram (80.10 %), captan (69.72 %) and chlorothalonil (62.22 %). The combi- products viz., carboxin 37.5 % + thiram 37.5 % WP and carbendazim 12 % + mancozeb 63 % WP showed cent per cent inhibition of mycelia at all concentrations (250, 500, 750 and 1000 ppm).

The growth parameters viz., seed germination (98.01 %), shoot length (20.51 cm), root length (17.22 cm), shoot weight (1.72 g) and root weight (0.99 g) was maximum in seed treatment with tebuconazole at 2 g/kg of seed which was on par with seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed over the control. However, there was no significant difference in number of branches between the treatments in pot experiment.

The lowest DRR disease incidence was recorded in seed treatment with tebuconazole at 2 g/kg of seed (12.44 %) which was followed by seed treatment with tebuconazole 2 g + carbendazim 1 g/kg of seed (20.19 %) and the highest per cent disease incidence was recorded in control with 50.50 per cent in pot culture experiment.

Dry root rot disease was effectively managed during *rabi* and *kharif* season under field condition by seed treatment with tebuconazole (2 g/kg of seed) which recorded lowest DRR disease incidence of 8.70 per cent with highest yield of 726.05 kg/ha and C: B (cost benefit) ratio (1:2.37), followed by seed treatment with tebuconazole (2 g/kg) + carbendazim (1 g/kg) which recorded diseases incidence of 15.66 per cent with yield of 679.45 kg/ha and C:B ratio (1: 2.20) compared to control which recorded 43.91 per cent disease incidence with yield of 284.70 kg/ha and C:B ratio of 1:0.95.

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Screening of Chickpea Genotypes for Dry Root Rot Resistance under Laboratory and Field Conditions

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ABSTRACT

A total of one hundred and forty seven chickpea genotypes were screened against dry root rot incidence caused by *Macrophomina phaseolina* under lab condition using blotter technique. None of the genotypes showed resistant reaction, while twenty three genotypes viz., BG 3055, GNG 2258, Phule G 12107, GJG 1202, GJG 1205, IPC 2010-112, DIBG 201, GJG 1207, Phule G 0611, BG 372, GJG 1208, IPC 2010-134, BG 3054, NDG 13-21, JGK 30, JGK 31, Phule G 12404, JGK 29, HK 4, GNG 1969, GNG 2207, BG 3044 and RSG 931 showed moderately resistance reaction. Out of two hundred and three genotypes screened under sick plot condition for dry root rot incidence, none of the genotypes showed resistance reaction. Whereas ten genotypes viz., ICCV05534, IPC 2010-72, IPC 2010-112, Phule G 0302, BG 3056, PG 071, BG 3059, ICCV 08412, ICCV08114 and IC 83325 showed moderate resistance.

CHICKPEA (*Cicer arietinum* L.), also known as Gram or Bengal gram, is the second most important pulse crop in the world, India accounting for 60 to 75% of the world's chickpea production. Chickpea seeds contain high quality easily digestible protein (25 per cent) and carbohydrates (20 per cent) making it an important source of protein. Thus it is also called as "Poor man's meat." In India it is grown in an area of about 9.51 million ha with a production of 8.83 million tonnes and a productivity of 929 kg/ha. In Karnataka it is grown in 0.8 million ha with production of 0.38 million tonnes and productivity of 473 kg/ha (Singh, 2013).

The crop is vulnerable to a number of diseases, some of which may be devastating. Chickpea suffers from about 172 pathogens consisting of fungi, bacteria, viruses and nematodes. Dry root rot of chickpea caused by *M. phaseolina* is a major constraint in chickpea production as it is emerging as a potential threat to chickpea cultivation in semi-arid regions because the host plant is predisposed to infection by moisture stress and high temperatures during the flowering to pod filling stage. The annual yield loss due to this disease alone is 10–20 per cent (Vishwadhar and Chaudhary, 2001). Use of host plant resistance is the most economical approach for the management of dry root in chickpea. A few chickpea lines with field tolerance to dry root rot have been identified, but high levels of resistance

are scarce in cultivated genotypes (Reddy *et al.*, 1991). Keeping this in view the present investigations were undertaken to evaluate performance of chickpea genotypes against dry root rot by using blotter paper technique and field screening.

The culture of *M. Phaseolina* was inoculated to 250 ml Potato dextrose broth and incubated for seven days at room temperature. The mycelial mat from flask was removed and macerated by adding 100 ml of distilled water. Seven days old seedlings of different genotypes including susceptible checks (BG212 and L550) were uprooted, roots were immersed in the inoculum of *M. phaseolina* for a minute and placed in between blotter papers. Later, incubated at 35° C for eight days under 12h of light and dark condition, 80 per cent RH. The blotters were moistened with sterile water everyday. On the eighth day, the seedlings were examined for lesions on root and scored for the disease severity. Disease severity was scored by using 1-9 scale (Nene *et al.*, 1981). Based on the disease score the genotypes were grouped as mentioned below (Table I).

A total of two hundred and three chickpea genotypes were screened in *M. phaseolina* sick plot field at G.K.V.K., Bengaluru, during *rabi* season of 2014-2015. Each genotype was sown in five meter row length. After every five test entries one line of

TABLE I
Categorization of chickpea genotypes into different disease reactions

Rating	Category	Symptoms
1	Resistant	No infection on roots
3	Moderately resistant	Very few small lesions on roots
5	Moderately susceptible	Lesions on roots clear but small, new roots free from infection
7	Susceptible	Lesions on roots many, new roots generally free from lesions
9	highly susceptible	Roots infected and completely discoloured

susceptible check L550 was sown. At the time of sowing giant culture of *M. phaseolina* of Bangalore isolate was incorporated to the soil to improve sickness

$$\text{Per cent disease incidence} = \frac{\text{Number of plants infected}}{\text{Total number of plants observed}} \times 100$$

of the soil. Observations on per cent dry root rot incidence were recorded from pod formation to maturity stage. The following formula was used to calculate dry root rot disease incidence. After estimating disease incidence, the entries were categorized into different disease reactions as given by Riyaz Ahmad Khan *et al.* (2013).

Among one hundred and forty seven chickpea genotypes screened for host plant resistance against dry root rot incidence under lab condition using blotter technique, none of the entries showed resistant reaction, while twenty three entries *viz.*, BG 3055, GNG 2258, Phule G 12107, GJG 1202, GJG 1205, IPC 2010-112, DIBG 201, GJG 1207, Phule G 0611, BG 372, GJG 1208, IPC 2010-134, BG 3054, NDG 13-21, JGK 30, JGK 31, Phule G 12404, JGK 29, HK 4, GNG 1969, GNG 2207, BG 3044 and RSG 931 showed

moderately resistant reaction (Table II). The present findings are well supported by the field and blotter screening., Saifulla *et al.* (2011) screened chickpea entries for dry root rot under field condition and found that 21 entries *viz.*, GNG 1861, Phule G 07112, BGD1056(R), RSG 931, Phule G 07101, Vijay, RSG 888, GJG 0825, PG064, GNG 1947, JG 1307, GBC 6 (AVT-1), GNG 2002, GNG 1936, CSJ 313, JG 18, HIR 60, JG 11, ICCV 07107 and ICCV 08323 were resistant and 36 entries were found moderately resistant to dry root rot out of 196 genotypes screened. Out of forty seven lines screened against dry root rot of chickpea, three genotypes were resistant, 22 were moderately resistant, 19 susceptible and three highly susceptible under blotter paper technique (Pande *et al.*, 2004).

Out of two hundred and three genotypes screened under sick plot condition for dry root rot incidence, none of the genotypes showed resistance reaction. While, ten genotypes *viz.*, ICCV05534, IPC 2010-72, IPC 2010-112, Phule G 0302, BG 3056, PG 071, BG 3059, ICCV 08412, ICCV08114 and IC 83325 showed moderately resistance reaction (Table III). Similarly Nagamma and Saifulla (2012) screened sixty four Kabuli genotypes against dry root rot and found six

TABLE II
Disease reaction of chickpea entries for dry root rot under blotter paper technique

No. of genotypes	Reaction	Genotypes
0	R	—
23	MR	BG 3055, GNG 2258, Phule G 12107, GJG 1202, GJG 1205, IPC 2010-2012, DIBG 201, GJ 1207, Phule G 0611, BG 372, GJG1208, IPC 2010-134, BG 3054, NDG 13-21, JGK 30, JGK 31, Phule G 12404, JGK 29, HK 4, GNG 1969, GNG 2207, BG 3044 and RSG 931
65	MS	—
45	S	—
14	HS	—
Total No. of genotypes		147

TABLE III
Disease reaction of chickpea genotypes for dry root rot under field condition

Disease incidence	Reaction	No. of genotypes	Genotype name
0-10%	Resistant	0	—
11-20%	Moderately Resistant	10	ICCV05534, IPC 2010-72, IPC 2010-112, Phule G 0302, BG 3056, PG 071, BG 3059, ICCV 08412, ICCV08114 and IC 83325
21-30%	Moderately Susceptible	20	—
31-50%	Susceptible	57	—
51-100%	Highly Susceptible	116	—
Total No. of genotypes		203	

genotypes viz., Phule G 04305, IPCK 07-62, RVSSG 12, HK 08-212, Phule G 09305 and AKG 2002-1K showed resistant reaction, nine genotypes viz., HK 94-134, GNG 1888, HK 06-152, GNG 2112, HK 06-171, Phule G 09316, CSJK 74, JGK 13 and CSJK 70 showed moderately resistant reaction. Cultivation of resistant varieties is an economical approach for the management of dry root rot of chickpea, but, only a few sources with low level of genetic resistance are available. Management can be made feasible and cost effective by identification of new resistant sources as this is the only alternative method so as to combat the disease, even though chemical control plays an important role in modern agriculture for disease management (Gupta *et al.*, 1997).

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Management of dry root rot disease in chickpea (*Cicer Arietinum* L.) caused by *Macrophomina Phaseolina*

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Abstract: A field experiment was carried out during *Rabi* and *Kharif* season at ZARS, UAS, GKVK, Bangalore under sick plot soil condition. Among the 15 treatments employed in management of dry root rot (DRR) during the *Rabi* season seed treatment with tebuconazole (2g kg⁻¹ of seed) recorded significantly lowest dry root rot incidence of 7.12 % with highest yield of 754.74 kg ha⁻¹, followed by seed treatment with tebuconazole (2g kg⁻¹) + carbendazim (1g kg⁻¹) recorded 14.87 % disease incidence with a yield of 721.05 kg ha⁻¹ compared to control which recorded 42.18 % diseases incidence with yield of 309.86 kg ha⁻¹. During *Kharif* season also seed treatment with tebuconazole (2g kg⁻¹ of seed) recorded lowest dry root rot incidence of 10.28 % with highest yield of 697.36 kg ha⁻¹, followed by seed treatment with tebuconazole (2g kg⁻¹) + carbendazim (1g kg⁻¹) recorded 16.44 % disease incidence with yield of 637.86 kg ha⁻¹ compared to control which recorded highest disease incidence of 45.64 % with yield of 259.55 kg ha⁻¹. Mean data of *Rabi* and *Kharif* seasons indicated that seed treatment with tebuconazole (2g kg⁻¹ of seed) recorded least disease incidence (8.70 %) with highest yield of 726.05 kg ha⁻¹ and C: B ratio (1:2.37), followed by seed treatment with tebuconazole (2g kg⁻¹) + carbendazim (1g kg⁻¹) recorded 15.66 % diseases incidence with yield of 679.45 kg ha⁻¹ and C: B ratio of 1:2.20 compared to control which recorded 43.91 % diseases incidence with yield of 284.70 kg ha⁻¹ and C: B ratio of 1:0.95.

Key words: Chickpea, *Macrophomina phaseolina*, Management, DRR

Introduction

Chickpea (*Cicer arietinum* L.), is the second most important pulse crop in the world. It is also known as Gram or Bengal Gram. In India, accounts around 60 and 75 per cent of the world's chickpea production. Chickpea seeds contain high quality easily digestible protein (25%) and carbohydrates (~20%) making it an important source of protein for the vegetarians of the country and thus it is also called "Poor man's meat." In India it is grown in an area of about 9.51 m ha with production of 8.83 million tones and productivity of 929 kg ha⁻¹. In Karnataka it is grown in 0.8 m ha with production of 0.38 million tones and productivity of 473 kg ha⁻¹ (Singh, 2013).

Chickpea plays a significant role in improving soil fertility by fixing the atmospheric nitrogen. Chickpea meets 80 % of its nitrogen (N) requirement from symbiotic nitrogen fixation and can fix up to 140 kg N ha⁻¹ from air. It leaves substantial amount of residual nitrogen for subsequent crops and adds plenty of organic matter to maintain and improve soil health and fertility. Because of its deep tap root system, chickpea can withstand drought conditions by extracting water from deeper layers in the soil profile (Gaur *et al.*, 2010). The per hectare production is low in spite of high yielding varieties and new agronomic practices. The reasons of low yields are so many,

apart from other reasons the main cause of low yields of this crop is the incidence of diseases. The crop is vulnerable to a number of airborne and soil-borne diseases, some of which may be devastating. Chickpea suffers from about 172 pathogens consisting of fungi, bacteria, viruses and nematodes. Soil borne diseases such as wilt (*Fusarium oxysporum* f. sp. *ciceris*), dry root rot (*Macrophomina phaseolina*), black root rot (*Fusarium solani*), collar rot (*Sclerotium rolfsii*), and stem rot (*Sclerotinia sclerotiorum*) are important yield reducers. The foliar diseases, which may seriously damage chickpea under favourable conditions are blight caused by *Ascochyta rabiei* and grey mould caused by *Botrytis cinerea*. Dry root rot of chickpea caused by *M. phaseolina* is a major constraint in the chickpea production as it is emerging as a potential threat to chickpea cultivation in semi-arid regions because the host plant is predisposed to infection by moisture stress and high temperatures during the flowering to pod filling stage (Sharma *et al.* 2010). The annual yield loss due to this disease alone is 10-20 % (Vishwadhara and Chaudhary, 2001).

The dry root rot disease generally appears around flowering and podding stage. The disease may also appear at seedling stage, however, the susceptibility of the plant increases

with age. The disease generally appears when day temperature is more than 30°C and soil moisture content of 60 %. Drooping of petioles and leaflets is confined to those at the very top of the plant. Sometimes when the rest of the plant is dry, the top most leaves are chlorotic. The leaves and stems of affected plants are usually straw colored. The lower portion of the tap root usually remains in the soil when plants are uprooted. The tap root is dark, shows signs of rotting and is devoid of most of its lateral and finer roots. Dark, minute sclerotial bodies can be seen on the roots exposed or inside the wood (Nene *et al.*, 2012). *M. phaseolina* is primarily seed and soil-borne fungal pathogen. In chickpea, infected seeds and microsclerotia surviving in the soil are the major source of primary inoculum. The pathogen also has wide host range. The improved practices for disease management will pave a way to mitigate losses caused by dry root rot.

Materials and Methods

A field experiment was carried out during Rabi and Kharif season at ZARS, UAS, GKVK, Bangalore under sick plot soil condition during 2014-15. Three replications were maintained for each treatment with RCBD design. The % disease incidence and yield was recorded in individual treatments and the data was statistically analyzed. The treatment details are as follows.

- T1: Seed treatment with *Tricoderma viridae* @ 10g kg⁻¹ of seed
- T2: Application of *Tricoderma viridae* 5 kg in 50 kg of FYM ha⁻¹
- T3: Application of *Tricoderma viridae* 10 kg in 50 kg of FYM ha⁻¹
- T4: Seed treatment with carbendazim 2g + thiram 1 g kg⁻¹ of seed
- T5: Seed treatment with carbendazim 2g + captan 1 g kg⁻¹ of seed
- T6: Seed treatment with tebuconazole 2g kg⁻¹ of seed
- T7: Seed treatment with tebuconazole 2g + carbendazim 1g kg⁻¹ of seed
- T8: Seed treatment with vitavax power 2g kg⁻¹ of seed
- T9: Seed treatment with vitavax power 2g + *Tricoderma viridae* 5g kg⁻¹ of seed
- T10: Neem cake 250 kg ha⁻¹
- T11: Pongamia cake 250 kg ha⁻¹
- T12: Neem cake 250 kg ha⁻¹ + *Tricoderma viridae* 5 g kg⁻¹ of seed
- T13: Pongamia cake 250 kg ha⁻¹ + *Tricoderma viridae* 5 g kg⁻¹ of seed
- T14: Sorghum 1:3 ratio
- T15: Control

Results and Discussion

Among the treatments employed in management of DRR during the Rabi season of 2014-15, seed treatment with tebuconazole (2g kg⁻¹ of seed) recorded significantly lowest dry root rot incidence of 7.12 % with highest yield of 754.74 kg ha⁻¹, followed by seed treatment with tebuconazole (2g kg⁻¹) + carbendazim (1g kg⁻¹) recorded 14.87 % disease incidence with a yield of 721.05 kg ha⁻¹ compared to control which recorded 42.18 % diseases incidence with yield of 309.86 kg ha⁻¹ (Table-1). The disease reduction (%) was superior (83.12 %) and % yield increase over control was also superior (58.95 %) in seed treatment with tebuconazole (2g kg⁻¹ of seed), followed by seed treatment with tebuconazole (2g kg⁻¹) + carbendazim (1g kg⁻¹) which recorded 64.75 % disease reduction over control and 57.03 % yield increase over control.

During Kharif 2014-15, seed treatment with tebuconazole (2g kg⁻¹ of seed) recorded lowest dry root rot incidence (10.28 %)

with highest yield of 697.36 kg ha⁻¹, followed by seed treatment with tebuconazole (2g kg⁻¹) + carbendazim (1g kg⁻¹) recorded 16.44 % disease incidence with yield of 637.86 kg ha⁻¹ compared to control which recorded highest disease incidence of 45.64 % with yield of 259.55 kg ha⁻¹ (Table-1). The disease reduction (%) was superior (77.48 %) and % yield increase over control was also superior (62.78 %) in seed treatment with tebuconazole (2g kg⁻¹ of seed), followed by seed treatment with tebuconazole (2g kg⁻¹) + carbendazim (1g kg⁻¹) which recorded 63.98 % disease reduction over control and 59.31 % yield increase over control.

Mean data of both Rabi and Kharif indicated that seed treatment with tebuconazole (2g kg⁻¹ of seed) recorded least disease incidence of 8.70 % with highest yield of 726.05 kg ha⁻¹ and C: B (cost benefit) ratio (1:2.37), followed by seed treatment with tebuconazole (2g kg⁻¹) + carbendazim (1g kg⁻¹) recorded diseases incidence of 15.66 % with yield of 679.45 kg ha⁻¹ and C: B ratio (1: 2.20) compared to control which recorded 43.91 % disease incidence with yield of 284.70 kg ha⁻¹ and C: B ratio of 1:0.95 (Table-2).

These findings are in confirmation with Vijay Mohan *et al.*, (2006) which showed that carbendazim (0.2 %) and Etaconazole (0.1 %) used as seed treatment, soil drenching and seed treatment plus soil drenching recorded lowest disease incidence of 15.6 % and 18.2 % highest grain yield of 19.2 and 18.9 q ha⁻¹ respectively, during rabi 2001-2002 and 2002-2003 crop seasons. The above treatment recorded 57.2 and 52.4 % higher yield over control with per rupee return of 12.78 and 11.80 respectively in fungicidal management of dry root rot of chickpea caused by *R. bataticola*. Manjunatha, (2014) reported that seed treatment with tebuconazole 2g kg⁻¹ recorded lowest % disease incidence of 26.61, with a highest yield of 722.81 kg ha⁻¹, that is on par with seed treatment with tebuconazole 2g + carbendazim 1g kg⁻¹ which recorded 38.59 % disease incidence with a yield of 596.74 kg ha⁻¹ followed by seed treatment with carbendazim 2g + captan 1g kg⁻¹ of seed recorded 46.27 % disease incidence with a mean yield of 519.92 kg ha⁻¹. They are significantly superior to control which recorded 95.77 % disease incidence with a lowest mean yield of 362.02 kg ha⁻¹.

It has been shown that tebuconazole was highly effective against *Fusarium graminearum* (Spolti *et al.*, 2012). Tebuconazole is a systemic fungicide of the triazole group, and the primary mode of action is the inhibition of ergosterol biosynthesis in fungi (Hewitt, 1998). Even though different triazole fungicides have a similar mechanism of action, they may show marked differences in their activity against different fungal pathogens (Buchenauer, 1987; Scheinpflug and Kuck, 1987). This broad-spectrum, relatively new triazole fungicide is being used for its effectiveness against soil-borne and foliar fungal diseases in nut, fruit, cereal and vegetable crops worldwide (Munoz-Leoz *et al.*, 2011). Results of the present study are in agreement with Brenneman *et al.* (1991) who reported the antifungal activity against *S. rolfsii* and *R. solani* *in-vitro*. Kanwal *et al.* (2012) evaluated the potential of tebuconazole against three problematic soil-borne fungal phytopathogens namely *M. phaseolina*, *F. oxysporum* f. sp. *lycopersici* and *S. rolfsii*. Different concentrations of this fungicide viz. 35, 70, 105 and 140 ppm were evaluated against the target fungal pathogen using food poisoning

Table-1: Management of dry root rot disease in chickpea during Rabi 2014-15

Treatments	Rabi				kharif			
	% disease incidence	% disease reduction over contro	Yield kg ha ⁻¹	% yield increase over control	% disease incidence	% disease reduction over contro	Yield kg ha ⁻¹	% yield increase over control
T1	29.43 (32.85) ^a	30.23	510.70 ^a	39.33	30.51 (33.53) ^a	33.15	484.00 ^a	46.37
T2	29.59 (32.95) ^a	29.85	507.93 ^a	39.00	30.69 (33.64) ^a	32.76	481.39 ^a	46.08
T3	27.85 (31.85) ^a	33.97	550.17 ^a	43.68	29.22 (32.72) ^a	35.98	500.28 ^a	48.12
T4	18.11 (25.19)	57.06	699.40 ^c	55.70	19.97 (26.54) ^a	56.24	600.45 ^c	56.77
T5	20.31 (26.79)	51.85	671.88 ^d	53.88	23.08 (28.71) ^b	49.43	565.57 ^d	54.11
T6	7.12 (15.48) ^j	83.12	754.74 ^b	58.95	10.28 (18.70) ^b	77.48	697.36 ^b	62.78
T7	14.87 (22.68) ^j	64.75	721.05 ^b	57.03	16.44 (23.92) ^b	63.98	637.86 ^b	59.31
T8	24.18 (29.45) ^j	42.67	600.36 ^j	48.39	26.08 (30.71) ^j	42.86	521.63 ^j	50.24
T9	20.92 (27.22)	50.40	641.03 ^a	51.66	23.33 (28.88) ^b	48.88	550.54 ^a	52.86
T10	32.42 (34.71) ⁱ	23.14	480.24 ⁱ	35.48	32.83 (34.96) ^j	28.07	433.77 ^j	40.16
T11	34.28 (35.84) ^e	18.73	461.03 ^m	32.79	35.41 (36.52) ^c	22.41	422.16 ^k	38.52
T12	30.60 (33.58) ^{ef}	27.45	490.82 ⁱ	36.87	31.70 (34.27) ^{bc}	30.54	475.61 ^h	45.43
T13	32.11 (34.52) ^{bc}	23.87	485.91 ^k	36.23	32.62 (34.83) ^j	28.53	461.28 ^j	43.73
T14	36.50 (37.17) ^b	13.47	440.86 ^j	29.72	38.04 (38.08) ^b	16.65	400.50 ^j	35.19
T15	42.18 (40.50) ^a	0.00	309.86 ^j	0.00	45.64 (42.50) ^a	0.00	259.55 ^m	0.00
S.Em ±	0.37	—	0.45	—	0.26	—	3.54	—
CD@5%	1.08	—	1.29	—	0.77	—	10.24	—

Table-2: Management of dry root rot disease of chickpea during Rabi and Kharif 2014-15 (Pooled Data)

Treatments	% disease incidence			Yield kg ha ⁻¹			C:B Ratio
	2014-15(Rabi)	2014-15(Kharif)	Mean	2014-15 (Rabi)	2014-15 (Kharif)	Mean	
T1	29.43 (32.85)	30.51 (33.53)	29.97 (33.19) ⁱ	510.70	484.00	497.35 ^h	1:1.64
T2	29.59 (32.95)	30.69 (33.64)	30.14 (33.30) ⁱ	507.93	481.39	494.66 ^h	1:1.53
T3	27.85 (31.85)	29.22 (32.72)	28.54 (32.29) ^j	550.17	500.28	525.23 ^a	1:1.62
T4	18.11 (25.19)	19.97 (26.54)	19.04 (25.87) ^j	699.40	600.45	649.93 ^{cc}	1:2.12
T5	20.31 (26.79)	23.08 (28.71)	21.70 (27.76) ^j	671.88	565.57	618.73 ^{cd}	1:2.00
T6	7.12 (15.48)	10.28 (18.70)	8.70 (17.15) ^j	754.74	697.36	726.05 ^a	1:2.37
T7	14.87 (22.68)	16.44 (23.92)	15.66 (23.31) ^j	721.05	637.86	679.45 ^b	1:2.20
T8	24.18 (29.45)	26.08 (30.71)	25.13 (30.09) ^j	600.36	521.63	561.00 ^{ef}	1:1.82
T9	20.92 (27.22)	23.33 (28.88)	22.13 (28.06) ^j	641.03	550.54	595.78 ^{be}	1:1.93
T10	32.42 (34.71)	32.83 (34.96)	32.63 (34.84) ^j	480.24	433.77	457.01 ^{hi}	1:0.39
T11	34.28 (35.84)	35.41 (36.52)	34.85 (36.18) ^j	461.03	422.16	441.60 ⁱ	1:0.36
T12	30.60 (33.58)	31.70 (34.27)	31.15 (33.93) ^{bc}	490.82	475.61	483.21 ^{gh}	1:0.41
T13	32.11 (34.52)	32.62 (34.83)	32.37 (34.68) ^j	485.91	461.28	473.59 ^{hi}	1:0.38
T14	36.50 (37.17)	38.04 (38.08)	37.27 (37.63) ^j	440.86	400.50	420.68 ^j	1:1.35
T15	42.18 (40.50)	45.64 (42.50)	43.91 (41.50) ^j	309.86	259.55	284.70 ^k	1:0.95
S. Em ±	—	—	0.39	—	—	14.57	—
CD @ 5 %	—	—	1.18	—	—	44.21	—

Treatments are: T1: ST with *T. viridae* @ 10g kg⁻¹ of seed; T2: Application of *T. viridae* 5 kg in 50 kg of FYM ha⁻¹; T3: Application of *T. viridae* 10 kg in 50 kg of FYM ha⁻¹; T4: ST with carbendazim 2g + thiram 1 g kg⁻¹ of seed; T5: ST with carbendazim 2g + captan 1 g kg⁻¹ of seed; T6: ST with tebuconazole 2g kg⁻¹ of seed; T7: ST with tebuconazole 2g + carbendazim 1g kg⁻¹ of seed; T8: ST with vitavax power 2g kg⁻¹ of seed; T9: ST with vitavax power 2g + *T. viridae* 5g kg⁻¹ of seed; T10: Neem cake 250 kg ha⁻¹; T11: Pongamia cake 250 kg ha⁻¹; T12: Neem cake 250 kg ha⁻¹ + *T. viridae* 5 g kg⁻¹ of seed; T13: Pongamia cake 250 kg ha⁻¹ + *T. viridae* 5 g kg⁻¹ of seed; T14: Sorghum 1:3 ratio; T15: Control

*Figures in the parenthesis are arcsine transformed values, * ST - Seed Treatment

technique. All the concentrations of tebuconazole completely arrested the growth of all the three fungal species.

Out of 8 treatments tested against *Rhizoctonia bataticola*, the lowest % disease incidence (4.33) was recorded in seed treatment with carbendazim @ 2 g kg⁻¹ of seed + seed treatment with *T. viride* @ 4 g kg⁻¹ of seed + soil application of FYM fortified with *T. viride* and highest yield of 893 kg ha⁻¹ compared to control with highest % disease incidence (35.03) and least yield of 660 kg ha⁻¹. The next

best treatment was recorded in seed treatment with *T. viride* @ 4 g kg⁻¹ of seed + soil application FYM fortified with *T. viride* with diseases incidence of 8.21 % and yield of 850 kg ha⁻¹ followed by seed treatment with *T. viride* @ 4 g kg⁻¹ of seed which recorded diseases incidence of 12.45 % and yield of 813 kg ha⁻¹ Nagamani *et al.*, 2011.

Management of soil borne diseases with host plant resistance and cultural methods have limited value in chickpea. Thus there is a need to formulate effective disease management module by using

chemicals as a first barrier for controlling the plant disease development by treating the seed with chemicals while sowing. The improved practices for disease management will pave a way to mitigate losses caused by dry root rot and improves livelihoods of the poor farmers. In this view seed treatment with tebuconazole 2g kg⁻¹ of seed was found to be a best management practice for the control of dry root rot followed by seed treatment with tebuconazole 2g + carbendazim 1g kg⁻¹ of seed.

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